Credit Risk

EU Rules & ECB / EBA Regulations

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Chapter 1

Reference regulatory framework

1.1 Hierarchy of sources and scope

The reference regulatory framework for credit risk management in financial institutions operating in the European Union is structured across multiple hierarchical layers and brings together prudential, accounting, and supervisory provisions. At the top sit the Union's primary and secondary legislation (regulations and directives), followed by the technical standards and guidelines issued by the European supervisory authorities, as well as guidance issued by the competent authorities. At national level, the transposition measures of the directives and the supervisory instructions ensure effective implementation and alignment with the specific features of the domestic market. This overall framework governs the entire credit lifecycle and the measurement of risk, with implications for methodologies, governance arrangements, reporting processes, and compliance profiles.

1.1.1 European sources

The main European sources are:

- The **standards issued by the Basel Committee on Banking Supervision** form the foundation of the international prudential framework. They define the minimum capital requirements, capital conservation buffers, and the methods for calculating risk-weighted assets (RWA), thereby ensuring the soundness and comparability of banks' balance sheets at global level. In the European Union, these principles are incorporated through the CRR/CRD framework, which translates their provisions into binding rules for financial intermediaries.
- **The European banking prudential framework**, commonly referred to as the *CRR/CRD package*, constitutes the legislative transposition within the European Union of the standards defined by the Basel Committee. It sets the **minimum capital requirements** that banks must hold in relation to the risks they take on, governs the **methods for measuring and controlling credit risk**, and lays down the **conditions for using** either standardized approaches or internal models (*Internal Ratings Based*, IRB). The framework also includes **corporate governance and internal control rules** designed to ensure prudent and transparent risk management, as well as the **market disclosure obligations** under *Pillar 3*, which promote comparability and market discipline among intermediaries.
- **The technical standards and guidelines of the European Banking Authority (EBA)** are the main operational tool for ensuring consistent and harmonized implementation of prudential rules across the European Union. The EBA issues **Binding Technical Standards** — the *Regulatory Technical Standards (RTS)* and the *Implementing Technical Standards (ITS)* — together with **Guidelines** (*GL*) and **Recommendations**, which are

applied according to the "*comply or explain*" principle and provide a uniform point of reference for national authorities and supervised intermediaries.

- **The international accounting framework (IFRS)** — adopted in the European Union for the **presentation and measurement of financial instruments** — governs the measurement of expected credit losses in line with the *Expected Credit Loss* (ECL) principle introduced by IFRS 9. This approach requires **banks to estimate expected losses on a forward-looking basis over the entire life of the exposure**, integrating **historical, current, and forward-looking information**. Ensuring consistency between the accounting and prudential frameworks is essential in order to align managerial, accounting, and regulatory metrics and to avoid misalignments in risk assessment processes and in the determination of absorbed capital.

- **The acts of the European Central Bank (ECB)** occupy, in the hierarchy of sources, an implementing and supplementary level with respect to the Union's primary legislation (CRR/CRD) and the standards issued by the EBA. In exercising its **direct banking supervision functions**, the **ECB** issues various types of acts: **Regulations**, which are binding and of general application; **Decisions**, addressed to specific entities such as individual significant banks; and **Guidelines** and **Recommendations**, which steer the interpretation and uniform implementation of prudential rules. These instruments operationalize supervisory requirements by setting qualitative expectations on governance, risk management, ICAAP/ILAAP, stress testing, and reporting, thereby helping to **ensure the consistency and effectiveness of the Single Supervisory Mechanism (SSM)**.

1.1.2 National sources

The national provisions issued by the competent authorities constitute the implementation layer of the European regulatory framework, ensuring the consistent application of EU directives and regulations within the country's economic and operational context. In Italy, these functions are exercised primarily by the **Bank of Italy**, as the prudential supervisory authority, and by **Consob**, for transparency-related aspects and the protection of the financial market. The Bank of Italy issues **Supervisory Instructions**, which set out the operational, organizational, and methodological rules that banks must comply with for the safe and prudent management of credit risk and for the proper execution of **internal model validation**. In particular, they govern:

- **Organizational structure and governance**: identification of roles and responsibilities, internal control safeguards, independence and expertise of the risk management, validation, and audit functions;

- **Internal models**: requirements for authorization, development, calibration, *backtesting*, and continuous monitoring of model performance, as well as the methods for documentation and periodic review;

- **Regulatory and statistical reporting**: reporting obligations to the Bank of Italy, in line with European reporting frameworks (COREP, FINREP, AnaCredit) and with the information requirements of the Single Supervisory Mechanism;

- **Relations with the supervisory authority**: procedures for **authorization**, **notification**, and **engagement** with the competent authorities, including *on-site* and *off-site* inspections, aimed at ensuring the reliability of risk management and measurement processes.

1.1.3 Authorities' guidance and official interpretation

Interpretative guidance in prudential matters is based on three main layers of direction and supervision, which ensure consistency and uniformity in the application of European and national regulations.

- The **European Banking Authority (EBA)** issues *Guidelines* (GL) and official *Q&A* that provide clarifications on methodological, calculation, and regulatory reporting aspects, fostering convergence of supervisory practices among Member States. EBA Q&As, published on an ongoing basis, constitute the main operational interpretative reference for authorities and intermediaries.

- The **European Central Bank (ECB)**, through *guides*, *supervisory manuals*, and *letters of expectations* addressed to significant institutions, defines reference criteria and methodologies for the use and supervision of internal models (e.g., the *ECB Guide to Internal Models*), for the *Supervisory Review and Evaluation Process (SREP)*, and for standards of governance and internal control. These instruments outline the framework of the supervisor's qualitative expectations, ensuring a consistent assessment of risks and capital adequacy.

- The **national competent authority**, in Italy the **Bank of Italy**, transposes and operationalizes the European requirements through *Communications*, *Supervisory Provisions*, and *Explanatory Notes*, adapting them to the specific features of the domestic financial system. These acts define the implementation arrangements, internal controls, and reporting obligations in line with the European framework.

Hierarchy of sources. In the European regulatory framework, **Regulations** and **Directives** prevail, in legal terms, over **Guidelines**, **Q&A**, and other guidance documents. However, **official interpretations** issued by European and national authorities constitute an essential point of reference to ensure the uniform application of prudential rules, promote consistency in reporting processes, and ensure the proper convergence of supervisory practices within the Single Supervisory Mechanism (SSM).

1.1.4 Operational, compliance, and reporting implications

The European and national regulatory framework on credit risk has transversal and systemic effects across the entire chain of processes, models, and internal controls, imposing stringent requirements for consistency, traceability, and governance. These implications concern several operational areas:

- **Data governance** — the regulations require that the data used for credit risk measurement and reporting be **complete, accurate, consistent, timely and stable, unique, valid, accessible, traceable, and representative**. It is necessary to clearly define *data ownership*, ex-ante quality controls and ex-post checks, as well as to ensure full reconstructability of the *data lineage* and the *audit trail*, in line with the principles of *BCBS 239* and the *EBA Guidelines on data quality*.

- **Internal modelling** — the development and use of models for estimating risk parameters (*PD*, *LGD*, *EAD*) must comply with rigorous **methodological criteria** for **calibration**, independent **validation**, and continuous **monitoring**. Performance tests (discrimination, calibration, stability), *override* policies, the management of model changes (material and non-material), and complete, traceable technical and functional documentation are required.

- **Governance and internal controls** — the governance system must ensure the separation of roles between the **first line** (model development and use), the **second line** (risk management and validation), and the **third line** (internal audit). A solid *Model Risk Management Framework* is required, including the model inventory, risk classification, remediation plans, and the involvement of strategic oversight bodies.

- **Regulatory compliance** — institutions must ensure **alignment between internal models and prudential requirements**, guaranteeing consistency between managerial and regulatory measurements (*use test*). They must also oversee reporting obligations to supervisory authorities and ensure a continuous flow of regulatory and methodological updates.
- **Reporting and disclosure** — banks are required to produce harmonized information flows according to European standards (**COREP**, **FINREP**) and **Pillar 3** disclosures relating to exposures, credit quality, internal models, and risk metrics. Reconciliation with accounting and managerial data, consistency checks, and traceability of the assumptions and methodological choices feeding regulatory reporting must be ensured.

1.2 European regulations

1.2.1 Legal nature and direct applicability

European Union regulations are normative acts of general application and are binding in all their elements, as established by Article 288 of the *Treaty on the Functioning of the European Union* (TFEU). They differ from directives because they do not require any transposition measures by the Member States. Their entry into force produces immediate and direct legal effects for the addressees, whether public authorities or private parties. **Direct applicability** therefore means that regulatory provisions become immediately operative under the terms and conditions laid down by the regulation itself, ensuring uniformity and consistency in the implementation of EU law. However, the Union legislator may introduce certain elements of flexibility to facilitate the transition and tailor the application to different national contexts. In particular, EU legislation may provide for:

- **deferred application periods** or **transitional measures**, aimed at enabling intermediaries and competent authorities to gradually adjust processes, systems, and capital requirements;
- **national options and discretions**, exercisable within expressly prescribed limits, which allow supervisory authorities or the national legislator to tailor the application of specific prudential requirements (e.g., regarding buffers, risk weights, or materiality thresholds);
- **secondary-level acts** — such as *delegated regulations* and *implementing regulations* — which specify technical aspects and set out the operational modalities for implementing the main regulations (the so-called *Level 1 acts*).

1.2.2 Classification of the main regulations for the banking and financial sector

The regulatory framework of the European Union that governs prudential supervision, risk management, and the stability of financial markets is structured into several directly applicable regulations. They constitute the legal foundation of the European banking system and define, in a harmonized manner, the requirements for capital, liquidity, governance, and crisis resolution. Below is a classification of the main legal acts, grouped by the relevant functional areas.

Prudential supervision and risk management

The European **prudential supervision** framework defines the fundamental rules to ensure the soundness, stability, and resilience of the European Union's banking and financial system. It sets out the principles, quantitative and qualitative requirements, and the responsibilities of supervisory authorities and intermediaries, with the aim of preserving confidence in the market and preventing systemic risks. Prudential supervision is based on a harmonized set of directly applicable regulations that govern minimum capital requirements, integrated risk management, liquidity buffers, exposure limits, and crisis management and resolution mechanisms.

- **Capital Requirements Regulation (CRR) — Reg. (EU) No. 575/2013** and subsequent amendments (CRR II and CRR III): constitutes the central pillar of the prudential framework, setting the minimum *First Pillar* capital requirements for credit, market, and operational risks, as well as the leverage ratio, limits on large exposures, and liquidity requirements. The regulation is supplemented by delegated and implementing acts that specify, among other things, the *Liquidity Coverage Ratio* (LCR), the *Net Stable Funding Ratio* (NSFR), and the regulatory reporting templates (COREP and FINREP).
- **Single Supervisory Mechanism (SSM) Regulation — Reg. (EU) No. 1024/2013**: assigns the **European Central Bank** specific prudential supervisory tasks over significant banks in the euro area. The objective is to ensure consistent and integrated supervision, reducing regulatory fragmentation and ensuring uniform application of the prudential framework across Member States.
- **Single Resolution Mechanism Regulation (SRMR) — Reg. (EU) No. 806/2014**: establishes the regulatory framework for the orderly resolution of banking crises within the Banking Union. It defines the roles and powers of the **Single Resolution Board** (SRB), governs resolution planning, and introduces minimum requirements for own funds and eligible liabilities (*MREL*) consistent with the international *Total Loss-Absorbing Capacity* (TLAC) standards.
- **Securitisation Regulation — Reg. (EU) 2017/2402**: establishes a uniform framework for securitisation transactions, defining the criteria of simplicity, transparency, and standardisation (*STS*) and linking those requirements to the prudential treatment of exposures under the CRR. It aims to strengthen investor confidence and to support the financing of the real economy.
- **Investment Firm Regulation (IFR) — Reg. (EU) 2019/2033**: introduces a specific prudential regime for investment firms, with requirements proportionate to their risk profile and business model. The regulation interacts with the CRR/CRD package to ensure consistency of treatment within banking groups that also include such intermediaries.

Market infrastructures and financial instruments

The European regulatory framework on market infrastructures and financial instruments defines the rules governing the functioning of markets, the transparency of trading, the safety of settlement systems, and investor protection. These regulations aim to ensure the integrity of the single capital market, the mitigation of systemic risks, and the confidence of market participants through harmonized standards of supervision, governance, and disclosure.

- **EMIR Regulation — Reg. (EU) No. 648/2012**: establishes obligations for central *clearing*[1] and risk mitigation for *over-the-counter* (OTC) derivative contracts, as well as

[1] Clearing is the set of actions that follow the agreement to trade between two parties but precede the actual exchange of cash and securities.

reporting requirements to *trade repositories*. It introduces *risk management* standards and prudential requirements for central counterparties (*CCPs*), in order to reduce counterparty risk and the propagation of shocks in the financial system.

- **MiFIR Regulation — Reg. (EU) No. 600/2014**: complements the MiFID II directive, strengthening pre- and post-trade transparency, regulating access to trading venues, and defining the product intervention powers of competent authorities. The provisions directly affect market liquidity, the efficiency of price formation mechanisms, and intermediaries' market risk management.
- **CSDR Regulation — Reg. (EU) No. 909/2014**: governs the functioning of *Central Securities Depositories* (CSDs) and the settlement cycle of transactions in financial instruments. It introduces *settlement discipline*[2] obligations, measures to prevent *fails*, and operational requirements for risk control and the resilience of post-trading infrastructures.
- **MAR Regulation — Reg. (EU) No. 596/2014**: aims to prevent and sanction *market abuse* (insider trading and price manipulation), strengthening internal control, monitoring, and reporting arrangements. Banks and intermediaries must establish *market surveillance* procedures and compliance systems adequate to detect anomalous behaviour.
- **Benchmarks Regulation — Reg. (EU) 2016/1011**: governs the governance, methodology, and integrity of *benchmarks* used for price determination in financial contracts and investment products. It imposes transparency obligations on index providers and quality controls on input data, with significant impacts on pricing, contractual documentation, and risk management.
- **Prospectus Regulation — Reg. (EU) No. 2017/1129**: sets the disclosure requirements for public offerings of financial instruments and for their admission to trading on regulated markets. The objective is to ensure that investors have clear, complete, and comparable information on the risks and characteristics of the products offered.
- **PRIIPs Regulation — Reg. (EU) No. 1286/2014**: introduces the obligation to prepare a *Key Information Document* (KID) for retail investment products. This standardised document provides concise information on risks, costs, and potential returns, strengthening transparency and *product governance* to protect non-professional investors.

Sustainability and data

The European regulatory framework on sustainability and personal data protection has introduced new obligations and standards that profoundly affect banking processes, risk models, and information systems. Financial institutions are required to integrate environmental, social, and governance (*ESG*) factors into their decision-making, reporting, and risk management processes, while ensuring the protection and proper management of personal data. The main reference regulations are the following:

- **SFDR Regulation — Reg. (EU) No. 2019/2088**: introduces sustainability disclosure obligations both at the entity level and at the financial product level. Banks and investment firms must disclose how sustainability risks and ESG factors are integrated into decision-making, investment, and risk management processes, as well as publish impact indicators and mitigation strategies. SFDR disclosures strengthen market transparency and support convergence towards sustainable finance.

[2] Settlement discipline is the set of rules, measures, and mechanisms (organizational, operational, and sanctioning) introduced to prevent and reduce settlement fails in transactions involving financial instruments and to manage failed deliveries when they occur.

- **EU Taxonomy Regulation — Reg. (EU) 2020/852**: defines a unified classification system for environmentally sustainable economic activities, based on technical criteria for substantial contribution to environmental objectives and for the absence of significant harm (*Do No Significant Harm* - DNSH). This taxonomy directly affects non-financial reporting, environmental risk metrics, and supervisory reporting requirements, embedding sustainability into strategic planning and prudential reporting.

- **GDPR Regulation — Reg. (EU) No. 2016/679**: establishes the legal framework for the protection of personal data and for their management in accordance with the principles of lawfulness, fairness, transparency, minimisation, and security. In the banking and financial context, the GDPR imposes specific obligations regarding *data governance*, the definition of legal bases for data processing, access controls, and process traceability. This also concerns the use of data in risk models and statistical analyses, requiring that each processing operation be carried out in compliance with proportionality and with the protection of the data subject's rights.

1.2.3 Obligations and impacts on processes, controls, and capital

The *self-executing* nature of European regulations implies that supervisory provisions apply directly to banks and intermediaries, generating immediate effects on operational processes, control systems, and capital planning. This mechanism requires continuous organizational, technological, and methodological adjustment to ensure full compliance with harmonized prudential and accounting requirements. In particular:

- the **timely adjustment of risk metrics** and of the **measurement systems to the common European definitions** of exposure, default, risk mitigation, and concentration is required, ensuring uniformity of calculation and comparability of data across institutions;

- the **calculation of capital requirements** for credit risk must be carried out by choosing between the standardized approach or the internal ratings-based (*IRB*) approach, with direct impacts on *Risk-Weighted Assets* (RWA) and on the composition of regulatory capital (*CET1*, *AT1*, *T2*);

- the **implementation of leverage and liquidity constraints** (LCR and, where applicable, NSFR) requires integration with *Asset & Liability Management* (ALM), treasury, and funding planning processes, ensuring structural balance and financial resilience;

- **internal control and compliance safeguards** are necessary to ensure the traceability of regulatory sources, the reproducibility of calculations, and the robustness of end-to-end processes, through *data lineage* tools, data quality controls, and segregation of operational functions;

- the **adaptation of product governance and reporting processes** to the MiFIR, MAR, PRIIPs, SFDR, and Taxonomy regulations strengthens transparency towards the market and the consistency of disclosures to the various stakeholders.

In this context, capital and liquidity planning must incorporate stress scenarios, prudential limits, and regulatory buffers, ensuring alignment between risk strategies, internal metrics, and European constraints.

1.2.4 Internal credit risk models and governance requirements

The **Capital Requirements Regulation (CRR)** governs the use of internal models for regulatory purposes for the measurement of credit risk, allowing institutions to directly estimate the main risk parameters — *Probability of Default (PD)*, *Loss Given Default (LGD)*, and *Exposure at Default (EAD)* — within the *Internal Ratings Based (IRB)* approach. This possibility is subject to meeting stringent qualitative and quantitative requirements aimed at ensuring the reliability, comparability, and prudential soundness of the model results. The adoption of internal models entails:

- **prior authorization** by the competent authority (ECB or national authority), as well as ongoing adherence to the compliance and performance requirements set out in the CRR and in the *EBA Guidelines on PD and LGD estimation and treatment of defaulted exposures*;
- a robust **model governance** framework, providing for a clear separation of roles between development, independent validation, approval, and periodic review, in line with the *EBA Guidelines on Model Risk Management*;
- **ongoing performance monitoring**, including *backtesting*, stability analysis, and controls over model changes (*model change management*), as well as objective and consistent criteria for default classification and for the management of *margins of conservatism*;
- **comprehensive technical and functional documentation**, enabling traceability of assumptions, data, and methodologies used, as well as reproducibility of results and the management of model risk within the *Model Risk Management Framework*.

1.2.5 Operational integration, stakeholders, and periodic updates

The implementation of the European regulatory framework requires an integrated and coordinated approach among the various corporate functions involved in risk management, regulatory compliance, and internal controls. To ensure methodological consistency, operational efficiency, and continuous alignment with regulatory developments, it is necessary to adopt a structured system for the governance of processes and information.

- A **single, integrated reference framework** is defined, which coordinates European regulatory sources, modelling practices, and internal governance requirements, for the benefit of the *Risk Management*, *Model Validation*, *Compliance*, and *Internal Audit* functions. This reference serves as the common basis for consistency among policies, methodologies, and operational tools.
- **Clear roles and responsibilities** are formalised for the development, validation, control, and approval of risk models and methodologies. **Periodic regulatory update** cycles are also established, with traceability of changes, centralised management of regulatory sources, and coordination with risk and capital committees.
- The **uniformity of calculation methodologies**, risk metrics, and reporting procedures is promoted, reducing operational ambiguities and fostering comparability over time and across the group's different entities. Such harmonisation makes it possible to ensure consistency among management, accounting, and regulatory data.

An operational integration model of this kind makes it possible to systematically verify compliance with European requirements, strengthen the **internal validation of models**, and improve **decision quality** in risk management and capital allocation.

Chapter 2

Capital Requirements Regulation (CRR I, II, III)

2.1 Definition, scope, and placement within the prudential framework

The Capital Requirements Regulation (CRR) is the European regulation that governs the capital and prudential requirements of credit institutions and investment firms in the European Union, integrating the Basel three-pillar framework: minimum capital requirements (Pillar I), the supervisory review and evaluation process (Pillar II), and market discipline (Pillar III). The CRR is part of the Single Rulebook and works in conjunction with the Capital Requirements Directive (CRD) on organizational and supervisory aspects.

2.1.1 CRR I and the definition of the European prudential framework

The **Regulation (EU) No 575/2013** (**Capital Requirements Regulation I, CRR I**) constitutes the foundation of the modern European prudential framework, introducing a uniform framework that is directly applicable to all credit and financial institutions of the Union. It has codified the fundamental principles on capital adequacy, capital quality, liquidity management and macroprudential discipline, strengthening the soundness of the banking system after the global financial crisis.

Main structural updates.

- **Defined the structure of own funds** (*Own Funds*) into three qualitative tiers of loss-absorbing capacity:
    - **Primary Class 1 Capital (Common Equity Tier 1, CET1).** Constitutes the highest-quality component of capital. Includes fully paid-up common shares, share premium, retained earnings and reserves, net of deductions, prudential filters, and holdings in financial institutions. Must be fully subordinated and freely available to absorb losses on a *going concern* basis, with no redemption clauses or incentives to call.
    - **Additional Class 1 Capital (Additional Tier 1, AT1).** Includes perpetual subordinated instruments that provide the institution with full discretion over the payment of coupons or dividends and the ability to absorb losses through *conversion into equity* or *write-down*. It strengthens *going concern* loss-absorbing capacity, ranking hierarchically after CET1.

  - **Class 2 Capital (Tier 2, T2).** Includes subordinated liabilities with a minimum maturity (at least five years) and the ability to absorb losses in *gone concern*. It is subject to regulatory amortization during the final five years of residual maturity and contributes to meeting the overall capital requirement.

- **Codified the total capital ratio** as a cornerstone measure of capital adequacy:

$$\text{Capital Ratio} = \frac{\text{Own Funds}}{\text{Risk-Weighted Assets (RWA)}},$$

  where RWA represent the value of exposures adjusted for credit, market and operational risk.

- **Established the minimum First Pillar capital requirements**, by setting minimum thresholds relative to risk-weighted assets (RWA):

$$\text{CET1 ratio} \geq 4{,}5\%, \qquad \text{Tier 1 ratio} \geq 6{,}0\%, \qquad \text{Capital ratio} \geq 8{,}0\%.$$

  The denominator of capital ratios is represented by **Risk-Weighted Assets** (RWA), which summarise the overall level of risk taken by the bank and determine the amount of regulatory capital to be held against the different risk profiles. RWA aggregate the contributions from the main risk categories covered by the First Pillar of the CRR. On top of these ratios, **capital buffers** must be held, entirely in CET1.

- **Defined the methods for calculating the capital requirement for credit risk.** The CRR sets out the various methods that banks may use to calculate capital requirements for **credit risk**, distinguishing three main approaches provided for by the regulation:

  - **Standardised Approach (STD).** Baseline method applicable to all banks. Requirements are determined by applying predefined risk weights based on the type of counterparty and, where available, on external ratings provided by recognised agencies ($ECAI$). Banks do not develop internal models but use the regulatory tables published by the European legislator.
  - **Foundation Internal Ratings-Based (F-IRB).** Banks internally estimate the *Probability of Default (PD)*, while the remaining parameters (*Loss Given Default (LGD)*, *Exposure at Default (EAD)* and maturity factors) are set or constrained by regulatory values. The use of this approach requires prior authorisation by the competent authority, pursuant to Articles 143 and 144.
  - **Advanced Internal Ratings-Based (A-IRB).** In addition to PD, banks internally estimate LGD, EAD and the effective maturity (M). It is the most sophisticated method and allows maximum risk sensitivity, but entails more stringent requirements in terms of governance, validation, internal control and supervision.

  The **IRB approach** (in its two versions, F-IRB and A-IRB) therefore represents an evolution compared with the standardised approach, enabling banks to calibrate capital requirements on the basis of their internal models, provided they demonstrate statistical reliability and full compliance with the qualitative requirements on validation and control.

- **Introduced the Leverage Ratio** as a non-risk-based metric to be calculated and reported (Articles 429–430), with subsequent regulatory review; the binding minimum threshold was to be introduced in subsequent legislation.

- **Outlined liquidity standards** to safeguard financial resilience, introducing the *Liquidity Coverage Ratio (LCR)*, defined as:

$$\text{LCR} = \frac{\text{High Quality Liquid Assets (HQLA)}}{\text{Net Cash Outflows (30 giorni)}} \geq 100\%,$$

  which measures the bank's ability to withstand cash outflows under short-term stress scenarios, ensuring the availability of high-quality liquid assets.

2.1.2 CRR II: structural updates and securitisations

The **CRR II** (*Reg. (EU) 2019/876*), applicable as of 28 June 2021, continues the evolution of the prudential framework in line with the Basel finalisation, reducing unwarranted variability in requirements and strengthening comparability across banks. CRR II introduced a set of technical and methodological innovations that significantly update the European prudential framework, making it more consistent with the international Basel standards. These changes cover both standardised approaches and internal models, with a focus on counterparty, market and securitisation risks, and the management of leverage and liquidity, as well as the large exposures framework and resolution requirements.

Main structural updates.

- **Counterparty and derivatives.** The new calculation method *SA-CCR (Standardised Approach for Counterparty Credit Risk)* has been introduced, replacing the previous *Current Exposure Method (CEM)*. This approach allows a more accurate estimation of credit exposure for derivative contracts and for securities financing transactions (*Securities Financing Transactions, SFT*), taking into account collateral margins and market volatility. The objective is to represent the actual risk more realistically and promote prudent counterparty management.

- **Market risk.** The new *Fundamental Review of the Trading Book (FRTB)* has been launched, a set of rules that redefine how banks calculate capital requirements for positions held in the trading book. In the initial phase, FRTB is applied in *reporting only* mode, i.e. without direct effects on capital requirements, but with the obligation to provide detailed information to supervisory authorities to verify the soundness of the new methods.

- **Leverage and liquidity.** The *Leverage Ratio* has been made mandatory, i.e. the minimum ratio of Tier 1 capital to total exposure not risk-weighted, set at 3%, defined as:

$$\text{Leverage Ratio} = \frac{\text{Tier 1 Capital}}{\text{Total Exposure Measure}} \geq 3\%,$$

  which ensures a minimum level of capital relative to total exposure not risk-weighted, preventing excessive structural leverage. Moreover, the *Net Stable Funding Ratio (NSFR)*[1] — which measures the stability of funding sources over the medium term — has become binding, together with further refinements to the *Liquidity Coverage Ratio (LCR)* and the definition of *High Quality Liquid Assets (HQLA)*. These rules ensure that banks maintain a balanced and sustainable liquidity profile.

[1]The *Net Stable Funding Ratio* is an indicator introduced by Basel III and made binding by Regulation (EU) No. 2019/876 (CRR II), aimed at ensuring that banks maintain a stable funding structure consistent with the maturity profile of their assets and liabilities. The ratio compares the amount of *Available Stable Funding (ASF)* — funding sources considered stable over the medium term — with the *Required Stable Funding (RSF)*, i.e. the stable funding requirement determined by the liquidity and maturity characteristics of assets. The minimum requirement is an NSFR of no less than 100%, ensuring a structural balance between assets and funding sources.

- **Large exposures.** The regulation on risk concentrations has been strengthened by anchoring the basis for calculating large exposures to *Tier 1* capital (Art. 395(1) CRR as amended by CRR II: *An institution does not expose to a client or to a group of connected clients an amount exceeding 25% of its Tier 1 capital*). In this way, the risk of excessive dependence on individual counterparties or economic groups is limited, reducing the potential domino effect in the event of default.

- **Capital and resolution.** CRR II has incorporated into its framework the regime relating to *Total Loss Absorbing Capacity (TLAC)* and eligible liabilities for the resolution of banking crises, in coordination with the *BRRD II* directive and the *Minimum Requirement for Own Funds and Eligible Liabilities (MREL)*. This ensures that, in the event of a crisis, banks have adequate instruments to absorb losses without resorting to public support.

Securitisations and the STS framework. A further area of reform concerns **securitisations**, i.e. transactions by which a pool of credit assets (such as mortgages or loans) is transformed into tradable securities. CRR II complements **Regulation (EU) 2017/2402**, which established the *STS (Simple, Transparent and Standardised)* securitisation framework. This new system aims to make securitisation transactions safer and more understandable, reducing complexity and improving the quality of the information available to investors. The regulation introduces a **harmonised hierarchy of approaches** for calculating the capital requirements associated with securitisation positions:

- **SEC-IRBA.** Approach based on the internal models of the underlying portfolio, which uses internally estimated risk parameters: *PD (Probability of Default)*, *LGD (Loss Given Default)* and *EAD (Exposure at Default)*;

- **SEC-SA.** Standardised approach, based on regulatory coefficients and structural characteristics of the tranches (repayment priority, thickness, credit enhancement);

- **SEC-ERBA.** Approach based on external ratings issued by recognised credit assessment agencies (*External Credit Assessment Institutions, ECAI*);

- **Fallback 1250%.** Residual requirement, applied in the absence of the conditions for using the previous approaches, with a fixed risk weight of 1250%. The 1250% amount is defined as the inverse of 8%, so that when calculating the minimum regulatory capital, the RWA (1250% of the exposure) are multiplied by 8%, equal to 100% of the exposure.

In practice, the use of the internal-model-based method (*SEC-IRBA*) is often limited by the lack of granular and complete data on the underlying portfolios. For this reason, many banks and investors opt for the **standardised approach**, which, while more conservative, is simpler to implement and more transparent. Overall, the new rules on securitisations strengthen market confidence, improve the quality of issuances, and contribute to the stability of the European financial system.

2.1.3 CRR III: Basel III finalization, scope and main developments

The **CRR III** completes the European transposition of the Basel III finalization, with publication in the Official Journal of the EU on 19 June 2024, entry into force on 9 July 2024, and largely applicable from **1 January 2025**. The aim is to reduce unwarranted variability of requirements, strengthen comparability among banks, and make the link between risk and capital more robust.

Main structural updates.

- **Output floor.** A **capital output floor** is introduced that limits the benefits of internal models by anchoring *RWA* to at least **72,5%** of the level calculated using standardized approaches. The output floor is subject to a **transitional phase** (from 50% in 2025, with a progressive increase up to 72,5% in 2030) and operates mainly at the consolidated level, ensuring a comparable minimum capital level across institutions. The mechanism, applied at an aggregate level rather than for each individual risk, limits unwarranted variability in capital requirements and ensures a minimum level of regulatory absorption.

- **Credit risk.** The CRR III strengthens consistency between the **standardized approach** and the internal ratings-based (**IRB**) approaches, in order to reduce unwarranted variability in capital requirements. The updates include greater granularity in exposure categories, new risk weights for corporates, SMEs and retail, as well as tighter constraints on internally estimated parameters in IRB models (in particular *LGD floor*, *PD minimum* and *input floors*). These measures make credit risk measurement more homogeneous across banks, improving the comparability of requirements and strengthening the reliability of internal models. Qualitative preconditions regarding data quality, model governance and independent validation remain fundamental.

- **Market risk (FRTB).** The *Fundamental Review of the Trading Book (FRTB)* represents the most profound overhaul ever introduced in the market risk framework. The reform redefines the criteria that determine which instruments must be included in the *trading book* and which in the *banking book*, with the aim of avoiding regulatory arbitrage between the two books. The concept of **trading desk** is introduced as the minimum unit of risk management and measurement, each subject to eligibility tests (*modellability test*) and validation by the supervisory authorities. Two methodologies are provided for calculating requirements:

  - the **standardized approach (SA)**, which applies risk weights based on portfolio sensitivities to risk factors (rates, credit, FX, equity, commodities);
  - the **internal model approach (IMA)**, which uses conditional value-at-risk simulations (*Expected Shortfall*) with a stress-time horizon and daily validation (*backtesting and P&L attribution test*).

  In the European context, FRTB has been transposed via CRR III; the full **binding application of capital requirements** has been further postponed through delegated acts of the European Commission, with an effective date after 1 January 2026, in order to ensure alignment with international timelines and preserve the global *level playing field*.

- **Operational risk. CRR III** also introduces a thorough overhaul of operational risk measurement, completing the transition to a single standardized approach in line with the Basel finalization. The previous methodologies — *Basic Indicator Approach (BIA)*, *Standardised Approach (TSA)* and *Advanced Measurement Approach (AMA)* — are repealed and replaced by a single **Standardised Approach (SA-Op)** applicable to all banks, with the aim of simplifying the regulatory framework and reducing unwarranted variability in requirements. Therefore, the capital requirement for operational risk is determined as:

$$\mathbf{K_{OpRisk}} = \mathbf{BIC} \times \mathbf{ILM},$$

  where:

  - the **Business Indicator Component (BIC)** represents a measure of the institution's operational scale, calculated on the basis of the main margins and revenues (interest, fees and other operating income);

- the **Internal Loss Multiplier** (ILM) represents the adjustment mechanism that calibrates the capital requirement as a function of the relationship between the **Loss Component (LC)** and the **Business Indicator Component (BIC)**. The **Loss Component** (LC) quantifies the average loss incurred by the institution as a result of operational events over a ten-year period. It is defined as fifteen times the average annual operational losses recorded over the last ten years:

  $$\mathrm{LC} \; = \; 15 \times \overline{L}_{10 \text{ years}}.$$

  This multiplier introduces a prudential factor that allows the institution's historical experience to be incorporated into the capital requirement. Therefore, if $x = \frac{\mathrm{LC}}{\mathrm{BIC}}$, i.e. the ratio between historical loss and the institution's economic scale, the ILM is a **monotonically increasing function of** x, such that:

  $$\mathrm{ILM} < 1 \text{ if } \mathrm{LC} < \mathrm{BIC}, \qquad \mathrm{ILM} = 1 \text{ if } \mathrm{LC} = \mathrm{BIC}, \qquad \mathrm{ILM} > 1 \text{ if } \mathrm{LC} > \mathrm{BIC}.$$

  In economic terms, a value of ILM > 1 implies an increase in the capital requirement, reflecting higher observed operational riskiness. Conversely, a value below one indicates a contained loss history relative to the scale of the business and results in a relatively lower requirement. In the context of **CRR III**, however, the European legislator has, at present, provided for the use of an **ILM equal to 1 for all banks**, with the consequence that the capital requirement depends solely on the **Business Indicator Component (BIC)** and not on the institution's history of operational losses.

The introduction of this framework entails significant operational and managerial impacts: it requires a **complete reconstruction of operational loss databases**, a **consistent reclassification of accounting items** and integration with risk management systems[2]. In addition, European supervision emphasizes the importance of maintaining **robust operational risk governance**.

- **Disclosure (Pillar 3). CRR III** significantly expands and harmonizes **Pillar 3 disclosures**, which constitute the main instrument of transparency and market discipline in the Basel prudential framework. The new provisions, updated in *Part Eight of the CRR* and in the *Implementing Technical Standards (ITS)* issued by the EBA, impose more comprehensive, consistent and comparable disclosure obligations across institutions, extending coverage to all relevant risk categories, including operational risk. Disclosures are structured on two levels:

  - **Qualitative disclosure**, aimed at describing risk management and governance policies, internal control processes, measurement methodologies and mitigation strategies adopted for each type of risk;
  - **Quantitative disclosure**, which reports in tabular and standardized form data relating to capital requirements, regulatory ratios and key risk indicators. In particular, for **credit risk** it includes disclosure on portfolios, IRB parameters (*PD*, *LGD*, *EAD*) and the impacts of the *output floor*; for **market risk** it includes reporting according to the new FRTB framework and *trading book* positions; for **operational risk** it includes the publication requirements for the *Business Indicator (BI)*, the *Internal Loss Multiplier (ILM)* and the associated capital requirement; in addition to **leverage indicators, capital buffers and ESG factors** of prudential relevance, which make it possible to assess the institution's overall sustainability.

[2]Risk management systems are the *Risk Control Self Assessment* (RCSA) and *Key Risk Indicators* (KRI).

The goal of the new *Pillar 3* framework is to strengthen **informational transparency** and **comparability** across banks and over time, improving the analytical capacity of investors, supervisory authorities and the public, and consolidating confidence in the European banking system through clear, standardized reporting consistent with COREP and FINREP supervisory data.

2.2 Impact on capital requirements, RWA and capital

The regulatory evolution introduced by the three **CRR** regulatory packages has progressively transformed the structure of European banks' capital requirements, affecting both the quality of regulatory capital and the determination of risk-weighted assets (*RWA*). The common objective of the reforms is to ensure that the capital held by banks more faithfully reflects the actual riskiness of exposures, **reducing asymmetries** and **improving comparability** across institutions.

Determination of RWA. The general formula for calculating risk-weighted assets is:

$$\text{RWA} = \sum_{i=1}^{n} \text{EAD}_i \cdot \text{RW}_i,$$

where:

- EAD_i represents the exposure at the time of default (*Exposure at Default*);
- RW_i is the risk weight associated with the exposure, a function of the nature of the counterparty and of the approach adopted (standardised or based on internal models).

Over time, the European legislator has strengthened the **standardised methods** to make them more sensitive to actual risk and has introduced **restrictions on the use of internal models (IRB)** in the presence of data shortcomings or insufficiently robust models. With CRR III, the introduction of the **output floor** (equal to 72.5% of standardised RWA) limits the possibility of excessively reducing requirements through internal methodologies, aligning capital levels to common and verifiable thresholds.

Overall capital requirement. The Pillar 1 capital requirement (K) is broken down into the three main risk categories:

$$K = K_{\text{credit}} + K_{\text{market}} + K_{\text{operational}},$$

which respectively represent:

- the capital required to cover unexpected losses arising from credit risk, assessed using standard or IRB approaches;
- the requirement related to market risk, the subject of a thorough overhaul with the introduction of the new *FRTB* framework;
- the requirement for operational risk, redefined in CRR III through a single standardised approach based on the *Business Indicator* and the *Internal Loss Multiplier*.

Structural and convergence effects. The introduction of the **output floor** and **FRTB** has an across-the-board impact on the overall level of required capital, reducing the dispersion of requirements across portfolios and *trading desks*. In this way, it limits the scope for methodological arbitrage between different approaches and promotes greater convergence in capital ratios between banks with internal models and those that use standardised methodologies. The expected outcome is a **harmonised and transparent** system, in which regulatory capital prudently reflects the actual risk and contributes to the overall stability of the European Union's financial system.

Prudential deductions and adjustments. To ensure that regulatory capital reflects only high-quality and genuinely available elements, the CRR sets out a series of **prudential deductions and filters** applied to CET1, including:

- **intangible assets**, including goodwill, fully deducted as they are not eligible to absorb losses;
- **deferred tax assets (Deferred Tax Assets, DTA)** dependent on the ability to generate future profits, subject to deduction thresholds or reduced risk weights;
- **holdings in financial institutions** and instruments counted in third parties' capital, deducted or limited according to concentration thresholds set out by the CRR;
- **shortfall of value adjustments** relative to expected losses, deducted from CET1, and **excess provisions** eligible in T2 within regulatory limits;
- **prudent valuation adjustments (Prudent Valuation Adjustment, PVA)** for fair value positions.

Transitional regimes and prudential filters are also provided to manage the introduction of new accounting standards, and **grandfathering** mechanisms for instruments not compliant with the new capital definitions, in line with the regulatory phase-out schedule.

Prudential consolidation. Capital requirements apply both on an **individual** basis (for each authorised entity) and on a **group-consolidated** basis, so as to fully capture risks at the aggregate level. Within the scope of consolidation:

- the **intragroup exposures** are eliminated and capital requirements and RWAs are aggregated;
- specific treatments are provided for **financial investments, vehicles and related companies**, in order to avoid double counting of capital and to ensure a prudent representation of the consolidated capital position.

Second Pillar requirements (Pillar 2). In addition to minimum requirements, the competent authorities may impose additional obligations through the **Supervisory Review and Evaluation Process (SREP)**:

- **Pillar 2 Requirement (P2R)**: binding requirement, additional to regulatory minima, aimed at covering risks not fully captured by First Pillar requirements (e.g., concentration, interest rate risk in the banking book, model risk);
- **Pillar 2 Guidance (P2G)**: non-binding guidance that expresses the expected resilience under adverse stress conditions. Although not mandatory, P2G is monitored by the supervisor and serves as a reference for capital planning.

For **P2R**, the same qualitative capital composition as for minimum requirements applies, namely predominantly CET1; although non-binding, **P2G** is likewise normally expected to be met primarily with CET1 capital.

Cross-cutting impacts. The combined effects of the CRR reforms and the evolution of prudential standards generate structural impacts on methodologies, processes, and governance, which can be summarised along four lines:

- **Regulatory and modelling alignment.** Need for consistency among regulatory sources, modelling approaches, and risk metrics across the entire data chain (from the estimation of PD, LGD, EAD, and CCF through to the calculation of ECL and RWA);
- **Strengthening of quality and validation safeguards.** Consolidation of data-quality controls, internal validation procedures, and the reconciliation between regulatory and accounting approaches, including those *forward-looking*;
- **Enhancement of monitoring and reporting.** Greater granularity in the analysis of migrations across risk stages, of the impacts on capital ratios, and on the overall capital position;
- **Updating thresholds and early-warning tools.** Review of approval limits and preventive surveillance mechanisms to anticipate deteriorations in creditworthiness and to manage exposures most sensitive to market or operational conditions.

Overall, these measures strengthen **capital resilience**, improve **strategic planning capability**, and promote integrated management across risk, performance, and sustainability, in line with European prudential supervision standards.

Chapter 3

Capital Requirements Directive (CRD IV, V, VI)

3.1 Evolution and regulatory framework

The **Capital Requirements Directive** (**CRD**), together with the **Capital Requirements Regulation** (**CRR**), is the cornerstone of the European transposition of the Basel III standards. The directives define the qualitative and organisational supervisory framework, while the regulation governs its quantitative and directly applicable aspects. The **Directive 2013/36/EU (CRD IV)** represented the first pillar of the post-financial-crisis reform, introducing a harmonised system of rules for prudential supervision and risk management within the European Union.

3.1.1 CRD IV – Directive 2013/36/EU

Transposed starting in 2014, CRD IV established the European **Single Rulebook** and introduced a series of structural innovations:

- **Corporate governance and internal controls**: requirement for institutions to adopt proportionate governance arrangements, with a clear separation between executive and control functions, and requirements of independence and professionalism for the members of management and supervisory bodies;

- **Internal risk assessment and management processes**: introduction of the **ICAAP (Internal Capital Adequacy Assessment Process)** and the **ILAAP (Internal Liquidity Adequacy Assessment Process)**, aimed at ensuring that banks maintain capital and liquidity adequate to their risk profile;

- **Prudential review and evaluation process (Supervisory Review and Evaluation Process - SREP)**: granting supervisory authorities the powers to assess the overall soundness of institutions, with the possibility to impose additional capital requirements and corrective measures;

- **Remuneration policies**: introduction of rules on variable compensation and its proportionality, including limits on the ratio between fixed and variable components and obligations of *deferral* (deferral of a payment or an economic inflow/outflow to a future period) and *clawback* (recovery clause: allows the company to take back money already paid if certain conditions are not met after payment);

- **Capital buffer framework (Combined Buffer Requirement)**: definition of conservation, countercyclical and systemic buffers to be held in *CET1*, in coordination with the CRR;

- **Role of the national competent authorities**: strengthening of supervisory, authorisation and intervention powers, assigning to the ECB (in the context of the Single Supervisory Mechanism, SSM) direct supervisory responsibilities over significant institutions;

- **European cooperation**: creation of a harmonised framework for cooperation among supervisory authorities, aimed at promoting convergence and consistency in application across the Union.

3.1.2 CRD V – Directive (EU) 2019/878

The **CRD V**, approved in 2019 and transposed in the Member States starting in 2021, represented an evolutionary step in building the European prudential framework, with the aim of strengthening consistency between qualitative supervisory requirements, corporate governance practices, and capital management. It introduced a series of structural refinements intended to increase the effectiveness of supervision and to make the system more proportionate, transparent, and convergent with international standards.

- **Proportionality and simplifications**: the directive strengthened the application of the proportionality principle in prudential supervision and, in coordination with **CRR II**, provided that *small and non-complex institutions (SNCI)* may benefit from simplified administrative and reporting requirements, while still maintaining minimum standards of soundness. This reduced the regulatory burden for less systemic institutions, fostering more targeted supervision of the risks that are actually material.

- **Remuneration, diversity and independence**: CRD V updated the framework for **remuneration policies**, strengthening the limits on the variable component (maximum 1:1 ratio to the fixed component, unless waived) and introducing criteria for *deferral*, *retention*, and *clawback*. It also introduced obligations regarding **gender diversity** in corporate bodies and **independence** requirements for internal control functions, in line with the *EBA Guidelines on internal governance* (EBA/GL/2017/11 and subsequent revisions).

- **Clarification of the Second Pillar**: one of the main innovations of CRD V was the formal distinction between the **Pillar 2 Requirement (P2R)**, which is binding and counts toward the overall capital requirement, and the **Pillar 2 Guidance (P2G)**, which is non-binding but subject to supervisory monitoring. This separation made the composition of overall requirements more transparent, improving predictability and comparability across institutions.

- **Third-country groups and Intermediate Parent Undertakings (IPU)**: to strengthen supervision over non-EU banking groups operating in the single market, CRD V introduced the obligation to establish an **Intermediate EU Parent Undertaking (IPU)**[1] for third-country groups with activities exceeding EUR 40 billion in the Union.

- **Coordination with resolution and loss-absorbing capacity**: the directive harmonised the framework of interaction between prudential supervision and resolution authorities, in particular with **BRRD II (Directive (EU) 2019/879)** and the **Total Loss-Absorbing Capacity (TLAC)** regime. The relationships between capital requirements and the

[1]The Intermediate EU Parent Undertaking (IPU) is a holding/intermediate parent undertaking in the European Union belonging to a third-country banking/financial group, set up when the non-EU group has significant activities in Europe above certain regulatory thresholds. It serves to bring under a single European "head" all (or almost all) of the group's banking/investment entities present in the EU and to make the group's structure, consolidated supervision, and prudential requirements (capital, liquidity, governance) at EU level clearer and more controllable for the authorities.

requirements for own funds and eligible liabilities (**MREL**[2]), in order to ensure that systemic banks can be resolved in an orderly manner without recourse to public funds.

3.1.3 CRD VI – Directive (EU) 2024/1619

The **CRD VI**, published in 2024 and largely applicable from 2026, continues the integration of the *Single Rulebook* by strengthening governance safeguards, supervisory convergence and coordination with the Basel finalisation package transposed in **CRR III**. In particular, in addition to the enhancement of **fit & proper** and the integration of **ESG risks** into governance processes, ICAAP and SREP, notably through the new **Article 87-bis** and the amendments to **Articles 74**, **76** and **98** of Directive 2013/36/EU, the directive introduces and specifies:

- **Suitability requirements (fit & proper) and key functions**: CRD VI significantly strengthens the framework for controls on the suitability of corporate officers, formally extending these requirements also to **holders of key functions** (*key function holders*) through the introduction of **Article 91-bis**. This innovation addresses the need to ensure that all persons with responsibilities relevant to the institution's management, control or risk maintain over time adequate **honourability**, **professional experience**, **technical and behavioural skills** and **effective time availability** to perform the role.

  The article introduces the obligation to conduct **periodic suitability assessments** not only at the time of appointment, but also during the term of office, with frequency and criteria set by the supervisory authorities or by the institution's internal policies. Institutions must have formal, traceable and documented assessment procedures, including checks for potential conflicts of interest, multiple mandates and skills updates.

  Where **suitability deficiencies** are identified, the directive requires the activation of **timely remediation measures**, which may include targeted training, redistribution of tasks, suspension or, in the most serious cases, replacement of the person concerned. These safeguards are intended to strengthen the **individual and collective accountability** of the bodies and to improve the **continuity and effectiveness of governance**, in line with the *fit & proper* standards already set out by the EBA in its *Guidelines on the assessment of the suitability of members of the management body and key function holders* (EBA/GL/2021/06).

- **Mapping of roles and responsibilities**: obligation to maintain a **formal mapping of functions** and **lines of responsibility** (functional organisation charts, delegations, control safeguards), to be kept up to date and made readily available to the competent authorities for supervision and *enforcement* purposes.

- **ESG within the prudential perimeter**: integration of **environmental, social and governance risks** into **governance** processes over short/medium/long-term horizons, into **ICAAP** and **SREP**, with expectations on *risk identification*, metrics, limits and scenarios, as well as accountability of the bodies regarding the oversight of materially relevant ESG profiles.

[2]The Minimum Requirement for Own Funds and Eligible Liabilities (MREL) was introduced by the Bank Recovery and Resolution Directive – BRRD (Directive 2014/59/EU), Art. 45, as a minimum requirement of own funds and eligible liabilities needed to make the application of bail-in credible and to ensure the resolvability of institutions. Within the Single Resolution Mechanism, the requirement was transposed and applied at euro-area level by the Single Resolution Mechanism Regulation – SRMR (Reg. (EU) No 806/2014), which assigns to the Single Resolution Board the setting of MREL for significant institutions and cross-border groups/LSIs in the euro area (cross-border LSIs). The framework was then strengthened and detailed in 2019 with BRRD II (Directive (EU) 2019/879) and SRMR II (Reg. (EU) 2019/877), as part of the "Banking Package", introducing more granular criteria (e.g., subordination, internal MREL, categories of banks).

- **Harmonisation of sanctions and cooperation**: strengthening of **sanctioning powers** and **cross-border cooperation** mechanisms among authorities (including **third-country branches**), to ensure a more uniform and timely application of supervisory safeguards across the Union.
- **Mandates to the EBA for application convergence**: delegation to the **EBA** for the development of **Regulatory Technical Standards (RTS)** and **Guidelines** aimed at defining: minimum contents of *fit & proper dossiers*, criteria on *time dedicated*, honourability, *diversity*, organisational and documentation requirements, as well as common supervisory practices (*Q&A*, information templates).
- **Alignment with Basel III finalisation**: coordination with the innovations of **CRR III** (output floor, FRTB, SA-Op, disclosure), ensuring consistency between qualitative requirements (CRD VI) and quantitative profiles (CRR III) on capital, liquidity and risk.

3.1.4 Scope of application and national transposition

As **directives**, the **CRD IV/V/VI** require **transposition into national law**, through which each Member State adapts the European provisions to its own institutional and regulatory framework. They apply primarily to **credit institutions** and, for certain prudential aspects, also to **investment firms** of significant size or with high systemic relevance. Transposition entails defining procedures, roles and responsibilities for the **national competent authorities**, which exercise prudential supervisory powers according to principles of proportionality and consistency with Union law. The main functions assigned to the national authorities include:

- the granting of **authorisations for banking activity**, including changes of control and extraordinary transactions (mergers, acquisitions, transfers of business units);
- the conduct of the **Supervisory Review and Evaluation Process (SREP)**, with the setting of **Pillar 2 Requirements (P2R)** and **Pillar 2 Guidance (P2G)**, as well as the adoption of any **corrective measures** or **early intervention**;
- **ongoing supervision of governance, internal controls and remuneration policies**, ensuring consistency with capital, liquidity and risk management requirements;
- the **application of administrative** and disciplinary sanctions, as well as the activation of mechanisms for **appeal, review and judicial protection** to safeguard the rights of supervised entities;
- **cross-border cooperation** with other European authorities and participation in the **colleges of supervisors** established within the **Single Supervisory Mechanism (SSM)**, coordinated by the **ECB** or the **EBA**.

CRD VI further strengthens supervisory convergence and cooperation between authorities, providing a **coordinating role for the EBA** in the development of common technical standards, in cross-border supervision and in the harmonisation of **sanctioning powers**. It also extends the supervisory scope to the **assessment of ESG risks**, integrating them into the prudential framework and risk analysis processes, with the aim of making the management of sustainability more uniform within the Banking Union.

3.1.5 Interaction with the CRR and relevant regulatory differences

The **Capital Requirements Directive (CRD)** and the **Capital Requirements Regulation (CRR)** form an integrated regulatory framework, but with distinct and complementary scopes. The **CRR**, directly applicable, governs the prudential requirements of the **First Pillar**, defining the **mandatory minimum capital requirements**, the methods for calculating **risk-weighted assets (RWA)**, **leverage ratio limits**, **liquidity requirements** and the quantitative rules on **internal models** for credit, market, and operational risk. In addition, **Third Pillar (Pillar 3)** transparency obligations fall predominantly within the CRR, with specific EBA ITS on the content and frequency of the *disclosure*.

The **CRD**, qualitative in nature and to be transposed at the national level, instead oversees **Second Pillar** aspects: governance, risk management, **ICAAP**, **ILAAP**, **SREP**, remuneration policies, and the capital buffer framework (*Combined Buffer Requirement*) as well as restrictions on profit distribution (*Maximum Distributable Amount*, MDA).

3.1.6 Organizational and governance requirements

The **CRD** requires institutions to adopt **robust and proportionate organizational structures**, commensurate with their nature, complexity, and operational scale. This includes:

- a **clear structure of responsibilities**, with defined reporting lines and separation between operational and control functions;
- **effective processes for the identification, management, monitoring, and control of risks**;
- **independent control functions** — *Risk Management*, *Compliance*, *Internal Audit* — equipped with a clear mandate, adequate resources, and direct access to the corporate bodies;
- for significant institutions, the establishment of a **Risk Committee** with adequate technical expertise and an advisory role on risk appetite and risk strategy.

The **management body** retains ultimate responsibility for the governance framework, approves risk policies and strategies, verifies their implementation, and ensures overall consistency with the corporate strategy and the capital plan.

3.1.7 Second Pillar prudential cycle

The **CRD** framework defines the functioning of the **Second Pillar** of the European prudential framework, based on the interaction between institutions' internal assessments and the reviews carried out by supervisory authorities. This cycle ensures that capital and liquidity requirements are not determined solely by regulatory formulas but dynamically reflect the institution's actual risk profile and managerial soundness. The two main components are:

- **Internal processes for the assessment and maintenance of capital and liquidity (ICAAP/ILAAP)**: each institution must have an **Internal Capital Adequacy Assessment Process (ICAAP)** and an **Internal Liquidity Adequacy Assessment Process (ILAAP)**, integrated into strategic planning and the internal control system. The objective is to ensure that the bank:
  - consistently identifies and quantifies **all material risks** (credit, market, operational, concentration, reputational, climate-related, etc.);
  - maintains over time **adequate levels of capital and liquidity** aligned with its risk appetite and business strategy;
  - considers stress scenarios and multi-year horizons to assess the institution's **forward-looking resilience**;

  - ensures **active involvement of the corporate bodies** (approval, review, oversight) and full traceability of methodological assumptions and the data used.

  ICAAP and ILAAP are not mere documentary compliance exercises, but integrated capital and liquidity planning tools, instrumental to day-to-day management and to sustainable growth strategies.

- **Supervisory Review and Evaluation Process (SREP)**: this is the **supervisory process** through which the competent authorities (ECB for significant institutions, national authorities for less significant ones) examine the reliability of internal processes and the institution's ability to maintain a risk profile consistent with prudential requirements. SREP is structured into quantitative and qualitative analysis phases, which include:

  - the assessment of the business model, governance, and control system;
  - the analysis of risk management and the quality of ICAAP/ILAAP processes;
  - the identification of vulnerabilities and weaknesses, with possible **targeted supervisory measures**;
  - the determination of **additional capital requirements (Pillar 2 Requirement, P2R)**, **non-binding guidance (Pillar 2 Guidance, P2G)** and **remedial or organizational strengthening plans**.

  Through SREP, supervision complements the regulatory perspective with a **holistic assessment** of the institution's soundness and sustainability over time, also in light of emerging risks and ESG factors.

Together, ICAAP/ILAAP and SREP constitute a **continuous prudential cycle**: institutions develop and update their internal assessment processes, while the authorities oversee, validate, and — where necessary — impose corrective actions or additional requirements. This mechanism strengthens the **overall resilience of the banking system**, promoting consistency between strategic planning, risk management, and capital strength.

3.1.8 Developments and supervisory priorities

The most recent developments in the framework (CRD V–VI) have introduced four axes of reinforcement:

- **Integration between capital planning, ICAAP and SREP**, with the definition and monitoring of the **Combined Buffer Requirement (CBR)**, **MDA** mechanisms and **CCyB** rates by jurisdiction and for each entity;

- **Strengthening of model governance**, with the management body and the risk committee responsible for approving and using internal models, ensuring data traceability, and conducting independent validation controls;

- **Alignment of remuneration policies** with the risk profile and governance and control arrangements;

- **Incorporation of ESG risks** into corporate strategies, management processes, and capital assessment over differentiated time horizons.

3.1.9 System of capital reserves and prudential supervision

Objectives and framework of reference. The **capital buffers framework** is a fundamental pillar of the European prudential framework set out by the **CRD** and the **CRR**, aimed at ensuring that credit institutions maintain an additional capital margin above the minimum Pillar 1 requirements. These buffers are preventive and countercyclical in nature: they serve to strengthen banks' ability to absorb losses under stress conditions and to curb the procyclicality of credit, thereby promoting the stability of the financial system.

Definitions and components. The **combined capital buffer** (*Combined Buffer Requirement, CBR*) is the aggregate of macroprudential and systemic buffers that institutions must hold entirely in **Common Equity Tier 1 (CET1)**. Its main components are:

- the **Capital Conservation Buffer (CCB)**, fixed at 2.5% of RWA and common to all institutions, aimed at preserving capital strength in normal times;
- the **Countercyclical Capital Buffer (CCyB)**, variable with the economic cycle and set by national authorities, to mitigate excessive credit expansion during boom phases;
- the buffers for **Systemically Important Institutions** — global (*G-SII*) or other (*O-SII*) — introduced to strengthen the resilience of intermediaries whose distress could have systemic effects;
- the **optional Systemic Risk Buffer (SyRB)**, applicable to specific macro-financial risks not covered by other requirements.

In summary:

$$\mathrm{CBR} = \mathrm{CCB} + \mathrm{CCyB} + \text{G-SII/O-SII} + \mathrm{SyRB}.$$

Automatic distribution restrictions (MDA). When an entity's actual capital falls below the level required by the supervisory authority, the automatic **Maximum Distributable Amount (MDA)** mechanism is triggered. This system limits the ability to distribute profits, pay dividends, coupons on AT1 instruments or bonuses to management until the buffer has been restored. The permissible reduction is determined by a coefficient $\alpha(\Delta_{\mathrm{CBR}})$ that depends on the degree of deviation from the regulatory threshold:

$$\mathrm{MDA} = \alpha(\Delta_{\mathrm{CBR}}) \times \Pi_{\mathrm{distribuibile}},$$

where $\Pi_{\mathrm{distribuibile}}$ represents the amount of profits or resources that can be freely distributed. The restrictions are proportional in nature: the further the entity deviates from meeting the CBR, the more stringent the limits on distributions become, with the aim of preserving internal capital and strengthening stability.

Chapter 4

EBA Guidelines

4.1 Scope, purpose and general principles

The **Guidelines** issued by the **European Banking Authority (EBA)** are *soft-law* instruments that aim to specify and harmonize the application of prudential regulation within the *Single Rulebook*. In the area of **credit risk**, they cover the entire life cycle of the exposure. The main objectives are: (i) to promote **consistent and reliable risk estimates**; (ii) to **reduce unwarranted variability** in capital requirements; (iii) to **strengthen governance and data quality**. In this context, among others, the guidelines on *definition of default*, *estimation of PD and LGD* (including *downturn LGD*), *loan origination and monitoring* and *management of NPE and forbearance* are particularly relevant.

4.1.1 Key EBA Guidelines

- **EBA/GL/2016/07 — *Application of the definition of default.*** Provides guidance on the consistent application of the **definition of default** for prudential purposes, with reference to classification criteria and evidence to be considered.
- **EBA/GL/2017/16 (updated 2020) — *PD, LGD estimation and treatment of defaulted exposures.*** Specifies the requirements for the **estimation and validation** of **PD** and **LGD** and the **treatment of defaulted exposures**, including the **calibration of LGD under downturn conditions**.
- **EBA/GL/2018/06 — *Management of non-performing and forborne exposures (NPE & Forbearance).*** Defines expectations on the **management of non-performing exposures** and **forborne exposures**, with guidance on classification, management strategies and *workout* processes.
- **EBA/GL/2020/06 — *Loan origination and monitoring.*** Sets out principles for **responsible lending** and **credit monitoring**, with a focus on *underwriting*[1], *pricing*, data collection and controls throughout the life of the loan.
- **EBA/GL/2021/05 — *Internal governance.*** Sets out organisational and **internal governance** requirements (structures, roles, information flows, control functions) to support sound and prudent risk management.
- **EBA/GL/2024/04 — *Guidelines on resubmission of historical data under the EBA reporting framework.*** Provides a common framework for the **resubmission of historical data** within the EBA reporting framework, setting criteria and expectations for error correction and for the quality of data reported to supervisory authorities.

[1] Underwriting is the process by which an intermediary (typically a bank or an insurance company) assesses, selects and assumes a risk by setting the price, terms and limits it is willing to take on for that risk.

These guidelines complement the binding Pillar 1 provisions (CRR) and the supervisory guidance in the SREP context, ensuring European consistency in the application of credit risk requirements.

4.2 Key requirements by area

The EBA Guidelines define specific technical and organizational requirements, requiring institutions to adopt standardized, documented processes proportionate to operational complexity, ensuring consistency across internal models, data, policies, and reporting.

4.2.1 Definition of default (EBA/GL/2016/07)

The application of the definition of default must be consistent and traceable across all risk processes. In particular:

- the **materiality thresholds** for payment delays must comply with regulatory limits (generally 90 days, extendable up to 180 days for specific types of exposures) and the counting rules for *past due days*;
- the **unlikely to pay** elements must be clearly identified, documented, and updated, including events such as *restructurings due to financial difficulty*, *insolvencies*, *revocations of credit facilities* or other evidence of deterioration;
- the **consistency of the scope** of application (at exposure or debtor level) must be ensured, as well as the traceability of the rules for *cure period*, *probation*, return to performing status and *override* of automatic classifications.

4.2.2 Estimation of PD and LGD (EBA/GL/2017/16)

Risk parameter estimates must be based on representative data, robust methodologies, and independent validations:

- portfolios must be **appropriately segmented** and based on **adequate historical time series**, with transparent handling of sampling biases and structural changes;
- the **PD** must be calibrated to the **long-run average of default rates** (*long run average*) and include **margins of conservatism** proportionate to data quality;
- the **LGD** must reflect realistic economic values, recovery costs and time to collection, with explicit documentation of assumptions and sources;
- an **independent validation** of the models is required and ongoing **performance monitoring** through *backtesting* and quantitative stability indicators;
- the **"use test"** must be demonstrated, i.e. the effective use of IRB models in decision-making processes (origination, pricing, capital allocation).

4.2.3 LGD in downturn conditions

Institutions must estimate the loss under adverse conditions (*downturn LGD*) in line with prudential principles:

- identify the **periods of macro-financial deterioration** relevant for each portfolio, based on historical time series of defaults and recovery rates;
- derive **add-on components** or model specifications that reflect the greater severity of losses during adverse phases of the cycle;

- document assumptions, drivers, and statistical sources, ensuring traceability and replicability of the estimates.

4.2.4 Management of non-performing exposures and exposures subject to forbearance (EBA/GL/2018/06)

The management of **non-performing exposures (NPE)** and **forborne exposures** constitutes a supervisory priority area:

- institutions must adopt **NPE strategies approved by the management body**, with quantitative targets and clear operational plans;
- the **workout units**[2] must be organized independently, equipped with dedicated resources and performance metrics consistent with the approved strategies;
- the **criteria for granting, classifying and monitoring forbearance measures** must be formalized and consistent with prudential and accounting regulations;
- there must be **probation and cure periods**, **provisioning and write-off**[3] policies that are transparent and proportionate to the residual risk;
- the **information flows** and **performance metrics** must enable continuous monitoring of progress against the objectives of the NPE plan.

4.2.5 Loan origination and monitoring (EBA/GL/2020/06)

The Guidelines on *loan origination and monitoring* strengthen the quality and accountability of the credit process:

- the **ex-ante assessment of creditworthiness** must be based on robust quantitative metrics (*debt service coverage ratio*, *loan-to-value*, debt sustainability);
- the **lending policies** must be consistent with the **risk appetite** and with risk-based pricing criteria (*risk-based pricing*);
- **collateral** must be valued by independent parties, with **periodic revaluations** and **data governance** controls to ensure their reliability;
- **continuous monitoring** of exposures is required with *early warning* indicators, periodic reviews, and minimum information packages for **internal reporting**.

4.2.6 Internal governance and organizational structure (EBA/GL/2021/05)

The Guidelines on internal governance specify the arrangements, processes and mechanisms that institutions subject to the CRD must implement to ensure effective and prudent management:

- the **management body in its management and supervisory function** retains ultimate responsibility for the internal governance framework, defining the organizational structure, responsibilities, the internal control framework, the institution's strategy and risk profile;

[2]The workout unit is the specialized organizational structure, separate from origination and day-to-day management functions, responsible for the management, restructuring and recovery of non-performing exposures (NPE), with dedicated expertise in: analysis of the debtor's repayment capacity, definition and assessment of restructuring options, collateral management, recovery/exit strategies, monitoring and specific reporting on the *workout process*.

[3]Write-off is the total or partial accounting derecognition of a loan when the bank deems it no longer recoverable.

- the **organization is based on the three lines of defense model**: business lines as the first line of defense, responsible for the operational management of risks; the *risk management* and *compliance* functions as an independent second line; the *internal audit* function as the third line;
- the **organizational framework** must be **documented**, **transparent** and **proportionate**; internal control functions must be independent from the operational lines.

4.2.7 Resubmission of supervisory historical data (EBA/GL/2024/04)

The Guidelines on the resubmission of historical data set out a common approach to the correction and re-transmission of data reported under the EBA reporting framework, in the presence of errors, inaccuracies or other material changes:

- when institutions identify **errors or inaccuracies in data already reported**, they must incorporate the corrections into the submitted data and **resubmit them without undue delay** to the competent or resolution authorities;
- should errors in the *current data*[4] also affect the historical data[5], the institution, in addition to correcting the current period, must **resubmit the historical data** affected for a **minimum horizon of one year**[6];
- there are cases in which the **resubmission of historical data is not required**;
- the **competent or resolution authorities** may request **resubmission for a number of dates** of reference **greater** compared with **the minimum set out** in the Guidelines and may ask institutions to **accompany the corrected data with explanations** as appropriate.

4.3 Compliance obligations and implementation timelines

The **EBA Guidelines** are *soft law* instruments issued pursuant to Article 16 of Regulation (EU) No 1093/2010 and are formally addressed to the **competent authorities**, which must apply them according to the *comply or explain* principle and ensure their implementation by **financial institutions**. For intermediaries, alignment takes place through operational implementation and national-level **supervision**. Each guideline specifies an **application date** and, where applicable, **transitional provisions**. At an operational level, credit institutions must:

- carry out a **gap assessment** against the stated requirements, identifying areas, priorities, and impacts;
- define and implement a **remediation plan** with **milestones**, responsibilities, and decision **traceability**;
- manage **material model changes** in accordance with the prescribed approval processes, providing **evidence** during supervisory reviews.

[4] Current data is the most recent data submitted

[5] Historical data refers to data relating to earlier reference dates

[6] In addition to the time horizon, the regulation requires a minimum frequency of reference dates to be reported: one date for annual data, two for semi-annual data, four for quarterly data and at least six for monthly data.

4.4 Supervisory expectations

The **supervisory expectations** linked to the EBA Guidelines focus on institutions effective ability to translate regulatory requirements into concrete, verifiable, and fully operational controls.

- First, the **full functionality of the governance system** is required, with a clear allocation of responsibilities among the business, risk, compliance, validation, and *internal audit* functions. Such organizational clarity makes it possible to avoid overlaps and ensure independence in controls.
- Supervisory authorities also expect that internal metrics and models are effectively used in business processes — the so-called **use test** — demonstrating that risk methodologies influence credit decisions, **capital allocation**, and the setting of **risk limits**.
- Another pillar concerns **data quality and governance**: institutions must be able to demonstrate the traceability of information (*data lineage*), the presence of consistency checks, and the reconciliation of data between regulatory reports and management reports.
- Finally, for **internal models**, supervision requires the existence of a **model risk management policy** (*Model Risk Management Framework*), an **independent validation**, and timely **remediation plans** based on the outcomes of *backtesting*, internal reviews, and audit findings. The set of these controls must demonstrate the institution's ability to maintain methodological consistency, robustness, and reliability over time.

4.4.1 Internal update procedures in the event of revisions

To ensure the **timely implementation** and **full compliance** with updates to the EBA Guidelines, institutions should establish a structured, cross-functional **regulatory change management** process.

- First, a centralized **regulatory watch** is needed, capable of continuously monitoring regulatory developments, assessing their impacts, and assigning priorities based on relevance and risk profile.
- To support operational implementation, it is advisable to establish a **cross-functional committee** that involves the main corporate areas, tasked with correctly interpreting the requirements, steering the approach to adoption, and ensuring consistency among policies, procedures, and operating practices.
- Implementation must be accompanied by a formalized **change management** process. At the same time, it is necessary to ensure **model updates**, proper oversight of **data quality** controls, and the re-performance of **validation** activities, integrating **targeted training**.
- Finally, all changes must be tracked through effective **versioning**.

Chapter 5

Data governance

5.1 Framework and objectives

The ability to **manage and aggregate risk data effectively** is an indispensable condition for **sound risk governance** and for making informed strategic decisions. This requirement applies to the data used for institutions' **strategic and operational management**, as well as to the data used in **risk, financial, and supervisory reports**.

Despite the importance of data quality, many institutions continue to exhibit **significant deficiencies** in terms of accuracy, integrity, completeness, timeliness, and adaptability of data, often due to a predominant focus on costs and on the complexities of implementing large-scale *remediation* projects. These weaknesses had already been identified during the 2016 *thematic review* conducted by the ECB on a representative sample of significant banks: none of them were fully compliant with the **BCBS 239** principles, which remain the international reference framework for *Risk Data Aggregation and Risk Reporting (RDARR).*

The ECB has announced that **RDARR assessments** will continue to be a priority of the 2025–2026 SREP cycle, with a more stringent level of **supervisory follow-up**, which will include periodic checks, comparisons across banks (*peer benchmarking*), and potential enforcement actions in cases of non-compliance. The stated objective is to make full compliance with the BCBS 239 principles a structural condition for the effectiveness of risk governance and for the overall credibility of the prudential framework.

BCBS 239 identifies **14 fundamental principles**, organized into four key areas:

i) **Governance and infrastructure**: clear responsibilities of the *management body* in overseeing data quality and adequate organizational and technological resources;

ii) **Risk data aggregation capabilities**: procedures, systems, and controls that ensure timely and accurate consolidation of data at group level;

iii) **Reporting practices**: consistency and clarity of information flows, with appropriate frequency, granularity, and dissemination to the relevant recipients;

iv) **Review and supervisory tools**: self-assessment mechanisms, *internal audit*, and oversight by the competent authorities, aimed at the continuous improvement of RDARR capabilities.

The **EBA Guidelines on internal governance** (EBA/GL/2021/05) and the **ECB Guide on RDARR** (May 2024) reiterate that the **management body** is responsible for the approval, periodic review, and oversight of risk management policies and strategies, in line with Articles 74 and 88 of **Directive 2013/36/EU (CRD)**. This entails defining organizational structures with transparent lines of responsibility and internal control mechanisms proportionate to the complexity of the institution.

In the context of the **Banking Union**, the ECB regards the quality and governance of risk data as a **structural supervisory priority**. The EBA, for its part, sets the **minimum supervisory expectations** for RDARR, requiring institutions to develop adequate processes and infrastructures to identify, manage, monitor, and report the risks to which they are exposed. Alignment with the BCBS 239 principles is not merely a regulatory compliance requirement, but an essential component of **organizational resilience** and of the institution's ability to preserve the integrity and reliability of its decision-making processes.

5.2 Data governance and responsibilities

The ECB applies the principles of **BCBS 239** according to the criterion of **proportionality**, assessing the complexity, size, and risk profile of supervised institutions. In this context, the **management body** plays a central role: it is responsible for the approval and oversight of risk management and data quality policies[1].

The **EBA Guidelines on internal governance** specify that the management body must ensure an adequate **data culture**, a clear **allocation of responsibilities** and a **functional separation** between development, control, and validation of information processes. In particular, the **Chief Data Officer (CDO)** or equivalent role must have adequate authority and resources to coordinate **data governance** activities, ensuring the adoption of common **data quality** standards, the definition of **metrics and control thresholds**, as well as the management of **remediation** plans in case of deviations from expected levels.

The **ECB Guide on RDARR** (May 2024) reinforces these expectations, clarifying that responsibility for data quality cannot be delegated exclusively to operational or IT functions, but must remain with the management body and senior management. The top-level bodies must ensure that the **data governance framework** is formalised, consistent with the group structure, and integrated into the overall risk management system.

The **data governance framework** must include roles and responsibilities (data owner, data steward, data office/IT), **control, verification, and reconciliation processes** across the various databases, **escalation mechanisms** and reporting to the governing bodies, and **remediation plans** with deadlines, priorities, and effectiveness indicators.

Institutions must be able to demonstrate the **end-to-end traceability** of data used in risk measurement processes, from the original source to regulatory reporting, ensuring **methodological consistency**, **comprehensive documentation**, and a **verifiable audit trail**.

5.3 Data architecture and data structure

A **solid data architecture** and an **adequate IT infrastructure** are essential preconditions for an effective *Risk Data Aggregation and Risk Reporting* (RDARR) system. An institution's ability to collect, integrate, and aggregate risk data consistently depends on the existence of a formally defined information structure, based on **common standards**, **controlled glossaries**, and a **data model** shared across functions and group entities.

The data architecture must ensure **completeness**, **accuracy**, **integrity**, and **timeliness** throughout the entire information lifecycle. This requires a structured integration between **data warehouse** systems, operational databases, and transformation processes (ETL/ELT), ensuring **reconciliation** among management, accounting, and regulatory sources. Each piece of

[1] Art. 88(1) CRD IV.

risk-relevant data must be traceable back to its origin, according to an end-to-end **data lineage** approach and **full auditability**.

ECB expectations stipulate that the IT infrastructure be **scalable**, **modular**, and designed to ensure resilience and operational continuity even under stress conditions. Systems must support the timely production of regulatory and management reports, avoiding critical dependencies on manual processes. The presence of **centralized repositories** and **thematic data marts** makes it possible to reduce information fragmentation, improve data quality, and make validation and control activities more efficient.

The **ECB Guide to internal models** (2025 edition) further emphasizes that the quality of the data architecture is a technical requirement for the **methodological robustness of internal models**. Institutions are required to ensure that the data used in modeling (PD, LGD, EAD) and calibration processes are **consistent**, **complete**, **verifiable**, and **stored in secure environments**, with automated checks on errors, anomalies, and duplicates.

Finally, a modern data architecture, based on **automation** and **vertical integration of data flows**, not only improves risk aggregation and reporting capabilities, but also enables a reduction in operating and IT costs over the medium term. The adoption of **data-driven** technologies, the progressive digitization of processes, and the rationalization of legacy systems are enabling factors for achieving **sustainable data governance** that is fully aligned with European supervisory expectations.

5.4 Data Quality (DQ): metrics and controls

Data quality is one of the priority areas of prudential supervision, as it affects institutions' ability to measure risks correctly, comply with regulatory obligations, and prevent material losses due to information errors. The ECB and EBA expectations require each entity to adopt a structured **Data Quality Management** framework, based on objective metrics and traceable controls throughout the entire data lifecycle.

The fundamental dimensions of **Data Quality (DQ)** — *completeness*, *accuracy*, *consistency*, *timeliness*, *uniqueness*, *validity*, *accessibility*, *traceability* — must be formally defined and translated into **Key Quality Indicators (KQI)**, consistent with the nature and materiality of the data processed. Each dimension is subject to tolerance thresholds, automated controls, and cross-checks for consistency, particularly across management, accounting, and regulatory sources. Controls must be **documented**, **periodically reviewed**, and **proportionate** to the entity's operational and IT complexity.

The **ECB Guide on RDARR** states that institutions must be able to identify and measure data quality deficiencies through a structured system of **Data Quality Issues (DQI)**, which includes:

- **classification of anomalies** by type and severity;
- definition of **responsibilities** and resolution **timelines**;
- **monitoring** of quality trends over time;
- corrective measures and **remediation** plans approved by the *management body*.

Such deficiencies are tracked and analyzed from a **root cause analysis** perspective, in order to identify systemic causes and prevent the recurrence of errors. The resulting information must flow into a **Data Quality Dashboard** or **Data Quality Report**, periodically submitted to the risk committee and the internal audit function.

5.5 Lineage, reconciliations and operational tests

Data lineage represents a cornerstone of the RDARR framework, as it makes it possible to ensure **end-to-end traceability** of risk data across the entire information lifecycle, from the original source to management and supervisory reports. It documents, in a structured way, the steps of transformation, aggregation, and reconciliation, enabling the verification of **consistency**, **completeness**, and **accuracy** of the information. An effective lineage system also allows timely identification of the causes of any anomalies and strengthens **operational accountability** for the data along the information chain.

The expectations set out in the *ECB Guide on effective RDARR* (2024) require credit institutions to develop a **comprehensive mapping of risk data lineage**, including:

- identification and description of the **original data sources** (operational systems, accounting ledgers, credit archives);
- representation of the **transformations and aggregations** applied along the information chain, with an indication of the logical rules and associated controls;
- periodic **reconciliations** between **management, accounting, and regulatory flows**, with evidence of deviations and the correction criteria;
- definition of **operational** and **control** responsibilities for each node of the data chain.

In parallel, *fire drill* exercises require institutions to demonstrate the ability to produce, within limited time frames, risk reports that are complete, accurate, and consistent, even under operational or informational stress.

5.6 Risk reporting

Risk reporting constitutes the concluding element of the RDARR cycle, translating data aggregation and data governance capabilities into informational tools useful for decision-making and supervisory processes. Institutions are required to establish a reporting system that ensures **accuracy**, **completeness**, **clarity**, and **timeliness**, while at the same time safeguarding the **confidentiality** of sensitive information and its dissemination to the appropriate recipients.

Reporting must provide a consistent representation of the institution's overall risk profile, supporting both operational management and strategic oversight by the corporate bodies. To this end, the information must be:

- **accurate**, with systematic reconciliation and validation controls across managerial, accounting, and regulatory sources;
- **complete**, able to cover all portfolios and the main risk categories (credit, market, interest rate, liquidity, operational);
- **clear and concise**, structured to provide recipients — management body, committees, control functions, and authorities — with a comprehensible, comparable, and decision-oriented view;
- **timely**, produced with a frequency proportionate to the materiality of risks and intensified under stress conditions or in the presence of operational anomalies.

The **ECB guidance** and the **EBA Guidelines** specify that risk reports must provide **coverage** of key **exposures** and **metrics**, indicating the main changes, the underlying **causes**, and the

corrective actions taken. The reporting architecture must ensure the traceability of information, enabling the **reconstruction of source data** and of the transformations applied along the information chain.

The **internal reporting policy** must define the **roles** and **responsibilities** of the functions involved (risk management, finance, IT, internal audit); **rules on frequency, format, review, and sharing** of reports; **validation and approval procedures** by the competent bodies; and **data security** requirements.

Moreover, institutions must demonstrate that risk reports effectively support decision-making processes and strategic oversight, incorporating **performance** and **early-warning** indicators consistent with the *Risk Appetite Framework (RAF)*. ECB supervision also requires that reporting be based on data subjected to quality controls (DQ), automatic reconciliations, and independent periodic reviews, so as to ensure the full **reliability** of information flows and the **consistency** of risk metrics over time.

5.7 Resubmissions and historical consistency of reporting

The **Guidelines on resubmission of historical data under the EBA reporting framework** establish that, in the presence of errors, omissions, or inaccuracies in prudential reporting, financial institutions are required to proceed with the **resubmission** of corrected data, in order to ensure the **historical consistency**, **traceability**, and **informational completeness** of regulatory reporting flows. The resubmission must cover not only the current date, but also historical dates for at least the last calendar year, with the exception of monthly flows, unless otherwise requested by the competent authority.

This obligation derives from the principle, set out in the *EBA Guidelines on the resubmission of historical data* (2024), according to which the information communicated to the supervisory authorities must maintain continuity and consistency over time, regardless of when the errors are identified. Supervisory authorities may therefore request resubmission over a broader time scope where this is necessary to ensure the reliability of analyses or to correctly recalculate capital and liquidity requirements. The decision regarding the extension of the resubmission scope takes into account the **materiality** of the errors, their impact on regulatory indicators, and the frequency with which they recur.

Institutions must have a structured process for **error management** and **data remediation**, including **formal procedures** for the **identification**, **classification**, and **correction of errors** in reporting data; as well as objective criteria for the **assessment of the materiality** of the impact of inaccuracies; and a **centralized register of corrected reports** and resubmissions carried out. The **documentation** must contain **all changes made and the checks performed**.

In line with the principle of **data accountability**, entities are responsible for the quality of the data submitted, even in the case of flows produced by third parties or consolidated at group level. Each resubmission must be accompanied by a **data quality statement** that explains the nature of the error, the corrective actions undertaken, and the expected impacts on supervisory and management reports.

5.8 CRR II/CRR III impacts on data and reporting

The evolution of the European regulatory framework has had a direct and growing impact on the **data structure**, on **prudential reporting**, and on **information transparency**. With **Regulation (EU) 2019/876** (*CRR II*) and the subsequent **Regulation (EU) 2024/1623**

(*CRR III*), the legislator has progressively extended and refined the obligations of **data governance**, **harmonized reporting**, and **public disclosure**, strengthening the comparability and quality of data supporting prudential processes.

CRR II introduced substantial changes to **regulatory reporting** (*supervisory reporting*) and **public disclosure** (*Pillar 3*), with the aim of increasing data granularity and improving comparability among institutions. The main innovations concerned:

- the **expansion of the scope of reported data**, with greater detail on portfolios, collateral, concentrations, and risk metrics;
- the **standardization of COREP and FINREP reporting formats**, in coordination with EBA XBRL modules;
- the **alignment between managerial, accounting, and prudential data**, through systematic reconciliations and internal consistency checks.

CRR III, in force from 1 January 2025, has further strengthened the prudential information system by introducing the **output floor** mechanism, which sets a minimum bound on capital requirements calculated using internal models. This constraint stipulates that the capital requirement derived from IRB approaches shall not fall below 72.5% of that resulting from standardized approaches. The objective is to reduce the **unwarranted variability** of RWA, strengthen the **comparability of capital ratios** across banks, and restore the **credibility of the IRB framework** in the eyes of supervisors and markets.

From a data standpoint, the introduction of the **output floor** requires greater **integration between IRB and STD datasets**, as well as a reporting system capable of simultaneously presenting the results of both approaches for each portfolio and level of consolidation. This implies the need to maintain **technical and semantic consistency** across the different information levels — individual, sub-consolidated, and consolidated — in accordance with EBA and ECB specifications.

The innovations of CRR III also extend to the area of **disclosure (Pillar 3)**, introducing more granular and harmonized templates, with standardized indicators for credit risk, market risk, leverage ratio, data quality, and the impacts of the output floor. The information must be **comparable over time and across entities**.

Chapter 6

Internal Governance and Supervision

6.1 Framework and objectives

Directive 2013/36/EU (CRD IV), subsequently amended by Directives **(EU) 2019 /878 (CRD V)** and **(EU) 2024/1619 (CRD VI)**, establishes that each institution must have **robust governance arrangements**, including:

- a clear and transparent organisational structure, with well-defined and consistent lines of responsibility[1];
- effective processes for the identification, management, monitoring and reporting of risks;
- adequate internal control mechanisms, including sound administrative and accounting safeguards, as well as remuneration policies consistent with sound risk management.

For significant institutions, the regulations also provide for the establishment of a dedicated **risk committee**, tasked with supporting the management body in strategic decisions and risk assessments[2]. The members of the management body must also possess **adequate knowledge, skills and experience** to understand the institution's business, main risks and operating models[3].

The **EBA Guidelines on internal governance (EBA/GL/2021/05)** — applicable from 31 December 2021 — translate these principles into **operational requirements** concerning the composition and responsibilities of the management body, the independent control functions, the organisational structure, the risk culture and the internal reporting mechanisms. They specify that the **Internal Audit** function must maintain hierarchical independence and direct access to the management body, ensuring an autonomous and periodic assessment of the effectiveness of the governance framework.

In parallel, the **ECB Guide on risk data aggregation and risk reporting (RDARR, 2024)** defines the **minimum supervisory expectations** regarding data governance, information quality and timeliness of reporting flows. It requires institutions to ensure robust processes to **identify, monitor and report risks**, in line with the BCBS 239 principles and with Article 74 of the CRD, recognising the **direct responsibility of the management body** for the oversight of these processes.

Finally, the **ECB Guide to internal models** (2025 edition) complements the provisions of CRR III and clarifies the **joint responsibilities of the management body and senior**

[1] Arts. 74, 76, 88 CRD IV.
[2] Art. 76 CRD IV.
[3] Art. 91 CRD IV.

management in the preparation and validation of internal models, as well as in verifying the quality of the underlying data.

Taken together, these instruments outline an **integrated regulatory ecosystem** that ties the quality of internal governance to the institution's ability to produce, aggregate and communicate reliable risk data, ensuring that business and supervisory decisions rest on accurate, timely and verifiable information.

6.2 Bodies and responsibilities

6.2.1 Management body

The **management body** represents the entity's decision-making apex and is responsible for the **sound and prudent management** of the business, as well as for the overall effectiveness of the **internal governance and control system**. Pursuant to Articles 74, 76 and 88 of **Directive 2013/36/EU (CRD IV)** and the subsequent amendments introduced by **CRD V** and **CRD VI**, this body must ensure:

- the approval and monitoring of **business strategies**, risk policies, and operational plans;
- the establishment of a **clear organizational structure**, with documented lines of responsibility consistent with the entity's complexity;
- the existence of effective **internal control mechanisms**, including the risk management, compliance, and internal audit functions;
- the promotion of a consistent **risk culture**, supported by accurate and timely information flows.

The **EBA Guidelines on internal governance** (EBA/GL/2021/05) specify that the management body must receive **clear, complete, and up-to-date information** on material risks and on the performance of control systems, in order to exercise effective oversight based on reliable data. Furthermore, it must ensure that corporate policies are consistent with the approved risk appetite and that control functions have adequate resources and direct access to decision-making bodies.

CRD VI has strengthened the management body's **individual and collective accountability** obligations, introducing periodic **fit & proper assessment** requirements and **remediation** mechanisms in case of deficiencies[4].

6.2.2 Senior management

Senior management is responsible for the operational implementation of the strategies and policies approved by the management body, as well as for the day-to-day management of risks and the maintenance of an adequate level of internal control. According to the **ECB Guide to internal models** (ed. 2025) and **Delegated Regulation (EU) 2022/439**, management responsibilities must be **formally defined, documented, and consistent** with the institution's governance framework. In particular, senior management must:

- ensure the **proper implementation** of the approved risk policies and methodologies;
- ensure the **quality and integrity of the data** used in decision-making and reporting processes;

[4] Article 91 bis CRD VI.

- oversee the **performance of internal models**, including validation and backtesting processes, in line with supervisory expectations;
- maintain **effective information channels** to the management body, ensuring the timely communication of issues and deviations from risk limits.

The delineation of roles between the management body and senior management is a prerequisite for effective governance: the former exercises the **strategic steering and oversight function**, the latter ensures **operational execution and compliance** with the approved policies. This separation must be supported by **formal documentation** (mandates, organizational charts, policies, and information flows) and by a reporting system that enables **continuous and verifiable oversight** by the management body.

6.3 Organizational structure and control functions

6.3.1 Control lines

The three lines of defense model distributes tasks and powers along the process chain, safeguarding independent judgment and the effectiveness of controls:

- **First line (business and operational functions).** The units responsible for origination and position management take on and monitor risks in accordance with approved policies, implement first-level controls, and ensure the quality of the data used in processes and reports.
- **Second line (risk management and compliance).** The risk management function defines the methodological framework, verifies the consistency of limits, and oversees risk *aggregation* and the risk *reporting chain*; the compliance function assesses the adequacy of controls with respect to regulations and promotes corrective actions. In the credit area, the *Credit Risk Control Unit* operates independently of the origination/renegotiation units, with a formal mandate for verifying rating systems, consistency of metrics (PD, LGD, EAD), and traceability of model changes.
- **Third line (internal audit).** The internal audit function, with unrestricted access to systems and documentation, provides independent *assurance* on the effectiveness of the governance and internal control framework. The audit plan is risk-based and includes, at defined intervals, the assessment of: governance arrangements, data quality and completeness, methodological integrity of internal models (assumptions, limits, performance), adequacy of the *model validation* and *change management* processes.

The organizational separation among the first, second, and third lines is both substantive and formal: the control functions do not participate in risk taking nor in the operational design of the processes they oversee; they have the resources, expertise, and information channels adequate to report directly to the management body and the relevant committees. At group level, the parent company ensures the coordination and alignment of controls across subsidiaries, including entities in third countries.

6.3.2 Committees and delegations

For significant institutions, the committee structure strengthens the guidance and oversight by the governing bodies:

- **Risk Committee.** It supports the management body in defining and monitoring the *Risk Appetite Framework* (RAF), in approving risk policies, and in assessing periodic reporting (*risk profile*, limits, capital and liquidity metrics). Its composition ensures a majority of non-executive members and the presence of independent members; the mandate, reporting calendar, and the right of access to the control functions and internal audit are defined.

- **Internal models committees.** When established, they operate by delegation of the management body with responsibilities in *model governance*: they approve or propose to the body the modelling and validation policies, examine the results of *backtesting/performance monitoring*, decide on (or propose) conservatism thresholds and *model changes* according to predefined materiality criteria. Their operation is governed by *charters* that specify composition, quorum, information flows, and responsibilities.

- **Delegations and responsibilities.** Operational delegations preserve the *accountability* of the management body: roles, responsibilities, and powers are documented in internal documents (policies, regulations, *committee charters*), with a clear separation between executive and control functions. Reporting flows are accurate, complete, and timely, consistent with supervisory expectations on risk data aggregation and *risk reporting*.

The entire framework — lines of control, committees, and delegations — is subject to periodic effectiveness *review*: the outcomes feed improvement plans, methodological updates and, when necessary, corrective measures directed by the management body. This *feedback* loop ensures constant adherence to the requirements of sound and prudent management, proportionality, and the practices expected by the supervisor.

6.4 Reporting to the governing bodies

Reporting governance requires that the information intended for the governing bodies be *accurate*, *complete* and *timely*. Pursuant to the **CRD** (Arts. 74, 76 and 88), the management body must be able to exercise an adequate steering and oversight function on the basis of reliable information flows; this implies that the contents, frequency and levels of detail of the reporting are formalized in internal policies, proportionate to the institution's complexity and consistent with the approved risk appetite. In parallel, the **CRR** requires informational consistency between the data used internally and those produced for prudential purposes, including *supervisory reporting* and *Pillar 3 disclosure*, so as to ensure semantic alignment, systematic reconciliations and end-to-end traceability.

The process must ensure an integrated view of the risk profile, linking *management metrics* and *regulatory metrics* (e.g. capital, RWA, leverage, liquidity), and enable comparison over time through stable definitions and model versioning. Under stress conditions or in the presence of anomalies, the frequency of reports must increase, ensuring timely **escalations** and informed decisions on limits, buffers and remediation plans. The structure of the deliverables must distinguish between an *executive summary* for immediate decisions and technical appendices for methodological and data quality verification. In application of the EBA Guidelines on internal governance (EBA/GL/2021/05), reporting to the governing bodies:

- makes explicit **assumptions**, **scope** and main **data sources**; highlights **reconciliations** between management, accounting and prudential flows (COREP/FINREP) and documents justified residual differences;

- includes **data quality** indicators with thresholds, trends and remediation plans; reports the outcomes of independent reviews (validation, audit);

- ensures **consistency** with external disclosure: internal contents on capital, risks and models are aligned with **Pillar 3** (CRR, Part Eight) and the requirements of *supervisory reporting*, avoiding semantic or numerical misalignments.

The **documentation and archiving** framework provides that each report is accompanied by: (i) references to sources and calculation rules; (ii) evidence of adjustments and *restatements* with effective date; (iii) logs of decisions taken and responsibilities, to ensure **reproducibility** of analyses, **accountability** of functions and continuous **alignment** with regulation and supervisory expectations.

6.5 Supervisory expectations

The **ECB Guide to internal models** (ed. 02/2024, updated 07/2025) defines a comprehensive framework of expectations on *internal governance*, *internal validation*, *internal audit* and *model use*, with the aim of ensuring the reliability of regulatory estimates and consistency with the decision-making remit of the management bodies. In line with the **CRD** (Arts. 74, 76, 88) and the **CRR** (Part Three: Pillar 1 requirements; Part Eight: public disclosure), institutions must:

- **formalise roles and responsibilities**, documenting the decision-making scope, delegations of authority and information flows that enable effective *oversight* of the model lifecycle;
- ensure the **independence of validation** from development/use, with an appropriate mandate, resources and reporting lines; all models and their related estimates are subject to **initial validation** and then **annual** validation;
- demonstrate **conformity with regulatory requirements** and ensure **proportionality and completeness of documentation** (methods, assumptions, data, limits and controls, **data traceability** and adequate **use tests**);
- maintain structured processes for **data governance** (definitions, ownership, controls) and for **data quality** (accuracy, completeness, timeliness), with evidence of *lineage*;
- put in place **remediation plans** with responsibilities, deadlines and effectiveness metrics when data or process deficiencies emerge; in case of material non-compliance, the supervisor may activate **SREP** tools (qualitative measures, P2R/P2G, follow-ups and inspections).

The ECB update of 07/2025, in line with **CRR III**, clarified the responsibilities of *senior management* and of the management body regarding the *readiness* of applications (completeness, quality of dossiers, governance of decisions) and strengthened expectations on *validation* and *audit*, with particular attention to data quality, consistency with *Pillar 3* disclosure and reconciliation with the standardised requirements (e.g. in the presence of an *output floor*).

Chapter 7

Definition of default and past due threshold

7.1 Scope

The reference framework derives from Article 178 of the *Capital Requirements Regulation* (CRR), which sets the conditions for classification as default either on the basis of a *material* past-due amount persisting for more than ninety consecutive days, or on the basis of qualitative *unlikely to pay* (UTP) indicators that make full repayment according to the contractual terms unlikely. The EBA Guidelines on the definition of default (EBA/GL/2016/07) specify the operational application of Article 178, including scope and level of application (at the obligor or credit-line level, depending on the portfolio), day-counting rules, materiality criteria (relative component and absolute component), and *cure* and *probation* conditions for returning to performing status.

The framework is common to the *Standardised* and *IRB* approaches and aims to ensure methodological consistency, data comparability, and process stability. For institutions adopting IRB, updating the definition of default entails adjustments to rating systems and to the *data history* (including *backfilling*[1] where possible), as well as the assessment of impacts on PD, LGD, and CCF and the potential application of transitional conservatism margins. The ECB guides on internal models clarify the organisational and procedural prerequisites (roles, responsibilities, roll–out plans, controls, and initial and periodic validation) and require comprehensive documentation of decision-making processes and modelling assumptions. This set of rules and expectations ensures uniformity of classification, information quality, and reliability of metrics, supporting managerial decision-making and compliance with prudential requirements.

7.2 Logical structure of the definition of default

The definition of *default* governed by Art. 178 of **Regulation (EU) No 575/2013** (**CRR**) is a cornerstone of the prudential framework, as it uniformly defines the conditions under which a credit exposure must be considered to be in regulatory default. Classification as default occurs when at least one of the following two families of conditions is met:

- **Past-due criterion** (*past due*): it applies when the amount owed by the obligor exceeds the *materiality* threshold set by the competent authority and this condition persists for more than **90 consecutive days** (or 180 days for certain exposures)[2].

[1] Backfilling is the operation of retrospectively reconstructing or supplementing a historical time series with missing data using alternative sources, proxies, or statistical methods, so as to obtain a complete and continuous dataset for analysis or modelling.

[2] Art. 178(1)(b) of the CRR.

- **Unlikeliness-to-pay criterion** (*unlikely to pay*, UTP): it is met when qualitative or accounting indicators make it unlikely, without resorting to the enforcement of collateral, that the debt will be repaid in full. These include: burdensome debt restructurings, the recognition of specific impairment charges, the suspension of the recognition of interest income (*non-accrual*[3]) or other objective evidence of the debtor's financial difficulty.

When the *past due* condition becomes *material* and persists for more than 90 consecutive days, classification as *default* is automatic and does not require any further discretionary assessment by the institution.

Pursuant to Art. 178(1) of the CRR and para. 52 of the EBA Guidelines (EBA/GL/2016/07), the definition may be applied at the **debtor** level or at the **credit facility** level, depending on the portfolio type:

- for **corporate, institutional, and other portfolios excluding aggregated retail**, classification in default at the *debtor* level entails extending the default condition to all exposures of the same debtor within the reference group;
- for **retail** portfolios (including residential mortgages and consumer credit), application at the level of the individual credit line is permitted, provided the methodology is consistent and duly documented.

The definition of default is an essential prerequisite for the proper measurement of regulatory risk parameters and for the consistency of credit risk metrics across standardized and internal approaches.

7.3 Past due *criterion*: counting and materiality threshold

7.3.1 What is counted and commencement of counting

Pursuant to Art. 178(1)(b) of **Regulation (EU) No. 575/2013** (**CRR**), a credit exposure is considered *in arrears* (*past due*) when the amount owed by the obligor (including principal, interest or other contractual components such as fees or ancillary charges) has not been paid on the contractual due date. The counting of days past due starts on the first day following the due date, with no tolerance threshold, and stops exclusively in cases where the condition of delay is removed through payment or another form of extinguishment of the debt.

Where the amortisation schedule is subject to a contractual modification that is valid under the applicable legislation (e.g. following a renegotiation or a legislative provision that suspends instalments) the counting of *past due* days is realigned to the new repayment schedule. Legal or sectoral moratoria, introduced by legislative or regulatory measures, suspend the counting for the duration of the measure, without this automatically resulting in a classification as *default.*

In the presence of a **formal dispute** relating to the existence or the amount of the obligation (e.g. in the event of legal challenges or court proceedings concerning the contract) the institution may suspend the counting of days past due until the dispute is resolved, provided that such a choice is consistent with an internal policy. Such positions must be monitored separately and be subject to periodic *judgemental review*, in order to ensure that the suspension is not used to circumvent the correct application of the definition of default.

[3]Non-accrual is the accounting practice whereby a bank stops recognizing in profit or loss the interest accrued on an exposure because it is deemed no longer recoverable or highly deteriorated.

7.3.2 Materiality threshold: components

The determination of the **materiality threshold** for past-due amounts is a central element for the uniform application of the *past due* criterion, pursuant to Art. 178(1)(b) and (2)(d) of **Regulation (EU) No. 575/2013** (**CRR**) and the delegated acts. This threshold represents the minimum level beyond which the past-due and unpaid amount must be considered *material*, and therefore relevant for default classification. The assessment of materiality is based on two components that must be met jointly:

- **Relative component**: expressed as the ratio between the total past-due and unpaid amount and the total value of the credit exposure to the obligor. It makes it possible to assess the proportional impact of the delay on the total contractual obligation, reflecting the principle of proportionality with respect to the size of the credit relationship.
- **Absolute component**: represented by a monetary threshold expressed in nominal value, which identifies the minimum amount below which the arrears is not considered material, regardless of its percentage impact on the overall exposure.

Under **Delegated Regulation (EU) 2018/171**, the national competent authorities set the threshold values for both components, differentiating between *retail* and *non–retail* portfolios. For the relative component, the threshold is typically set at 1% of the credit exposure, whereas for the absolute component the minimum amount is 100 euro for the retail portfolio and 500 euro for the non–retail portfolio, subject to different national determinations. Both components must be exceeded simultaneously for the arrears to be considered *material* for the purposes of default classification.

For certain types of credit products characterized by low-amount periodic payments — such as *bullet* or *interest–only* contracts — the non-payment of multiple instalments may be required for the relative component to exceed the prescribed threshold. In such cases, to ensure timely identification of credit deterioration, institutions are required to establish additional *unlikely to pay* (UTP) indicators that anticipate the recognition of a default situation, in line with the principle of prudence set out in the CRR.

The materiality threshold therefore represents a tool for balancing the need for regulatory consistency with the need to avoid an overly sensitive classification triggered by minor or technical delays.

7.4 Special cases and technical situations

7.4.1 Technical *past due*

Delays that can be classified as **technical past due** do not, in themselves, constitute a *default* event within the meaning of Art. 178 of **Regulation (EU) No 575/2013** (**CRR**), as they do not reflect an actual inability of the debtor to meet their obligations, but arise from purely operational or administrative circumstances. The following fall into this category:

- **errors in data or information systems** of the credit institution, resulting in the failure to record a payment correctly made by the debtor;
- **malfunctions in payment systems** (e.g. interbank delays or clearing anomalies) that cause temporary delays in the accounting recognition of the collection;
- **normal accounting processing or reconciliation times**, provided that the payment was made on time by the debtor according to the contractual terms.

In such circumstances:

- the **count of days past due** is suspended or corrected as soon as the proper accounting evidence is restored or the debtor's payment is demonstrated;
- the institution must establish **formal internal policies** governing the criteria for identifying technical cases, operational responsibilities, *remediation* procedures, and the need for **timely reporting** to control functions;
- each event must be **documented and traced** by means of objective evidence (e.g. payment receipts, system logs, error reports, and accounting reconciliations);
- for the purposes of **IRB** models, such cases are **excluded from default datasets**, from default rate calculations, and from calibration metrics, ensuring the consistency and reliability of the time series used.

The proper management of *technical past due* is an integral part of the **data quality** framework and of internal controls over credit risk. It helps avoid improper classifications and maintain full **information integrity** in prudential reporting and modelling processes.

7.4.2 Exposures to public administrations

For **exposures to public administrations**, **Regulation (EU) No. 575/2013** (**CRR**), in Art. 178(1)(b), recognizes the possibility of extending up to **180 consecutive days** the grace period before classification as *default*, in view of the specific operational procedures and administrative timelines that characterize public-sector payment processes. This differentiated treatment may be applied, in particular, to exposures arising from contracts for the supply of goods or services to central, regional, or local public entities, provided that:

- there are no **objective elements of unlikeliness to pay** (*unlikely to pay*), such as insolvency proceedings, suspension of payments, or distressed restructurings;
- the debtor administration maintains a **sound financial and credit capacity**, evidenced by adequate capital and liquidity indicators;
- the delays are **attributable to procedural or administrative factors**, and not to economic difficulties or material breaches by the debtor.

The credit institution is required to retain **dedicated documentation** justifying the application of the extended term, attesting that the delay is attributable to administrative circumstances and that the credit risk is deemed negligible. Exposures treated in this way must not be included in the **calculation of the materiality threshold** for the *default* of other positions relating to the same debtor. However, if the materiality threshold is nevertheless exceeded for additional exposures to the same counterparty, the entire set of positions is classified as *default*.

7.4.3 Trade receivables and *factoring*

Within the scope of **trade receivables** and **factoring**[4], the determination of the *past due* status and the consequent identification of any *default* situations must be consistent with the contractual and accounting nature of the transaction, as well as with the prudential requirements laid down by **Regulation (EU) No. 575/2013** (**CRR**), in particular in Articles 166 and 178, and by the **EBA Guidelines on the definition of default** (**EBA/GL/2016/07**).

[4] Factoring is the transaction whereby a company assigns its trade receivables to a factor, who manages, advances, or guarantees them, in order to obtain immediate liquidity and reduce collection risk.

- In **factoring with purchased receivables recognized on the balance sheet**, the counting of *past due* days starts from the **contractual due date of the individual assigned invoice**. The delay is measured against the payment date provided for the assigned debtor and not against the flows between factor and assignor. The assessment of the *materiality* of the arrears must take into account both the nominal amount of the invoice and any concentration of risk on individual debtors, so as to ensure consistency with the regulatory threshold (*relative and absolute*).

- In **factoring without on-balance-sheet recognition** (*off–balance*), arrears are monitored on the customer's **factoring account**, periodically verifying compliance with the **contractual utilisation limits**, temporary excesses, and their duration. Persistent exceedance of the limits or failure to return within 90 days may constitute an indicator of *unlikely to pay* and, in the absence of corrective actions, lead to classification in *default*.

- When the transaction meets the prudential criteria set out in Art. 153(5) and 154(6) of the CRR for **purchased receivables**, the institution may apply the definition of *default* in a manner **aligned with the retail perimeter**, i.e. at the level of the individual exposure or line, ensuring consistency in the determination of *materiality* and in the traceability of the counts. This approach is consistent with the **ECB's expectations** on IRB models, which require uniformity of treatment and adequate documentation of the classification rules.

7.5 Return to performing (cure and probation)

The **return to performing** (*cure*[5]) is the process by which an exposure classified as *default* pursuant to Art. 178 of the **CRR** ceases to be considered as such when the conditions that had determined its classification have durably ceased. Reclassification to *non–default* generally requires a period of **at least three months** to elapse from the cessation of all default *triggers*, together with objective evidence of **restoration of repayment capacity**.

Institutions adopt internal criteria, formalised in policies approved by the competent corporate bodies, that ensure a consistent and documented application of the following minimum prerequisites:

- **Payment behaviour**: absence of *material past-due amounts* for the entire duration of the *cure* period and regular performance in line with the plan in place; any technical tolerances are not relevant for counting purposes if duly highlighted and corrected.

- **Financial sustainability**: updated assessment of the obligor's *financial situation* (cash flows, leverage, coverage), such as to exclude residual *unlikely to pay*; the scope and data sources must be traceable and consistent with management and regulatory systems.

- **Absence of further UTP indicators**: there must be no new qualitative or accounting elements of improbability of payment (e.g. further onerous concessions, unjustified suspensions, deterioration of the position).

For positions subject to **distressed restructuring** (treated as a default event pursuant to Art. 178 CRR), a longer **probation**[6] applies, defined in policies in line with the EBA Guidelines. During this period, the return to performing is subject, inter alia, to:

[5]The cure period is the minimum period in which, after the resolution of the causes of default/forbearance, the borrower must demonstrate regular performance and the absence of new signs of deterioration in order to be reclassified to a non-default status.

[6]The probation period is the period of enhanced monitoring following the exit from default/forbearance, during which the borrower must maintain consistent and sustainable behavior; any anomalies result in an immediate relapse into default/forbearance status.

- execution of at least one **material payment** under the new plan (not merely symbolic and defined ex ante by objective criteria);
- **absence of new material past-due amounts** and continuous compliance with post-restructuring due dates;
- confirmation, on the basis of updated credit analyses, of the absence of *unlikely to pay*.

If the definition of default is applied at **debtor level** (*debtor level*), reclassification to *non default* occurs only when *all* exposures to the same obligor meet the *cure/probation* conditions. Institutions ensure the **traceability** of the process through decision logs, references to data sources, reconciliations with regulatory reporting flows, and retention of the evidence used. Internal validation verifies the adequacy of the criteria and controls; *internal audit* provides *assurance* on the effectiveness of the governance design and on the correct application of the approved policies.

7.6 Implementation aspects for IRB and interaction with the supervisory authority

The adoption and maintenance of the **IRB** approach under the **CRR** require a coordinated set of actions on data, processes, models, and governance, as well as a structured engagement with the competent authority within the assessment methodology set out in the *RTS on IRB Assessment Methodology* (Regulation (EU) 2022/439) and the *ECB Guide to internal models*.

Alignment of the default definition and data remediation. Entities adapt rating systems to the updated definition of *default*, ensuring: **formal mappings** between managerial and accounting events and the default status, with consistency across portfolios and application levels (debtor or line); **remediation of the data history** and *backfilling*, where technically possible, to obtain consistent series; **data quality controls** on key variables (arrears, UTP, distressed restructurings), with reconciliations against regulatory reporting flows.

Impact assessment on parameters and expected loss. The revision of the definition of default affects estimates and calibrations of **PD**, **LGD**, and **CCF**. Institutions: recalculate the *default rates* and adjust the **PD calibration** to the long-term average, documenting any **margin of conservatism** (MoC); update **LGD** (including the *downturn* component) and **CCF**, reflecting recovery processes, realisation timelines, and line utilisation dynamics; quantify the effect on **EL** and **RWA**, substantiating any prudential adjustment and the **impacts on capital and pricing**.

Internal validation and independence. All IRB models are subject to **initial validation** and then **periodic validation** (at least annually) by a function independent of *model development* and of managerial use. Validation: verifies the power of **discrimination**, **calibration**, and **stability**; performs **backtesting** and sensitivity analyses, assessments of MoC and of the key assumptions; issues a **formal opinion** with **documented conclusions and remediation plan**, shared with the management body. **Internal Audit** provides *assurance* on the effectiveness of the governance design, processes, and controls.

Model governance and inventory. In line with CRD/CRR, the EBA/GL on internal governance and the EBA/GL on *Model Risk Management*, the institution maintains: a **model inventory** (scope, purpose, versions, metrics, usage limits); a **policy** on development, use, **override**, conservatism, monitoring and **lifecycle management**; a clear **segregation of roles**

between first line (development/use), second line (risk management/validation) and third line (audit).

Change management and roll–out plan. Changes to IRB systems are classified according to the supervisory **change policy** (Delegated Regulation (EU) 2022/439): **material changes** subject to **prior authorisation**, and **minor changes** subject to **notification** to the supervisory authority; sequential **roll-out plans** with milestones, priority criteria, documentation readiness, and quality metrics. Each submission is accompanied by an **evidence package** (methods, data, results, governance, validation) and an analysis of the impacts on RWA, capital, and decision-making processes.

Managerial use, limits and reporting. The institution demonstrates the effective **use test** of IRB systems for credit, pricing, capital allocation and limits, with: model performance **KPI/KRI**, **early warning** thresholds and recalibration triggers; periodic **reporting** to governing bodies and committees, reconciliations to COREP, FINREP and *Pillar 3*.

Interaction with supervision. Within the SSM framework, the ECB and the national authorities apply the **Delegated Regulation (EU) 2022/439** and the *Guide to Internal Models* to: assess **governance**, **data** (coverage, quality, *lineage*) and **methods**; carry out **technical analyses** (*model investigations*, requests for supplementary data) and *on/off-site* inspections; impose **qualitative/quantitative measures** (inclusion of additional MoCs, usage limits, remediation plans) and, where appropriate, require *resubmission* of data. The management body and *senior management* retain **ultimate responsibility** for compliance, data quality and methodological soundness; decisions are documented and made traceable for validation and supervisory review purposes.

Chapter 8

PD, LGD, ELBE and downturn

8.1 Purpose and scope

The correct identification and management of **downturn conditions** (*recessionary phases*) is an essential prerequisite for ensuring the prudential nature of risk parameter estimates in **IRB** (*Internal Ratings-Based*) models. In particular, estimates of *Loss Given Default* (**LGD**) and *Credit Conversion Factors* (**CCF**) must incorporate the effects of an adverse economic phase, ensuring that the estimated values are adequately conservative relative to long-term averages.

Regulation (EU) No 575/2013 (*CRR*), in Articles 181 and 182, stipulates that institutions determine parameters "appropriate for a downturn". In this context, internal methodologies must make it possible to distinguish periods of economic contraction, analyse their impact on recovery rates and conversion factors, and apply appropriate adjustments to reflect losses observed during stressed phases of the credit cycle.

The techniques and methodologies for implementing these provisions were introduced by **Commission Delegated Regulation (EU) 2021/930**, which defines:

- the **nature** of the economic recession, understood as a combination of macroeconomic and credit indicators representative of the economic cycle of the portfolio under consideration;
- the **severity**, identified through the most adverse values observed over a 12-month period for each indicator within a historical horizon **of 20 years**, such as to cover **at least two economic cycles**, extending the analysis to earlier periods if the 20 years are not sufficiently severe;
- the **duration**, which must be consistent with the persistence of negative effects on the selected indicators, with a minimum of 12 months, allowing multiple distinct periods where the worst conditions emerge at different times for different indicators.

Commission Delegated Regulation (EU) 2021/930 further requires that the economic recession be identified **separately for each IRB exposure type** (corporate, retail, SME, etc.), since the impact of a downturn can vary depending on the characteristics of the portfolio and its geographical–sectoral distribution. For each exposure type, institutions must select representative indicators that include at least **GDP and the unemployment rate**, supplementing them—where available—with **default or aggregate loss rates** from reliable statistical or market sources.

As clarified in the recitals of Commission Delegated Regulation (EU) 2021/930, the identification of downturn conditions does not coincide with hypothetical supervisory stress test scenarios, as it must be based on **historically observed data** and not on forward-looking simulations. The aim is to ensure that **downturn LGD** and **downturn CCFs** reflect the actual loss observable in adverse economic phases. The identified conditions are subsequently applied to the estimation of **downturn LGD**[1] and **downturn CCFs**[2], when the downturn value is more prudent than the long-term historical average. Estimates must be based on historical data that include at least one period identifiable as recessionary, and must be supplemented by **sensitivity analyses** and **stress tests** aimed at verifying the **stability of the estimated factors**. Institutions are therefore required to document in a **complete and traceable** manner the process of identifying the recession, the information sources, and the representativeness of the data. The methodology and results must be **reassessed at least annually** and whenever new evidence emerges that may alter the nature or intensity of the recessionary conditions.

8.2 Concise definitions of parameters

PD – Probability of Default The **PD** represents the probability that a credit obligation will enter *default* within a one-year time horizon. PD is not subject to regulatory specification in terms of *downturn*.

LGD – Loss Given Default The **LGD** measures the percentage share of expected economic loss in the event that the debtor defaults, net of actual and discounted recoveries. Pursuant to Article 181, paragraph 1, letter b) of the CRR, for IRB approaches institutions must estimate an *LGD appropriate for a downturn phase* (*downturn LGD*) whenever it is more prudent than the long-term average. The estimates must consider:

- the **realisation value of collateral and guarantees**;
- the **direct and indirect recovery costs**;
- the **discounting of expected recovery cash flows**;
- the **variability of losses observed** during adverse economic conditions.

ELBE – Expected Loss Best Estimate The **ELBE** (*Expected Loss Best Estimate*) represents the best estimate of the expected loss for exposures already classified as in *default*, used for determining value adjustments and internal capital reserves. Although not directly governed by **Delegated Regulation (EU) 2021/930**, ELBE maintains a functional relationship with **downturn LGD** and **downturn CCFs**, since both components contribute to representing the overall expected loss under adverse economic conditions.

CCF – Credit Conversion Factor The **Credit Conversion Factor** (**CCF**) is defined as the ratio between the amount expected to be drawn at the time of default and the amount available but not yet drawn. It therefore represents the percentage conversion of off-balance-sheet credit lines into risk exposures and is applied to the *off-balance sheet* components to determine the corresponding *Exposure at Default* (**EAD**).

In line with **Delegated Regulation (EU) 2021/930**, the *downturn CCF* reflects the effects of an economic contraction on borrowers' use of credit lines, taking into account **changes in utilisation rates** and drawdowns during phases of macroeconomic stress; concentration and correlation dynamics among portfolio exposures; **any policies to reduce, revoke, or suspend** the lines by the bank.

[1]Art. 181(1)(b) CRR I.
[2]Art. 182(1)(b) CRR I.

EAD – Exposure at Default The **Exposure at Default** (**EAD**) represents the amount of exposure to credit risk at the time of *default*, including both balance-sheet components (*on-balance*) and off-balance-sheet components (*off-balance*). In accordance with Article 166 of the **CRR**, the EAD can be represented as the sum of:

$$\text{EAD} = \text{EAD}_{\text{on}} + \text{CCF} \times \text{EAD}_{\text{off}},$$

where:

- EAD_{on} is the carrying amount of already drawn exposures;
- EAD_{off} represents the undrawn portion of credit lines;
- the factor CCF converts the *off-balance* portion into a risk equivalent.

For **A-IRB** approaches, the EAD must consistently reflect observed drawdown and repayment behaviour, taking into account the contractual nature of the lines, the type of counterparty and the presence of cancellation clauses or *covenants*. In recessionary phases, the institution assesses the impact of economic conditions on utilisation rates and on the evolution of exposures, in order to determine values consistent with the prudential estimates of **downturn CCFs**. The EAD therefore constitutes the basis for calculating expected loss:

$$\text{EL} = \text{PD} \times \text{LGD} \times \text{EAD}.$$

8.3 Downturn: nature, severity, duration

The definition of *downturn* conditions is a central element for the construction of prudential estimates of IRB parameters. Recessionary phases must be identified on the basis of objective and historically observable criteria, taking into account the **nature**, the **severity** and the **duration** of the economic recessions relevant to each portfolio.

Separation by type of exposure. The economic downturn must be identified **separately for each exposure type relevant for IRB purposes**, as defined in the **CRR**. This distinction reflects the different sensitivities of portfolios and credit segments. Distinct identification ensures that the estimates of *downturn LGD* and *downturn CCF* are representative of the specific risk characteristics of each portfolio.

Nature of the recession. The **nature** of the recession is determined by a coherent set of *economic and credit indicators* relevant to the type of exposure, divided into **macroeconomic indicators** and **credit indicators**. The set of indicators must reflect the **geographical–sectoral combination** of the reference portfolio and may be enriched with segment-specific measures (e.g. real estate price trends for *secured* exposures, or SME default rates). The selection of indicators must be documented and justified, ensuring statistical coherence and economic relevance. Each indicator must be included at most once for each cluster. **Aggregations of areas or sectors** are allowed only when the time series exhibit **strong correlations in the movements of the indicators**, such as to ensure a homogeneous representation of economic conditions.

Severity of the recession. The **severity** of a recession is determined by identifying, for each indicator, the **worst value observed over a period of 12 consecutive months**. This period is calculated within a *historical* time span that includes at least **20 years** and a complete business cycle. If the twenty-year horizon does not contain sufficiently severe phases, the entity must extend the analysis to **earlier historical periods**, ensuring that the selected conditions represent an effective benchmark recession scenario. Indicators must be consistent over time and, where necessary, normalised or adjusted for methodological changes in statistical sources.

Duration of the recession. The **duration** of a recession period must be at least **12 months**, with the possibility of extending it when the more severe indicator values imply a more prolonged recessionary phase. If the worst indicator values extend over a longer period, the duration must be *proportionally extended*, so as to reflect the persistence of the negative economic effects. It is permissible to identify **multiple distinct recession periods** within the same time frame, provided that they do not overlap, are pertinent to the type of exposure considered, and represent historical phases genuinely characterised by a deterioration of the economic cycle.

Frequency and scheduling. The 12-month period used for the calculation of the severity of the recession can **start in any month of the year**, without calendar constraints. This flexibility enables a more accurate identification of periods of deterioration in economic and credit fundamentals.

8.4 Indicators and data

Minimum indicators. Regardless of the portfolio, the set of indicators must include:

- the **gross domestic product (GDP)**, as a summary measure of the performance of overall economic activity;
- the **unemployment rate**, as an indicator of the labour market cycle and debtors' repayment capacity.
- the **aggregate default rates** observed at system-wide or sector level;
- the **credit losses**, which make it possible to gauge the magnitude of the economic cycle's effects on credit exposures.

For **corporate**, **SME** and **retail** portfolios, institutions must consider additional segment-specific indicators consistent with the segment's economic cycle (e.g. real estate price trends for secured mortgages, corporate insolvency indices, or changes in household consumption). The overall set must reflect the relationship between the macroeconomic cycle and the credit quality of the portfolios, ensuring the statistical significance of the observed correlations.

Data. Institutions must use **adequate and reliable data sources**, capable of ensuring temporal consistency and methodological quality in the construction of time series. Sources may include:

- national central banks or **Eurostat**, for official macroeconomic indicators;
- supervisory authorities, banking associations, or rating agencies, for credit-related data;
- internal datasets, provided they are representative of sufficiently broad portfolios and of controlled quality.

Delegated Regulation (EU) 2021/930 clarifies that institutions must use **appropriate and reliable data sources**, but **are not required to acquire data for all available indicators where costs are disproportionate** to the materiality of the portfolio. Data quality must be subject to periodic *data quality checks* and internal validation, in line with the **data governance** requirements established by the supervisory authorities.

8.5 Application to LGD and CCF in recession

LGD in recession. The **Loss Given Default in recession ($\mathbf{LGD_d}$)** represents the estimate of the loss conditional on default in a context of macroeconomic deterioration. Pursuant to Article 181, paragraph 1, point (b) of the CRR, institutions using the IRB approach must estimate an LGD appropriate for a recessionary phase whenever it is higher than the long–run average (*long–run average LGD*, denoted as $\mathrm{LGD_{LRA}}$). Formally:

$$\mathrm{LGD}_d = \max(\mathrm{LGD_{LRA}}, \mathrm{LGD_{recession}}),$$

where the component $\mathrm{LGD_{recession}}$ is constructed by linking the estimate to the recessionary economic conditions previously identified in terms of nature, severity and duration. $\mathrm{LGD_d}$ estimates must:

- be **based on historical data observed during adverse economic phases**, or, in their absence, on statistical proxy approaches consistent with EBA requirements (e.g. macroeconomic sensitivity models);
- reflect **changes in recovery rates**, **collateral realisation times**, and **recovery costs** during recession phases;
- be **documented and internally validated**, with evidence of the calculation methodology and consistency checks between $\mathrm{LGD_{LRA}}$ and $\mathrm{LGD_d}$.

Downturn CCF The **Credit Conversion Factor in downturn ($\mathbf{CCF_d}$)** quantifies the percentage of the undrawn exposure that is expected to be actually used at the time of default, under macroeconomic stress conditions. In accordance with Article 182 of the CRR, the institution must estimate CCF values that take into account the potential increase in credit lines used during phases of economic contraction. Similarly to what is set out for LGD, the downturn version is defined as:

$$\mathrm{CCF}_d = \max(\mathrm{CCF_{LRA}}, \mathrm{CCF_{recession}}),$$

where $\mathrm{CCF_{LRA}}$ represents the long–run average and $\mathrm{CCF_{recession}}$ the estimate obtained by linking utilisation behaviour to the downturn conditions identified for the reference portfolio. Estimates of $\mathrm{CCF_d}$ must consider:

- the **greater propensity to use** credit lines (*drawdown*) in the presence of liquidity strains or deterioration in credit quality;
- **consistency with internal historical data** and, where available, with system-wide external sources;
- documentation of the empirical link between the **severity of the recession** and the **increase in the observed conversion factors**.

Delegated Regulation (EU) 2021/930 specifies that, where historical data do not allow a direct estimate of the downturn CCF, institutions must apply a **prudential approach based on qualitative and statistical evidence**, supported by adequate independent validation.

Consistency between LGD and CCF in recession The estimates of LGD_d and CCF_d must be consistent with each other, since both represent effects arising from the same adverse macroeconomic conditions. In particular, they must refer to the **same identified recession** (in terms of nature, severity and duration); they must use the **same portfolio segmentation** and the **same historical data structure**.

8.6 Documentation and review

Documentation and traceability. Institutions are required to prepare documentation that is complete and consistent with the principles of **data lineage** and **audit trail**, enabling the transparent reconstruction of all stages of the parameter estimation and validation process. In particular, the documentation must include:

- the description of the **framework of the rating systems**, indicating the methodologies used to determine the *downturn LGD* and the *downturn CF*;
- the **processes and criteria** adopted for identifying phases of economic recession, including their nature, severity and duration, as well as the rationale for the choices in the case of multiple alternative periods;
- the **information sources and datasets** used, specifying for each indicator the origin, update frequency, observation period and data quality validation procedures;
- the **methodology for building the estimates**, including the criteria for calculating long–run averages, the margins of conservatism applied, and any use of proxy approaches in the absence of complete historical data;
- the **results of internal and independent validations**, highlighting statistical tests, sensitivity analyses and consistency checks across portfolios.

Traceability must extend to the entire lifecycle of the model (from initial approval to subsequent reviews and changes). All versions of the models must be recorded in a dedicated *Model Register*, ensuring the availability of the information necessary for prudential supervisory activities.

Periodic review The estimates of the *LGD* and *CCF* parameters and the underlying data must be **reassessed at least annually** and whenever new relevant information emerges. The review must verify the **consistency of the estimates** with the most recent historical data; analyse the **statistical stability** of the relationships between macroeconomic indicators and loss rates; include an **independent validation** of the proposed changes; ensure **reporting to the management body and the supervisory authorities**.

Should the review lead to significant methodological changes, the institution must activate the procedure provided for in its *change policy*, classifying the intervention as a **material change** or **non–material change**. Every update must be submitted to **review by the internal models committee**, with subsequent approval by the management body, ensuring full traceability and alignment with supervisory requirements.

Chapter 9

CCF, EAD and bucket

9.1 Purpose, scope and references

The *Credit Conversion Factor* (CCF) represents the share of an undrawn exposure that, on average, is expected to be actually drawn at the time of the counterparty's default. The estimation of internal CCFs therefore contributes to the determination of EAD, understood as the amount of the credit exposure to the debtor at the time of default, including on- and off-balance-sheet components weighted according to their respective conversion factors.

Estimation methodologies must be consistent with the general principles of robustness, representativeness and prudence set out in the IRB framework. The approach must be based on solid empirical evidence and on a long-term horizon (*long-run average*), including periods of economic stress so as to reflect the cyclical pattern of credit line utilisation.

In the presence of limited data, institutions are required to apply margins of conservatism (*Margin of Conservatism*, MoC) adequate to address estimation uncertainties, as required by Article 179(1)(f) of the CRR. Overall, the regulatory provisions aim to ensure that internal estimates of CCF and EAD reflect historical experience and adverse macroeconomic conditions; are subject to independent validation and periodic review (Art. 185 of the CRR); and are consistent with their actual use in credit risk management processes.

9.2 Definitions and scope

9.2.1 EAD for off-balance-sheet exposures and definition of *CCF*

For off-balance-sheet exposures included within the scope of **IRB** models, the exposure for supervisory purposes (**Exposure at Default**, **EAD**) is determined as the sum of the on-balance-sheet and off-balance-sheet components, weighted by their respective **Credit Conversion Factors** (**CCF**). For the *off-balance* portion, the EAD is calculated as:

$$\mathbf{EAD}_{\text{off}} = \mathbf{CCF} \times \mathbf{U}$$

where **U** is the **committed but undrawn amount** at the reference date. The **CCF** represents the **share of the undrawn amount** that, on average, is expected to be actually used at the **time of default**. Operationally:

$$\mathbf{CCF} = \frac{\text{utilisation at default} - \text{utilisation at the reference date}}{\text{undrawn amount at the reference date}}$$

The reference **committed limit** is, as a rule, the ***advised*** one (formally communicated and binding). Where higher but effective ***unadvised*** limits exist, the **highest limit** is considered to avoid **underestimation** of potential exposure.

Internal estimates of CCF apply exclusively to off-balance-sheet items for which the **CRR** allows the use of internal parameters (Arts. 166(8), 181, 182), while the other categories remain subject to standardised **regulatory percentages** (Arts. 111(1), 166(10)). Examples of eligible scope:

- **credit lines** revocable/renewable with **continuous monitoring** and adequate **historical data**;
- **irrevocable commitments** to extend funds, financial guarantees, *stand-by* and letters of credit;
- **liquidity/credit facilities** to securitisation structures treated under IRB.

For supervisory purposes, internal CCF estimates must be **consistent** with origination/monitoring policies and with the **governance** that requires:

- **estimation units** and **scope** clearly defined;
- **historical series** *long and representative*, including *adverse* phases;
- **backtesting** and verification of **stability** of estimates over time;
- **comprehensive documentation** of assumptions/limits and **Margin of Conservatism** (**MoC**).

9.2.2 Estimation unit and granularity

The estimation of *Credit Conversion Factors* (CCF) must be carried out on **homogeneous segmentations** to ensure consistency in the usage behaviour of facilities and adequate **temporal representativeness** of the data. In the **IRB** context, the estimation unit may be defined by *product class*, *portfolio*, or *facility grade*, provided that the segmentation is supported by empirical evidence and a clear methodological justification.

The **homogeneity** of the segment is ensured by criteria that, as a rule, consider at least:

- **nature of the facility** (revocable vs. term; presence of cancellation clauses, covenants, usage conditions);
- **limit structure** (advised vs. unadvised; shared or stand-alone limits; relationships with other facilities on the same obligor);
- **customer characteristics** (retail vs. non–retail; risk profile; industry, geographical area);
- **collateral and mitigants** (*credit risk mitigation*) that influence drawdown behaviour and exposure management;
- **usage patterns and seasonality** (proximity to limit saturation; frequency and magnitude of drawdowns; calendar effects);
- **quality and depth of historical series** (coverage, integrity, consistency with the identified recession periods).

The adopted granularity must allow **local validations** of the calibration, so as to identify any *systematic deviations* (directional *patterns* or concentrations by bucket). It is good practice to accompany such evidence with *out–of–time* and *out–of–sample* tests to mitigate the risk of *overfitting* and to verify the **stability** of the estimates under different market conditions.

When **data availability** is **limited**, the bank may resort to *expert-judgment estimates* supported by **traceable documentation** and **MoC** proportionate to the residual uncertainty. In any case, the chosen estimation unit and granularity must remain **stable over time** and **aligned** with the policies for granting, management, and monitoring of facilities, so as to ensure *comparability*, *reproducibility*, and *market discipline* of the resulting metrics.

9.3 Data, measurement and comparison with historical experience

9.3.1 Historical measure (*LRA*) of CCFs

The **long–run average** (**LRA**) of the *Credit Conversion Factor* (**CCF**) is the reference measure for the prudential calibration of off-balance-sheet exposures. It quantifies, as a long-term average, the proportion of the undrawn amount that is actually drawn at the time of default, under economic conditions representative of multiple cycles.

The LRA is calculated as the **weighted average of realised CCFs** over a homogeneous set of observations (*facility grade*, *pool* or portfolio segment). Formally:

$$\overline{CCF}_{\text{LRA}} = \frac{\sum_{i=1}^{N} w_i \, CCF_i}{\sum_{i=1}^{N} w_i}$$

where:

- CCF_i is the conversion factor realised on exposure i;
- w_i represents the weight associated with observation i (e.g. number of *defaults* or economic relevance);
- N is the total number of historical observations.

For each year in the observation period, an **average annual value** of the CCF is calculated, reflecting the effective historical experience. The multi-year LRA is generally the **arithmetic mean of the annual values**, provided that each year represents, in a balanced way, different economic conditions. In line with Article 181 of the CRR and the related provisions on the observation period, the historical period must include at least one complete economic cycle, comprising both *ordinary phases* and *adverse phases* (*downturn*), so as to capture the **cyclical variability** of utilisation rates.

If the sample is **unbalanced**, with a prevalence of "good" or "bad" years, the LRA must be **adjusted prudentially** by rebalancing the sample. Where historical observations include values of $CCF_i > 100\%$, such cases must not be **arbitrarily truncated**. Exceeding 100% may result from **over–utilisation** of the authorised credit limit before default; **recording errors** or *lags* between the reference date and the default event; **anomalous management behaviours** (e.g. extraordinary credit extension immediately before classification as default). In such circumstances, the institution must conduct a **data quality analysis**, documenting anomalous cases and assessing whether to correct the CCF or not, specifying the rationale for the choice.

The approach described meets the **ECB supervisory expectations**, according to which:

- the LRA must be built on **historical series of sufficient length** (at least five years of actual data);
- the data must cover heterogeneous economic phases and reflect the **market conditions** typical of the portfolio;

- any **adjustments or expert judgments** must be **documented, justified and traceable**;
- methodological choices (sampling, exclusions, aggregations) must be **reproducible** and consistent with the institution's *data quality* policy.

9.3.2 Time weighting and consistency

Consistent with the **CRR** (in particular Articles 179 and 185) and with the *RTS 2022/439* on the *Assessment Methodology*, IRB parameter estimates must reflect actual long-term experience, ensuring **stability** and **consistency** over time; the application of **differentiated time weightings** to historical observations is permitted, provided they are adequately justified, documented, and subject to independent validation. In particular, for **retail** portfolios and for exposures with high operational dynamism, it may be appropriate to assign a greater weight to the most recent observations, provided that:

- the choice **increases the predictive power** of the model or **improves the representativeness** of current business conditions;
- the evolution of the **business model** or of the **origination strategy** makes more remote data less meaningful (e.g. due to structural changes in products, usage behaviours, or lending policies);
- time weighting is applied in a **stable and systematic** manner, consistent with the validation controls and with the criteria of **effective use** set out in the IRB framework.

The purpose of time weightings (*time weighting*) is to balance the representativeness of the past with the relevance of the present, avoiding the *long–run average* (**LRA**) being distorted by obsolete periods or by periods no longer aligned with current portfolio conditions. Weights may be linear or exponential, provided they are **predefined**, **documented**, and **justified** with respect to the characteristics of the portfolio. The application of a weight that increases over time must not generate **artificial instability** in the estimates nor contradict the regulatory objective of *prudence and persistence* of the IRB metrics.

In line with **ECB expectations** and with the specifications of the *RTS 2022/439* on the *Assessment Methodology*, the institution must demonstrate that the introduction of **time weightings does not alter the relative rank ordering** of risks nor the **average calibration** over time; meets the criteria of **transparency, replicability, and traceability** of the methodological choices; is subject to **independent validation**, which confirms its effectiveness and the absence of pro–cyclical distortions.

The application of time weightings constitutes a **prudential option** aimed at adapting the LRA of CCFs to the most recent economic and managerial conditions, while at the same time preserving regulatory consistency. Any change to this approach constitutes a **methodological change** subject to the IRB *change policy* and must be adequately **documented**, **approved**, and **notified to the competent authority**.

The verification of the calibration of CCFs requires a comparison between *estimated* values and *realised* values. To this end, the set of exposures is partitioned into continuous sub-intervals of the CCF or, where the data permit, down to the single contract. The aim is to verify that the **deviations** do not exhibit systematic patterns within the sub-intervals but are compatible with normal *statistical noise*.

Let $\{B_k\}_{k=1}^{K}$ be a partition of the range of predicted CCFs and, for each bucket B_k:

$$\widehat{c}_k \;=\; \frac{1}{n_k}\sum_{i\in B_k}\widehat{CCF}_i, \qquad \overline{c}_k \;=\; \frac{1}{n_k}\sum_{i\in B_k} CCF_i^{\text{real}}, \qquad e_k \;=\; \overline{c}_k - \widehat{c}_k,$$

where n_k is the number of observations in bucket B_k, $\widehat{CCF}_i$ the estimated value and CCF_i^{real} the realised value. A *well-calibrated* model requires $e_k \approx 0$ for all buckets, in the absence of directional patterns (e.g. $\bar{c}_k > \hat{c}_k$).

For a summary measure of accuracy at sub-interval level, weighted indicators are useful:

$$\text{WBias} = \sum_{k=1}^{K} w_k \, e_k, \qquad \text{WAE} = \sum_{k=1}^{K} w_k \, |e_k|, \qquad \text{WRAE} = \sum_{k=1}^{K} w_k \, \frac{|e_k|}{\hat{c}_k \vee \varepsilon},$$

with weights $w_k = n_k / \sum_j n_j$ and $\varepsilon > 0$ small, to avoid divisions by values close to zero. WBias captures any overall *imbalance*; WAE and WRAE measure the average absolute error in terms of *level* and *ratio*, respectively.

The check of the **absence of systematic patterns** can be supported by:

- *calibration curves* by bucket ($\hat{c}_k$ on the x-axis, $\bar{c}_k$ on the y-axis) with verification of *closeness* to the bisector;
- tests of *average consistency* at bucket level (null hypothesis $H_0 : e_k = 0$) and verification of the *symmetry* of deviations;
- **local monotonicity** checks: for ordered buckets, to avoid systematic reversals of the risk *rank ordering*;
- **stability** analyses of results for different granularities (K) and for independent periods (*out-of-time*).

When the calibration analysis highlights the presence of **biases** concentrated in specific risk intervals, the institution must act with a set of targeted corrective actions. First, it may be necessary to **review the calibration function** or the **model segmentation** (e.g. by changing classification thresholds, introducing new transformations of the drivers, or including relevant interactions). Second, **data quality must be checked**, identifying any *outliers*, recording errors, or accounting delays, and, where appropriate, proceeding with the removal or justified reassignment of anomalous observations. Finally, in sub-intervals characterised by greater uncertainty, it is possible to apply a targeted **Margin of Conservatism**, ensuring consistency with the principles of the IRB approach.

The entire *bucket* backtesting process must be **documented**, **replicable**, and subject to **validation**. The presence of persistent deviations or directional patterns constitutes a **recalibration signal** and, if it entails methodological changes, falls within the scope of the *change policy* with the required formalities (materiality classification, possible notification, authorisation by the competent authority).

9.3.3 Recession (*downturn*) and areas of instability

The calibration of **CCF** must include a prudential adjustment for **adverse conditions**, in accordance with Delegated Regulation (EU) 2021/930 and with Articles 181 and 182 of the **CRR**. For each type of exposure relevant for IRB purposes, institutions must identify periods of economic recession based on consistent macroeconomic and credit indicators. Usage profiles of credit lines must be linked to these recessionary phases, taking into account possible *lags* between the occurrence of macroeconomic shocks and the actual increase in drawdown rates.

Construction of the CCF in recession. Let CCF^{LRA} be the long-run average of realised CCFs, calculated at the minimum *facility grade*/pool level, and let $CCF^{\downarrow}$ be the value estimated for the identified recession. Prudential calibration is calculated as:

$$CCF_d \;=\; \max\!\big(CCF^{\mathrm{LRA}},\, CCF^{\downarrow}\big),$$

where $CCF^{\downarrow}$ is derived from the values actually observed during the recession period for the relevant exposure type, possibly adjusted for time *lags*. One operational approach is:

$$CCF^{\downarrow} \;=\; g\!\big(\mathbf{Z}_{t-\tau}\big), \qquad \mathbf{Z}_{t-\tau} = \text{vector of indicators lagged by } \tau,$$

with $\tau \in \mathbb{N}$ chosen on the basis of empirical analysis (e.g. utilisation peaks following GDP troughs with a delay of a few quarters). The function $g(\cdot)$ and the choice of τ are motivated by evidence from data and stability tests.

Areas of instability near credit line saturation. Defining the *utilisation rate* u_t as the ratio between the amount drawn and the amount granted at time t, areas close to **credit line saturation** ($u_t \to 1$) often exhibit *local instability* of the CCF: small increases in residual drawdown, expressed relative to the *undrawn* $(1 - u_{t-1})$, can generate:

$$CCF^{\mathrm{real}} \;=\; \frac{\mathrm{drawn}_{t_0} - \mathrm{drawn}_{t_0-1}}{\mathrm{granted}_{t_0-1} - \mathrm{drawn}_{t_0-1}} \;\; \text{with denominator} \approx 0 \;\Rightarrow\; CCF^{\mathrm{real}} \text{ high/volatile.}$$

To manage such **instability**, targeted methodological measures are adopted: **functional segmentation** by utilisation levels or by contractual characteristics (margins, covenants, availability clauses); **calibration specifications** (monotonic transformations of the drivers, local regularisation, piecewise functions); **monotonicity checks** and **backtesting**, so as to avoid systematic inversions of the risk ordering of buckets; targeted **Margin of Conservatism** in regions of high uncertainty.

9.3.4 Data scarcity, judgmental estimates and margins of conservatism

In the presence of **information scarcity** or **non-material exposures**, the IRB framework allows the use of **judgmental estimates** of *Conversion Factors* (CCF), provided such estimates are based on prudential, transparent, and fully traceable criteria. This possibility, envisaged by the CRR framework and by the EBA Guidelines on the estimation of risk parameters, is subject to strict requirements of **documentation, justification, and conservatism**.

The minimum conditions that must be met are as follows:

- **Methodological justification and traceability of assumptions:** each judgmental estimate must be based on an explicit and replicable methodological logic. The sources used must be documented in detail, allowing the **complete reconstruction of the process** and the independent verification of its consistency.

- **Application of margins of conservatism (*Margin of Conservatism*, MoC):** in accordance with Article 179(1)(f) of the CRR, the uncertainty arising from limited or lower-quality data must be offset by introducing an adequate **MoC**.

- **Definition of a prudent minimum threshold:** the final value of the *Credit Conversion Factor* (**CCF**), inclusive of the *Margin of Conservatism* (**MoC**), must reflect a level of prudence commensurate with the identified uncertainties. In the absence of robust statistical evidence demonstrating lower usage, the bank may, on a prudent basis, assume that the entire amount of undrawn credit lines can be fully drawn at the time of default.

In operational terms, judgmental estimates (*expert-based estimates*) must be: subject to **independent validation**, with sensitivity analysis and comparison with external benchmarks; included in processes of **ongoing monitoring** and periodic review; accompanied by a **documented rationale** that makes explicit the underlying assumptions and limits of use. The use of judgmental estimates must remain **exceptional** and **proportionate** to the materiality of the portfolio. Where such estimates become systematic or have significant impacts on regulatory parameters, the institution is required to: launch a **data enhancement plan**, with measurable objectives for quality and informational completeness; classify the update under the IRB *change policy* as material; ensure that every prudential choice is **documented**, **approved** by the competent committees and, where necessary, **notified to the competent authority**.

9.3.5 Model, validation and overfitting

In developing CCF and EAD models, institutions must ensure a balance between **predictive power** and **statistical robustness**, preventing *overfitting* and ensuring the **generalizability** of the results. The models must undergo thorough validation both during development and in regular use. The verification requirements include:

- ***Out-of-sample* tests**: the model's predictive capability must be verified on a data sample **not used** for calibration. This makes it possible to measure the model's robustness with respect to data not seen during estimation, preventing the algorithm from capturing sample-specific noise. Such tests can be conducted by means of:
  - *split-sample validation*, i.e. splitting the dataset into *training*, *validation*, and *test* subsets;
  - k-fold *cross-validation* techniques, with average performance and analysis of the variance across partitions;
  - comparison of mean squared errors, calibration deviations, and residual distributions.
- ***Out-of-time* tests**: the model must be tested on data belonging to **time periods different** from those used for calibration, to ensure the stability of the estimated relationships under varying market conditions. This check makes it possible to assess:
  - the **temporal robustness** of the calibration function (absence of drift in the coefficients or causal relationships);
  - the **ability of the model** to maintain satisfactory predictive performance across different cyclical phases;
  - the consistency of the input parameters with the underlying economic drivers.
- **Stability and sensitivity analysis**: the model must be subjected to **sensitivity stress** on the main drivers, in order to assess the impact of small changes in the input data on CCF/EAD forecasts. Excessive volatility of the results or the presence of non-monotonic patterns among adjacent variables are signs of instability and potential overfitting.
- **Cross-segment consistency check**: calibration must remain consistent across similar segments or portfolios (e.g. across classes of counterparties or products with similar behavioural profiles), preventing unjustified methodological or structural discrepancies.

To systematically mitigate the risk of overfitting, institutions adopt:

- **parsimony criteria** in the selection of explanatory drivers, limiting the use of weakly significant or highly correlated variables (*multicollinearity*);

- **monotonicity constraints** or regularisation of calibration functions, so as to preserve the economic coherence of the estimated relationships;
- **statistical regularisation techniques** (e.g. LASSO or Ridge penalties) and parametric stability tests;
- **intra-model** robustness checks, such as residual analysis and verification of the absence of systematic bias by *bucket* or CCF sub-intervals.

Out-of-sample and out-of-time checks, together with stability and consistency analyses, are an essential component of the **independent validation** required by Articles 179 and 185 of the CRR. These activities serve to confirm the model's predictive capability and soundness over time, including on data or periods not used during estimation.

9.3.6 Margin of Conservatism and monotonicity

The application of **Margin of Conservatism** (MoC) to CCF models is a fundamental requirement of the IRB prudential framework. The MoC must proportionately reflect the **residual uncertainties** arising from data limitations, methodological approximations, or the presence of **external drivers** that are not perfectly observable, ensuring that the final CCF estimates are prudent, stable, and consistent with the underlying risk hierarchy. In particular, the *Margin of Conservatism* (**MoC**) is structured into three main components, to be applied cumulatively depending on the specific sources of uncertainty identified:

- **$\mathbf{MoC_A}$ – Data–driven:** covers shortcomings related to the **representativeness** of the data used in development or calibration;
- **$\mathbf{MoC_B}$ – Model-related:** reflects the intrinsic limitations of the methodology, relating to **data quality**, the **functional specification of the model**, the **explanatory power of the drivers**, or the **variability of the estimates**;
- **$\mathbf{MoC_C}$ – Residual:** represents an additional margin, complementing the previous ones, aimed at capturing the remaining overall uncertainty arising from cyclicality or adverse scenarios.

Each *Margin of Conservatism* (**MoC**) must be defined in a transparent and verifiable manner. First, its value must be **quantified and traceable**, clearly specifying the source of uncertainty to be covered and the methodology used to estimate it. Second, the MoC must be **comprehensively documented**, so as to allow the *model validation* and *internal audit* functions to independently verify its consistency and soundness. Finally, the application must be **consistent with the various components of the model** (PD, LGD, CCF).

Monotonicity and consistency of risk ordering. The calibration of the *Credit Conversion Factors* (**CCF**) must preserve the **monotonicity** of risk along the scale of *grades* (rating classes) or *pools* (homogeneous sets of exposures), ensuring that higher levels of riskiness correspond, in average terms, to CCF values that are not lower. In formal terms, for two groups of exposures i and j, ordered according to the probability of default (PD), the following relation must hold:

$$PD_i < PD_j \quad \Rightarrow \quad CCF_i \leq CCF_j,$$

unless there are sound economic or statistical reasons that justify local deviations. Should a MoC cause an inversion or anomaly in the estimated risk levels, the institution is required to document the causes and, where possible, correct the distortion through a **recalibration**. If the correction is not technically feasible, the bank must demonstrate that the residual effect does not produce **significant impacts** on capital requirements.

Chapter 10

Credit Risk Mitigation (CRM)

10.1 Objectives, scope and principles

Credit Risk Mitigation (CRM) represents the coordinated set of contractual, legal, and operational techniques that make it possible to reduce the effective exposure to credit risk. CRM is an essential instrument for the prudent management of risk and for the proper calculation of regulatory capital requirements, as it allows the controlled recognition of the **effect of guarantees and financial collateral in internal models** and in standardised approaches.

Within the European Union regulatory framework, CRM is governed primarily by **Regulation (EU) No. 575/2013** (**CRR I**) and subsequent updates. Under Article 194 of the CRR, credit risk mitigation can be recognised only when the protection is legally valid, enforceable in all relevant jurisdictions, and fully effective under market stress conditions or insolvency. **Legal certainty** and **enforceability** of collateral and guarantee agreements therefore constitute indispensable requirements for the prudential recognition of the mitigation benefit. The regulation further requires that CRM techniques be:

- **operationally effective**, i.e. correctly recorded, monitored, and assessed over time, with margining processes[1] and collateral management consistent with the institution's risk policies;
- **measurable and verifiable**, by means of metrics that allow the estimation of the protection's effect on risk parameters;
- **consistent with the economic and accounting purpose** of the transaction, so that the mitigation effect recognised for regulatory purposes reflects a real reduction of risk and not a merely formal transfer.

A cornerstone principle of CRM is the prevention of **adverse correlation** (*wrong-way risk*), namely the possibility that the ability to realise the collateral decreases precisely when the counterparty becomes insolvent. For this reason, the CRR (Regulation (EU) No. 575/2013, Part Three, Title II, Chapter 4, Arts. 196–210), as amended by CRR II and CRR III, imposes specific checks on the correlation between obligor and guarantor, on the diversification of collateral, and on the segregation of assets.

From a **governance** standpoint, CRM must be integrated into the internal credit risk measurement models (IRB), the internal control system, and corporate decision-making processes, with a documented and traceable approach.

[1] Margining is the process by which an intermediary calls for or updates margins (initial/variation margin) to cover the exposure arising from financial transactions with counterparty risk. In practice, it is the continuous mechanism for calculating, calling for, and settling margins to keep the exposure within acceptable levels.

10.2 Categories of protection

Pursuant to Part Three, Title II, Chapter 4 of **Regulation (EU) No. 575/2013** (**CRR**), as amended, **Credit Risk Mitigation (CRM)** techniques are divided into two main categories, distinguished based on the nature of the protection and the ways in which it reduces credit risk: *funded protection (funded)* and *unfunded protection (unfunded).* This distinction is fundamental for determining the mitigation effect in the calculation of capital requirements.

Funded credit protection (Funded Credit Protection). Funded protections directly reduce credit exposure through the establishment of a security interest or a lien over liquid assets or financial instruments. The following fall into this category:

- **real collateral** and **financial collateral**, such as cash deposits, government securities, or high-credit-quality bonds, which can be enforced or liquidated in the event of the debtor's default;
- **set-off and contractual netting agreements**[2] (*on-balance* and *off-balance*), recognised by the regulator if legally binding and fully enforceable in all relevant jurisdictions.

Under Articles 197–199 of the CRR, a protection may be recognised for prudential purposes only if it meets specific legal and operational validity requirements. The **guarantee agreement** or **collateral agreement** must be **fully valid and enforceable**, ensuring the legal effectiveness of enforcement. The asset pledged as collateral must be capable of being reliably valued and be subject to prudential **haircuts** that take into account market volatility, foreign-exchange risk, and maturity. Margining, segregation, and independent custody procedures must also be in place so as to **ensure the effective availability of the asset in the event of the counterparty's default**. Finally, it is necessary to verify that there are no adverse correlations between the obligor and the collateral, to prevent situations of **wrong-way risk**, in which the value of the collateral decreases precisely when the debtor's credit quality deteriorates.

Unfunded protections (Unfunded Credit Protection). Unfunded protections, governed in particular by Articles 201–216 of the CRR, transfer credit risk to a third party without altering the nature of the underlying asset. They include:

- the **personal guarantees** (*personal guarantees*) and the **suretyships**, which bind the guarantor to payment in the event of default of the principal debtor; such guarantees must be irrevocable, unconditional, and on first demand;
- **credit derivatives**, in particular *credit default swaps (CDS)* and related instruments, provided that the contractual reference is consistent with the hedged exposure, that the payment triggers are clear, and that the protection provider has an adequate credit profile.

The prudential recognition of unfunded protections entails the application of the so-called **substitution mechanism** (*substitution approach*), whereby the *Probability of Default (PD)* or the risk weight of the exposure is replaced by that of the guarantor or protection provider. The mitigating effect is therefore subject to the guarantor's actual ability to fulfil its commitment and to the absence of adverse correlation between guarantor and debtor.

[2] Contractual netting is the mechanism provided for in a legally binding agreement that bilaterally offsets receivables and payables between two counterparties, turning them into a single net amount in the event of default or during ordinary operations.

Prudential equivalence principle. For both funded and unfunded protections, the European framework requires that the risk mitigation effect be **effective, documented, and verifiable**. The benefit in terms of a reduction in the capital requirement must be proportionate to the risk actually transferred and must be supported by quantitative and legal evidence.

10.3 Eligible instruments and operational limits

The instruments deemed eligible must exhibit legal soundness, operational effectiveness, and an objectively verifiable market value, as well as allow prompt liquidation in the event of counterparty default.

In line with Articles 197–215 of the CRR, the intermediary's internal framework establishes a **positive eligibility perimeter** (recognised instruments, counterparties, and contractual forms) and a set of **qualitative and quantitative limits** designed to prevent residual risks, concentrations, and adverse correlations. The eligible instruments and their operational conditions can be grouped into the following main categories:

Financial collateral. Eligible collateral includes cash, deposits with the bank or with highly creditworthy third parties, and government securities or investment-grade bonds characterised by high quality and liquidity. For prudential purposes, the value of the collateral is subject to regulatory **haircuts**[3] to take into account:

- price volatility of the asset posted as collateral (H_{vol});
- foreign-exchange risk for collateral denominated in a different currency (H_{fx});
- maturity mismatch between the protection and the exposure (H_{mat}).

The adjusted amount of collateral is calculated as:

$$C_{\text{adj}} = C \times (1 - H_{\text{vol}} - H_{\text{fx}} - H_{\text{mat}}),$$

where C represents the gross nominal value of the collateral. It is also required that the collateral be **segregated**, held independently, and subject to **periodic valuation** by functions independent from the operating lines.

Set-off agreements and *netting*. The prudential recognition of bilateral netting agreements (*netting* or *close-out netting*) is conditional on the existence of legally binding and enforceable contracts in all relevant jurisdictions, as well as on verification of their effectiveness in the event of insolvency. The institution must demonstrate the ability to execute netting promptly and with reliable outcomes, through periodic operational tests. Such agreements reduce net exposure by aggregating multiple positions with the same counterparty and enable the mitigation of overall risk.

[3] A **haircut** is a percentage reduction applied to the market value of the recognised protection (C) in order to take into account the possible decrease in its value under adverse conditions. *Haircuts* are applied to both the value of the exposure and the value of the collateral. *Haircuts* may be determined according to regulatory parameters set by the CRR (*supervisory haircuts*) or, subject to the approval of the competent authority, estimated internally using models. In the latter case, the institution must demonstrate the methodological robustness of the model and the availability of historical time series.

Personal guarantees. Personal guarantees (*personal guarantees*, suretyships, and standby letters of credit) fall within the unfunded credit protection recognised under Articles 201–215 of the CRR. They are eligible only if the guarantor:

- has **proven creditworthiness** and economic and financial independence from the obligor;
- undertakes obligations with **irrevocable** and **on first demand** clauses, which ensure immediate enforceability in the event of default;
- is subject to **ongoing monitoring** of its repayment capacity.

Any timing mismatches between the maturity of the guarantee and that of the exposure (*maturity mismatch*) are treated in accordance with the prudential rules set out in Articles 237–239 of the CRR, with a proportional reduction of the recognised benefit.

Credit derivatives. The regulation allows for the prudential recognition of **credit default swaps (CDS)** and analogous derivative instruments, provided that the contractual reference (*reference obligation*) coincides with the underlying exposure and that the risk transfer conditions laid down in Articles 214–215 of the CRR are met. The bank must prudently manage:

- the **basis risk**, i.e. the potential divergence between the performance of the hedge and that of the underlying credit;
- the **counterparty risk** towards the protection provider, applying appropriate valuation adjustments (*Credit Valuation Adjustment, CVA*).

Operational limits and concentration. To ensure genuine diversification and the robustness of the protection portfolio, the corporate framework establishes stringent management and control rules. **Concentration limits by guarantor**, **counterparty**, or **type of collateral** are set, in order to avoid excessive exposures to individual entities or assets.

The **use of hedges** on positions that exhibit a **direct correlation** with the debtor is prohibited, in order to prevent situations of *wrong-way risk*. The **reuse of collateral** (*rehypothecation*) is **subject to restrictions** to ensure full availability and effective segregation of collateral, while for portfolios valued at *mark-to-market* specific margining requirements and minimum thresholds of contractual standardisation are imposed.

10.4 Requirements for documentation, assessment, and monitoring

Based on Articles 194–215 of **Regulation (EU) No. 575/2013** (**CRR**) and the **EBA** guidelines, the recognition of **Credit Risk Mitigation (CRM)** techniques requires structured and documented controls, grounded in legal certainty, transparency, and methodological consistency. The effectiveness of the mitigation depends not only on the validity of the contractual guarantees, but also on the quality of the valuations and the ongoing monitoring of their value and performance.

Documentation and legal validity. All protection agreements (guarantee maintenance covenants, margining requirements, collateral substitution rights, and *netting*) must be supported by **standardised contractual documentation**, consistent with international practices. It is mandatory to have **up-to-date legal opinions** attesting to the enforceability and effectiveness against third parties of collateral and netting clauses. Legal validity is a necessary condition for the recognition of the prudential benefit under Article 194 of the CRR.

Operational evidence and control processes. Operational functions must ensure a documented and traceable oversight of the workflows for managing collateral and personal guarantees. In particular, the following must be in place:

- procedures for **margining** and daily data reconciliation, with a clear segregation between the front office, risk management, and operations functions;
- an **exceptions and collateral fails register**, enabling the monitoring of deviations from operational standards and any delays in the delivery or substitution of collateral;
- structured processes for **management of credit events**, including the prompt notification of default, the activation of contractual rights, and the assessment of the possibility of enforcement.

These records constitute the basis for verifying the actual availability and quality of the protections, both for regulatory purposes and for internal reporting.

Independent valuation and data quality. Collateral valuation policies must be formalised and provide for an **independent valuation**, carried out by functions not involved in trading or position management. The review frequency is proportionate to the **volatility of the underlying asset**, the **materiality of the exposure**, and the **credit quality of the guarantor**. Valuations must be based on verifiable market data, and any change in value must be recorded promptly. It is also necessary to maintain ongoing controls over the correct **mapping between exposures and protections**.

Continuous monitoring and robustness checks. The **periodic monitoring** of credit protections verifies that the value of the collateral remains adequate net of *haircuts*, that *margin calls* and substitutions occur in a timely manner, and that no adverse correlations or *wrong-way risk* situations emerge. The robustness of contractual *triggers* and the credit quality of guarantors are also checked. In addition, periodic **stress tests** are conducted to assess the effects of market shock scenarios on volatility, liquidity, and collateral liquidation timelines, together with checks on the operational validity of *netting* agreements and the enforceability of collateral in the event of default.

10.5 Incorporation into models

The framework distinguishes the treatment of credit risk mitigation according to the approach used for measuring credit risk: **standardised approach (SA)** and **internal ratings-based approaches (IRB)**.

Standardised Approach (SA). In the standardised approach, credit risk mitigation is directly reflected in the determination of **Risk-Weighted Assets (RWA)** through two main mechanisms:

- the **substitution mechanism** (*substitution approach*), applicable to personal guarantees and credit derivatives. In this case, the risk weight of the exposure is replaced with that of the guarantor or protection provider, provided that the latter meets the eligibility criteria and that the correlation with the obligor is not significant;
- the **recognition of financial collateral**, achieved by directly reducing the exposure net of prudential *haircuts*. The adjusted exposure is calculated as:

$$E^{*} = \max\big(0, E \times (1 + H_{\mathrm{e}}) - C \times (1 - H_{\mathrm{c}} - H_{\mathrm{fx}})\big),$$

where E is the nominal exposure, C is the value of the collateral, H_e and H_c are the *haircuts* on the exposure and on the collateral, and H_{fx} is the foreign-exchange risk adjustment.

Under the standardised approach, **maturity mismatches** (*maturity mismatch*) and currency mismatches (*currency mismatch*) proportionally reduce the recognised prudential benefit, according to the formulas in Articles 239–240 of the **CRR**.

Internal Ratings-Based (IRB) approaches. In the **Internal Ratings-Based (IRB)** approaches, CRM is integrated directly into the key risk parameters, in ways that reflect the degree of sophistication of the authorised internal model. In particular:

- the obligor's **PD** may be replaced by that of the guarantor, where the coverage is recognised as fully effective and legally enforceable;
- the **LGD** may be differentiated between secured and unsecured exposures, incorporating the effects of collateral and recognised security interests in rem;
- the **EAD** takes into account the effects of *netting* and contractual set-off, reducing the nominal exposure based on bilateral set-off agreements valid under Article 205 of the CRR;
- **double default risk** (*double default risk*) — that is, the probability that the obligor and the guarantor default simultaneously — is managed according to conservative criteria, as provided by Article 153, paragraph 3 of the CRR, and by the *EBA Guidelines on Credit Risk Mitigation*.

Within the IRB framework, the prudential recognition of mitigation requires that the estimation of internal parameters consistently reflect the characteristics of the protection and that the traceability of the link between exposure, guarantee, and protection provider be ensured.

10.6 Concentration risk

In addition to the volatility and liquidity of the collateral, regulation requires considering **concentration risk** arising from excessive exposure to a few guarantors or types of collateral. This risk can amplify potential losses in the event of a simultaneous deterioration in collateral values or guarantor default. A commonly used indicator to measure the level of concentration is the **Herfindahl-Hirschman** index, defined as:

$$H_{\text{conc}} = \sum_i w_i^2,$$

where w_i represents the share of the value covered by guarantor or asset i over the total amount of protections. High values of H_{conc} indicate strong concentration and entail, for management and capital calculation purposes, the introduction of **operational limits**, **prudential add-ons**, or **additional haircuts** aimed at compensating the residual risk.

10.7 Pricing, governance and metrics

The **pricing** policy for credit risk hedges must reflect, in a *traceable* manner, the economic and regulatory benefits associated with the protection, balancing them against explicit and implicit costs.

- On the **economic side**, the primary benefit is the **reduction in expected loss**, where CRM affects LGD (guarantees/collateral) and, where permitted, PD via *substitution*.

- On the **prudential side**, the benefit is the **reduction in regulatory capital** through the decrease in *Risk-Weighted Assets*, with effects on the absorption of CET1/AT1/T2 and on the opportunity cost of capital.

The *full cost* of the hedge includes **explicit costs** (guarantee fees, credit derivative premia) and **implicit costs** (operational costs of collateral and margining, liquidity/funding frictions, *encumbrance* of liquid assets). The **net present value of the hedge** ($\mathrm{NPV}_{\mathrm{CRM}}$) measures the economic benefit of risk mitigation, taking into account capital savings and associated costs:

$$\mathrm{NPV}_{\mathrm{CRM}} = \mathrm{PV}(\Delta\mathrm{EL} + \Delta K \cdot r_{\mathrm{cap}}) - \mathrm{PV}(\text{fees} + \text{operational costs})\,.$$

In the context of the formula, PV denotes the **present value**, i.e. the current estimate of future benefit and cost cash flows discounted at a rate consistent with the risk. The parameter r_{cap} represents the cost of capital. The cost-effectiveness assessment is performed for each individual transaction and for the portfolio, analysing sensitivities to price, exchange rate, maturity, market, and *wrong-way risk* factors.

10.8 Governance and *change management*

Governance is based on approved and updated policies that define: (i) scope of eligibility for instruments and counterparties; (ii) *haircut* parameters and valuation methodologies; (iii) concentration limits and thresholds; (iv) approval, review, and *remediation* processes. **Change management** maintains the change log (eligibility lists, parameters, models, and limits), with traceability of the rationales and the impacts on EL, RWA, and capital.

Roles and segregation of duties. The allocation of **roles** ensures independence and cross-checks. In particular, the functions are broken down into: **Origination**: negotiation of hedging transactions; **Legal**: opinions on enforceability and effectiveness of collateral/guarantees, netting, and credit derivatives in the relevant jurisdictions; **Operations**: collateral management, margining, reconciliations, and segregation; **Risk Management**: methodologies, limits, monitoring of *haircuts*, *basis* and *wrong-way risk*, scenario analysis, and *stress test*; **Model Development & Validation**: design and independent validation of PD, LGD, and EAD, and of the effects of CRM; **Compliance & Internal Audit**: regulatory compliance, process reviews, and *audit trail*.

Chapter 11

Change policy

11.1 Purpose, scope and principles

The **change policy** for internal credit risk models (IRB and A-IRB) represents the coherent set of rules, responsibilities, and processes that govern the management of changes and extensions to rating systems, in line with the European regulatory framework.

The change policy aims to ensure that the credit institution continuously maintains full compliance with the qualitative and quantitative requirements for the use of the internal ratings-based approach (*Internal Ratings-Based approach*) pursuant to Articles 142–191 of the CRR. The change policy ensures that every change to IRB models is traceable, proportionate to its impact, and consistent with the principle of **continuity and integrity** of the risk measurement system, preserving the **reproducibility** of results and the transparency of the process.

The **scope of application** covers the entire life cycle of internal rating systems: **data quality**; **segmentation**; **development and calibration** of the models; **independent validation**; **authorisation and approval** by the competent authority, pursuant to Articles 143 and 144 of the CRR; **operation and periodic monitoring**; **review and update** in case of significant changes in portfolio, data, methodologies, or risk context. Also within the scope of the policy are:

- **extensions of use** of the models to new segments, products, or jurisdictions, including following extraordinary transactions (mergers, acquisitions, or portfolio integrations);
- the use, where authorised, of **permanent partial use (PPU)** provided for in Article 150 of the CRR, with an obligation to monitor and periodically review the conditions for continued use;
- **remediation** processes arising from supervisory findings or from evidence of significant deviations from performance standards.

The **regulatory rationale** of the change policy lies in the need to ensure that any methodological change or extension of the IRB scope does not compromise the prudential soundness, consistency, and quality of the models.

11.2 Classes of changes and extensions (taxonomy and materiality)

Based on Articles 143, 144 and 146 of **Regulation (EU) No 575/2013** (**CRR**) and **Delegated Regulation (EU) No 529/2014**, changes to IRB internal models are distinguished according to their **materiality**, that is, the significance and impact they have on risk parameters (PD, LGD, CCF, EAD), on capital requirements, and on management and control processes. The materiality-based classification makes it possible to determine whether a change requires prior authorisation from the supervisory authority (*ex-ante*) or merely subsequent notification (*ex-post*), as well as to determine the level of documentary detail, methodological evidence, and quantitative analyses to be provided.

Material changes (*material changes*). Material changes include variations that may materially affect model results or compliance with prudential requirements. Pursuant to Article 143, paragraph 3, of the CRR and Delegated Regulation (EU) 529/2014, such changes are subject to **prior notification** (*ex-ante*) and require assessment and authorisation by the competent authority (ECB or national authority). Included in this category are:

- **methodological changes** that significantly alter the estimation of the fundamental parameters (PD, LGD, CCF, EAD);
- **revisions of the segmentation or the reference population** that change the discriminatory power or the calibration of the model;
- **structural changes** resulting from extraordinary transactions (mergers, acquisitions, portfolio disposals) or from the introduction of new types of exposures, counterparties, or credit instruments;
- **changes** that entail **material impacts on capital requirements** or on the regulatory classification of IRB portfolios.

The decision to classify a change as material takes into account the **size of the quantitative impact** (*materiality threshold*) and the **qualitative relevance** (degree of alteration of the model relative to the approved version).

Non-material changes. Non-material changes (*non-material changes*) concern variations with limited or no impact on the structure and overall performance of the model. They are subject to **subsequent notification** (*ex-post*) according to time windows defined in the **internal change policy** and in the ECB guidelines. Such changes must be accompanied by documentary evidence demonstrating:

- the **continuity of the qualitative and quantitative requirements** set out in the CRR and EBA standards;
- the **stability of performance metrics** (discriminative *power*, calibration, parameter robustness);
- the **absence of significant effects** on capital absorption or on origination and monitoring processes.

Non-material changes are assessed and approved internally, with periodic reporting to the supervisory authority via the *model inventory* and *model change register* channels.

Extensions of the scope of application. The **extensions of the scope** include the application of an existing IRB model to new segments, products, jurisdictions, or legal entities. The regulatory treatment of such extensions is determined by the level of materiality and by consistency with the approved methodologies. In particular:

- for extensions considered **material** (e.g., new portfolios or acquired companies), **prior authorisation** (*ex-ante approval*) is required;

- for **minor** extensions, recognition may occur via *ex-post notification*, subject to internal compliance verification and preservation of model performance;
- in cases of post-M&A integration, the bank is required to conduct an analysis of **methodological convergence** and to ensure that the **data quality**, validation, and reporting processes are aligned with the group framework.

Internal taxonomy and decision criteria. The **internal taxonomy** specifies the decision-making criteria, the materiality thresholds, and the **classification triggers** that determine the category to which the change belongs. It links the *change policy* classes to the approval controls (internal committees, risk functions, and independent validation) and to the communication flows to the supervisory authority, ensuring consistency with the regulations.

11.3 Authorization process, notifications and procedural limits

The process for managing changes to internal credit risk (IRB) models ensures that each change is reviewed, approved, and communicated according to criteria of **transparency**, **proportionality**, and **traceability**.

11.3.1 Procedural phases

The operational cycle for the approval and notification of changes is structured into the following main phases:

1. **Proposal and compliance self-assessment**: the model development or management function prepares a change proposal, accompanied by a compliance *self-assessment* against the regulatory requirements and the internal *change policy*. This assessment includes materiality analysis, expected impacts on risk parameters, and consistency with the authorised scope.
2. **Internal decision and governance**: the proposal is submitted to the competent internal committees for approval, subject to the prior opinion of the **validation** function. The final decision defines the classification of the change (material/non-material) and the resulting regulatory process.
3. **Preparation of the documentation package**: the bank drafts the complete documentation, including a methodological note, quantitative analyses, performance evidence, data lineage, results of *backtesting* and monitoring, as well as a statement of compliance with the regulation. This package constitutes the *application file* for notification to the competent authority.
4. **Interaction with the competent authority**: the *Joint Supervisory Team* (JST) reviews the documentation received and may issue requests for additional information or clarification. Approval is subject to verification of compliance with the methodological, qualitative, and organisational requirements set out in the regulation.

11.3.2 Notification and approval

In line with Delegated Regulation (EU) 529/2014, changes are managed according to the following principle of proportionality:

- **substantial changes** (*material changes*) require **prior notification** (*ex-ante*) and formal approval by the competent authority before their implementation. The initiation of the process presupposes the availability of a complete information package and the independent validation of the results;

- **non-substantial changes** (*non-material changes*) are subject to **subsequent notification** (*ex-post*), according to timeframes and modalities defined in the internal policy, with periodic reporting to the supervisory authority;
- relevant **scope extensions** related to new portfolios, products, or legal entities, particularly those resulting from merger or acquisition transactions, require **prior authorisation** and a dedicated assessment by the competent authority.

11.3.3 Procedural limitations and mixed approach (PPU)

In the presence of a **mixed approach** (*Permanent Partial Use*, PPU), governed by Article 150 of the CRR, the bank is required to maintain **continuous monitoring** of the conditions that justify its use. Such monitoring includes:

- the periodic verification of the continued existence of the objective criteria that prevent the full application of the IRB approach;
- the analysis of convergence towards the gradual extension of the IRB perimeter, as required by Article 148 of the CRR;
- the consistency between the authorised IRB systems and the exposures treated under the Standardised Approach, in terms of data, controls, and decision-making processes.

The maintenance of the PPU is subject to demonstrating a credible strategy of progressive alignment, with documentary evidence and regular *reporting* to the competent authority.

11.3.4 Traceability and process control

All phases of the change and notification process must be tracked in a **centralised model change register**, which records for each event: the proposal date, the materiality level, the internal decision, the date of submission to the supervisory authority, the outcomes of the *review process*, and any corrective actions. This register serves as the reference tool for audits, *model inventory management*, and compliance checks. The entire authorisation process for IRB models is based on principles of **accountability**, **regulatory continuity**, and **European harmonisation**, so as to ensure that every methodological change or scope extension takes place in compliance with the rules of transparency and prudential soundness.

11.4 Documentation, evidence, and monitoring requirements

Any change or extension of an internal risk model must be accompanied by a structured set of **documentary evidence** demonstrating regulatory compliance and methodological soundness of the proposed solution. The **evidence package** is part of the authorisation or notification process and must ensure **replicability of results**, **traceability of assumptions**, and **transparency**.

11.4.1 Minimum contents of the documentation package

The package must include at least the following elements:

- **Use test**: evidence of the effective use of rating systems in business processes for credit granting and renewal, in the monitoring of exposures, in pricing determination, in value adjustments (*Expected Credit Loss*), and in the allocation of economic and regulatory capital. In line with Article 144 of the CRR, the model must support managerial decisions and not be limited to mere prudential compliance purposes.

- **Model performance and stability**: *backtesting* analyses and ongoing monitoring of discriminatory power (e.g., Gini index, AUC), calibration (observed/expected default rate ratio), parameter stability, and robustness across different phases of the economic cycle. Evidence must be provided on sample representativeness, data quality, and the statistical assumptions adopted.
- **Technical and functional documentation**: a complete description of the methodology, explanatory variables, estimation techniques, structural assumptions, and methodological alternatives considered, as well as the code used. The model's scope of application, segmentation logic, observation periods, and any usage limitations must be specified.
- **Operational and IT controls**: checks on data quality, completeness, and timeliness; consistency across operational, regulatory, and reporting environments; documentation of **data lineage** and end-to-end **audit trail** processes. Evidence of automated and manual validation controls, as well as anomaly management procedures, is required.

11.4.2 Post-implementation monitoring

Following the authorisation and implementation of the change or extension, the bank must establish **continuous monitoring** of the model's performance, aimed at verifying the **maintenance of qualitative and quantitative requirements** set out by regulation; the **absence of *model drift*** phenomena or **deterioration of parameters** estimated relative to the observed data; the **timely activation** of **remediation** and **recalibration** measures in the presence of material deviations; and consistency with the principle of **regulatory continuity**, ensuring that the changes do not compromise the historical comparability of risk metrics and capital requirements.

The monitoring process is subject to periodic review by **validation**, which assesses its effectiveness and alignment with supervisory expectations. The results of the reviews are communicated to the competent internal committees and are recorded in the **model change register**.

11.5 Materiality metrics and regulatory impacts

The **materiality** of a change or an extension of internal models is assessed through a coherent set of quantitative and qualitative indicators, aimed at measuring the expected impact on capital absorption, risk parameters, and decision-making processes. The measurement is carried out at multiple levels of granularity (portfolio, segment, model). In this framework, the internal policy sets homogeneous calculation criteria, decision thresholds, and explicit links with reporting obligations to the supervisory authority, in line with the CRR/CRD framework and with the assessment methodology of the competent authority.

From a *quantitative* standpoint, the main impact is measured on regulatory capital through the change in *Risk-Weighted Assets* (RWA) and the Pillar 1 requirement:

$$\Delta\mathrm{RWA} \;=\; \mathrm{RWA}^{\mathrm{post}} - \mathrm{RWA}^{\mathrm{pre}}, \qquad \Delta\text{Capitale minimo} \;=\; 0{,}08 \times \Delta\mathrm{RWA},$$

where 0,08 represents 8% of the minimum capital ratio. To allow cross-sectional comparability, a normalised measure is also used:

$$\mathrm{MI}_{\mathrm{RWA}} \;=\; \frac{|\Delta\mathrm{RWA}|}{\mathrm{RWA}^{\mathrm{pre}}},$$

interpreted as a *materiality index* on capital. For managerial purposes, this is complemented by the change in expected loss (EL) and in risk parameters:

$$\Delta\mathrm{EL} \;=\; (\mathrm{PD} \times \mathrm{LGD} \times \mathrm{EAD})^{\mathrm{post}} - (\mathrm{PD} \times \mathrm{LGD} \times \mathrm{EAD})^{\mathrm{pre}},$$

$$\Delta\overline{\mathrm{PD}} = \frac{\sum_i w_i\,\mathrm{PD}_i^{\mathrm{post}}}{\sum_i w_i} - \frac{\sum_i w_i\,\mathrm{PD}_i^{\mathrm{pre}}}{\sum_i w_i}, \quad \Delta\overline{\mathrm{LGD}},\ \Delta\overline{\mathrm{EAD}} \text{ defined in an analogous way,}$$

with weights w_i proportional to EAD or to the capital contribution, depending on the level of analysis. Performance measures are compared *pre/post* to evidence model stability under different cyclical conditions.

Materiality considers: (1) the scope of the **methodological change** compared with the approved version; (2) the **effects** on the **scope of application** (segmentation, products, jurisdictions); (3) the **implications for the processes** of origination, management and monitoring, pricing, value adjustments, and capital allocation; (4) the **consistency** with the governance arrangements, validation, and *use test*. These elements, together with the quantitative indicators, feed a *decision matrix* that determines whether the change falls within the set of *material* changes (with *ex-ante* notification obligation and prior assessment) or *non-material* changes (with *ex-post* reporting), as well as the possible need for prior approval for scope extensions.

The policy defines **decision thresholds** and **levels of analysis**, specifying: (1) the reference **datasets**; (2) the **observation time horizon**; (3) the **normalization and aggregation criteria**; (4) the **reconciliations with supervisory data**. The link with **reporting obligations** governs the methods and timelines for communication to the competent authority, ensuring consistency with the model change register and with the *model inventory*.

The outcomes of the assessment guide: (1) the **authorization or notification obligations**; (2) the ***roll-out* plans** (scope, phases, decision *gates*) or ***recalibration*** (parameters, samples, observation horizon); (3) the conditions for the **maintenance of or exit from *permanent partial use* (PPU)**; (4) the update of the **control arrangements** and model risk policies, including *remediation* actions and post-implementation monitoring requirements.

11.6 Use test, validation and performance

11.6.1 Principles and objectives

Pursuant to Articles 144, 145 and 185 of the **Regulation (EU) No 575/2013 (CRR)** and the provisions of the *ECB Guide to Internal Models* (ed. 2025), the **use test** represents an essential prerequisite for the authorisation and maintenance of the IRB approach. It requires that internal rating systems and risk parameters be effectively integrated into management and decision-making processes, and not limited to the mere determination of regulatory capital requirements.

From this perspective, the model must serve as an operational tool supporting the functions of credit **origination and renewal**, **monitoring** of the performance of exposures, **pricing** and setting of economic terms, calculation of **value adjustments** and **allocation of capital**, both economic and regulatory.

In parallel, **independent validation** is a fundamental requirement for verifying the methodological soundness, data quality, and adequacy of the results produced by the model. It is carried out by a function separate and independent from development and must cover the entire model life cycle: design, implementation, monitoring, and periodic review. The frequency of the checks is proportionate to the level of risk and the materiality of the model, but must in any case ensure a full review at least annually, in line with Article 185 of the CRR. Validation verifies:

- the **logical-mathematical correctness** of the formulas and underlying assumptions;
- the **statistical robustness** of the estimates and the **adequacy of the samples** used;
- the **stability of performance** over time and their **consistency with observed data**;

- the **compliance of the models** with governance policies, regulatory requirements, and the principles of *model risk management*.

The outcomes of the validation activities are formalised in specific **assessment reports** submitted to the competent committees and, where relevant, to the supervisory authority, accompanied by any *remediation plan* and *follow-up*.

11.6.2 Required evidence

The quantitative and qualitative evidence supporting the continued validity of the model must include at least:

- **Discriminatory power analysis** (*discriminatory power*): measurement of the model's ability to correctly distinguish non-defaulted counterparties from those in default, using indicators such as the Gini coefficient, the area under the ROC curve (*AUC*), or the correct classification rate.
- **Calibration analysis**: verification of the alignment between observed and estimated default rates, using indicators such as the observed/expected ratio (*O/E ratio*), the *Hosmer–Lemeshow test*, or the *Brier score*.
- **Sensitivity and robustness analysis**: assessment of the responsiveness of the estimated parameters to changes in input assumptions, explanatory variables, or the data scope. The goal is to ascertain that the model's performance remains stable under alternative scenarios or across different phases of the economic cycle.
- **Temporal stability analysis**: comparison of results over time to highlight any *model drift* or *data drift*, distinguishing between structural changes in the portfolio and statistical degradation of the model.
- **Data consistency checks**: quality, completeness, and timeliness controls of the datasets used for development and monitoring; traceability of transformations, exclusions, and applied filters, with explicit reference to the *data lineage*.
- **Sampling analysis**: verification of the representativeness of the samples used to estimate the parameters (*PD*, *LGD*, *EAD*), ensuring a time horizon adequate to capture different economic conditions, as required by Article 180 of the CRR.

The collected evidence must demonstrate that the model retains over time a level of **accuracy, stability, and consistency** compatible with regulatory and managerial use. Any issues identified during validation or monitoring must be subject to traceable action plans, with remediation timelines proportionate to the severity of the risk and with **periodic reporting** to corporate bodies and the competent authority.

11.7 Governance, roles and reporting

The governance of internal credit risk models is based on the principle of the **three lines of defense**. This arrangement ensures the segregation of functions, the traceability of responsibilities, and the independence of controls, safeguarding the overall quality of the modeling framework and its alignment with prudential requirements.

11.7.1 Roles and responsibilities

- **First line of defense** — Includes the *business units* and the operational structures that use the models in the processes of credit origination, monitoring, and management. They are responsible for the proper use of ratings and risk parameters (*PD*, *LGD*, *EAD*), as well as for performing first-level controls on input data and calculation outputs. They must ensure full adherence to internal credit and *use test* policies, ensuring consistency between managerial and regulatory use of the models.

- **Second line of defense** — It consists of the *risk management*, *compliance*, validation functions and the Credit Risk Control Unit (CRCU), as provided for by Articles 189 and 190 of the CRR. It oversees **methodological supervision**, data quality, consistency of the scope of application, and traceability of information flows. The CRCU defines and updates the *model policies*, assesses changes and extensions (*model change policy*), monitors model performance, and ensures compliance with regulatory requirements and supervisory expectations.

- **Third line of defense** — The *internal audit function* ensures an independent and periodic assessment of the effectiveness, completeness, and actual implementation of the model governance framework. It verifies the soundness of the development, validation, approval, and monitoring processes, as well as the adequacy of the controls implemented by the first two lines. The outcomes of the audits are reported to the corporate bodies and, where appropriate, communicated to the supervisory authority.

11.7.2 Governance processes

The governance framework is structured into policies and procedures approved by the bodies with a strategic oversight function (Board of Directors and Risk Committee), which establish the **criteria for approval** and the **review of internal models**; the **modalities** for managing **model changes** and **scope extensions**; the **reporting** and **escalation** flows.

11.7.3 Documentation

All documentation related to models, policies, validation evidence, and communication flows must be kept **complete, up-to-date, and verifiable**, ensuring the **version traceability** of the models and their approvals; the **preservation of evidence** of development, validation, and periodic review; the availability of a **model register** (*model inventory*) containing up-to-date information on scope of application, validation status, performance, materiality of changes, and interactions with the supervisory authority.

Chapter 12

Loan Origination & Monitoring and Outsourcing

12.1 Purpose and framework

The **Guidelines on loan origination and monitoring** (*Loan Origination & Monitoring*, LOM) promote a sound, transparent and comparable credit granting process within the Union, ensuring that credit decisions are based on reliable information and on controls proportionate to risk. They explicitly link the *risk appetite* to origination and monitoring processes, thereby ensuring alignment between business objectives and regulatory requirements.

There are four main objectives: (i) **to ensure the quality** of new originations through proportionate information and assessment standards; (ii) **to strengthen financial stability** through ex-ante and ex-post controls on creditworthiness and on the behaviour of exposures; (iii) **to protect customers** through clear, documented and non-discriminatory processes; (iv) **to harmonise practices** for operations and controls in the banking market, fostering comparability and market discipline.

Implementation requires an **internal governance framework** that integrates credit policies, operating limits and decision-making responsibilities with capital and liquidity safeguards, ensuring consistency between management and regulatory metrics. This framework is supported by a data infrastructure and information systems capable of providing a *single customer view* across the entire life cycle of exposures and collateral, in compliance with data quality requirements.

Completing the framework are the rules on **outsourcing** (*outsourcing*) and **business continuity**, aimed at managing operational, legal and ICT risks with appropriate contracts and controls also for activities entrusted to third parties, and a **reporting system** that is traceable and proportionate to the institution's complexity, capable of documenting decisions, approvals and risk levels, as well as promptly providing supervisory authorities with the information requested in line with transparency obligations.

12.2 Governance, risk culture and internal controls

12.2.1 Role of the bodies and decision-making structure

Credit governance must ensure prudent, transparent management consistent with the institution's *risk appetite*. Pursuant to Articles 74 and 88 of **Directive 2013/36/EU (CRD IV)** and subsequent updates, the management body in its supervisory function is responsible for approving the credit strategy (**Risk Appetite Framework (RAF)**) and for establishing a clear hierarchy of decision-making powers that ensures the effective segregation of duties among operational, control, and decision-making functions.

The **management body** sets overall exposure limits and concentration criteria by portfolio, sector, and counterparty, approves origination and monitoring policies, and periodically verifies the consistency between growth objectives, credit quality, and capital sustainability. The **decision-making structure** must ensure independence between:

- *Origination*: functions responsible for sourcing and preparing credit proposals;
- *Creditworthiness assessment*: technical units or risk officers responsible for the analysis of creditworthiness, cash flow, collateral and guarantees;
- *Decision*: decision-making bodies or credit committees, vested with powers consistent with the risk thresholds and autonomy levels provided for by the delegation framework;
- *Control and monitoring*: second-line functions (Validation, Risk Management and Compliance) and third line (Internal Audit), responsible for ex post control.

The system must include **continuous monitoring mechanisms** of the portfolio, **early warning** models, and periodic **stress tests**, aimed at assessing the resilience of credit quality in adverse scenarios and the impact on capital and liquidity (ICAAP and ILAAP). The internal control framework must be integrated and documented, including ex ante and ex post verification procedures, reporting flows to top management and the supervisory authorities, and risk-adjusted performance indicators.

Remuneration policies must be consistent with the risk strategy and structured so as not to create distortive incentives that could undermine credit quality or promote excessive risk-taking. Therefore, individual and collective objectives must include risk parameters, portfolio quality, and long-term sustainability.

12.2.2 Culture, conduct, ESG and AML/CFT

Risk culture constitutes the basis for healthy and responsible organisational conduct. In accordance with the *EBA Guidelines on Internal Governance (EBA/GL/2021/05)*, institutions must promote ethical standards, professional integrity and **widespread accountability**, ensuring that staff understand the prudential and reputational implications of their credit decisions. There must also be provisions for **regular training** initiatives and effective **whistleblowing** procedures, to safeguard transparency and the proper functioning of decision-making processes.

The integration of **ESG factors (Environmental, Social, Governance)** into credit origination and monitoring processes is required for both prudential and strategic sustainability reasons. Institutions must assess the impact of environmental and climate, social and governance risks on borrowers' solvency profiles and on the quality of collateral, applying more intensive analysis and monitoring to sectors and counterparties exposed to high risk. ESG factors must be incorporated into valuation models, *due diligence* processes and pricing policies, in line with the ECB's supervisory expectations (Guide on climate-related and environmental risks, 2020).

The information collected for the assessment of **solvency** must be consistent with the safeguards set out in the legislation on **anti-money laundering and countering the financing of terrorism (AML/CFT)**, ensuring consistency between creditworthiness checks and customer due diligence obligations. The institution must designate a **member of the management body** responsible for implementing AML/CFT policies and ensure that the granting, renewal and monitoring processes are coordinated with the controls for preventing money laundering and financial abuse risks.

12.3 Data infrastructure and MIS

The data infrastructure must ensure a **single customer view**, guaranteeing integration among management systems, risk engine, and regulatory reporting, in line with the capital and liquidity safeguards (ICAAP/ILAAP) required by regulation. The information architecture includes a coherent *data model*, an official *data dictionary* with metadata and validation rules, *staging/golden source* environments, and end-to-end **data lineage** and **audit trail** processes.

The **MIS** (Management Information System) supports decision-making, monitoring, *pricing*, rating/score, and risk-adjusted performance measures, providing multi-level views (portfolio, counterparty, position, etc.) and systematic reconciliations between managerial and regulatory data. Data quality is governed along the dimensions of **accuracy**, **completeness**, **timeliness**, **consistency**, and **traceability**, with ex-ante controls (admissibility rules, formats, mandatory fields) and ex-post controls (quality indicators, thresholds, remediation plans). The design of the information sets for origination and monitoring can be inspired by the EBA *templates* for NPLs, in order to capture the essential *drivers* (master data and economic group, cash flows, collateral, covenants, behavioural profile).

12.4 Origination: information, creditworthiness, decision, and pricing

12.4.1 Information and creditworthiness

Credit proposals are supported by an **information set**, proportionate to the risk, size, and complexity of the transaction, as well as complete and verifiable. Economic-financial and qualitative information enable the estimation of **debt sustainability**, including scenarios and sensitivities.

For *corporate* counterparties, **debt-service coverage** indicators are used to assess the firm's financial sustainability relative to cash flows generated by operating activities:

$$\text{DSCR} = \frac{\text{CFADS}}{\text{Debt Service}}, \qquad \text{ICR} = \frac{\text{EBITDA}}{\text{Interest Expense}},$$

where:

- DSCR (*Debt Service Coverage Ratio*) measures the firm's ability to cover the total amount due for principal and interest (Debt Service) with the cash flows available for debt service (CFADS[1]). A value of DSCR > 1 indicates that operating cash flows exceed financial commitments, signalling a contained risk profile.
- ICR (*Interest Coverage Ratio*) assesses the sustainability of the financial burden alone, by relating EBITDA (*Earnings Before Interest, Taxes, Depreciation and Amortization*, i.e. gross operating profit) to interest expenses (Interest Expense). Higher values of ICR denote a greater capacity of the firm to meet its interest payments, even in the event of profitability downturns or rising market rates.

Both indicators form part of the *creditworthiness* metrics set out in the EBA guidelines on loan origination and monitoring, and are central elements in the forward-looking assessment of solvency and in the definition of exposure limits.

[1] In the context of the DSCR, the term **CFADS** (*Cash Flow Available for Debt Service*) refers to the **cash flow available for debt service**. It is, in essence, the cash flow generated by the project or the firm over a given period which, once operating costs and other necessary outflows have been met, **remains available** to service the payment of **interest**, **principal** and any **other similar charges**.

For *retail* customers and, in particular, for *households*, creditworthiness analysis is based on indicators of individual **debt burden** and **financial leverage**, aimed at measuring debt sustainability relative to the debtor's earning capacity. The main metrics are:

$$\text{LTI} = \frac{\text{Loan Amount}}{\text{Annual Income}}, \qquad \text{LSTI} = \frac{\text{Debt Service}}{\text{Disposable Income}},$$

where:

- LTI (*Loan-to-Income*) expresses the ratio between the amount of the loan granted and the debtor's gross annual income, providing a measure of overall **financial leverage**;
- LSTI (*Loan Service-to-Income*) represents the weight of debt service (principal plus interest) on net disposable income.

High values of these indicators signal greater borrower vulnerability to income shocks or interest-rate changes, with implications for credit risk assessment and credit granting policies.

Risk profile assessment models and technologies are subject to **model risk governance**: traceability of variables and data transformations; statistical and *out-of-sample* robustness of estimates; performance monitoring (discrimination, calibration, temporal stability); safeguards against *bias* and *drift*; data quality requirements and documented *model change*. Decisions are recorded in an **auditable** manner, with evidence of the approved conditions, preconditions, and applied mitigants (including ESG profiles and consistency with AML/CFT safeguards).

12.4.2 Decision and documentation

Each credit proposal must be accompanied by a **documented rationale** and by a complete information set, including economic and financial analyses, qualitative assessments, ESG elements, AML/CFT profiles, the structure of credit protection and guarantees, and the outcomes of rating/score models. Decisions must document, in a traceable manner:

- the approved terms and any preconditions for disbursement;
- the applied mitigants (guarantees, covenants, *haircut*, repayment plans, *early termination* clauses);
- assessments of consistency with concentration limits, *risk appetite*, and sustainability metrics.

The **documentation** must be *comprehensive*, *consistent*, and *auditable*, so as to enable the full reconstruction of the decision-making process and alignment among management, control, and regulatory reporting systems. Each decision, amendment, or renewal must be recorded in the internal information systems, accompanied by evidence of the approval date, the powers exercised, and any deviations from standard policies.

12.4.3 Pricing and risk-adjusted profitability

The **pricing** of credit exposures must be consistent with the assumed risk profile, the counterparty's quality, and the characteristics of the transaction. In line with Article 79 of **Directive 2013/36/EU (CRD IV)**, as amended, and with the **EBA Guidelines on Loan Origination and Monitoring (EBA/GL/2020/06)**, pricing must explicitly reflect the main economic drivers of the transaction, and in particular:

- the **cost of capital** employed, determined on the basis of **regulatory capital absorption** (under CRR/CRD) and **internal capital absorption** estimated by the bank. In other words, the expected margin must adequately remunerate the capital that the institution is required to allocate to cover the risks taken;
- the **cost of funding**, which includes both the **cost of the funding** required to finance the transaction and the **liquidity and funds transfer charges** (*Funds Transfer Pricing, FTP*). The rate applied to the customer must therefore cover the effective cost at which the bank obtains and manages funds in the market;
- the **expected cost of risk**, estimated through the parameters of **probability of default (PD)**, **loss given default (LGD)**, and **exposure at default (EAD)**. Expected loss must be embedded in the price, so as to offset over time the losses that will materialize in the portfolio;
- **market conditions** and **target profitability**: pricing must be consistent both with the interest rate levels and market-observable spreads for comparable transactions, and with the **profitability targets** set by the bank, in line with pricing policies and the overall commercial strategy.

The bank uses **risk-adjusted performance metrics** to assess the expected profitability of transactions and their consistency with the assumed risk profile. These indicators make it possible to integrate measures of economic return with capital requirements and the opportunity cost of equity. The main metrics used are:

- RAROC (*Risk-Adjusted Return on Capital*) measures the risk-adjusted economic profitability by comparing the *risk-adjusted net margin* — i.e. the result net of *Expected Loss* and operating costs — with the *economic capital* absorbed to cover unexpected losses. Values above the cost-of-capital threshold indicate value-creating transactions.
- RORWA (*Return on Risk-Weighted Assets*) expresses net return in relation to RWA, providing a synthetic measure of regulatory profitability per unit of risk.
- EVA (*Economic Value Added*) assesses the creation of residual economic value as the difference between *Net Operating Profit After Taxes* (*NOPAT*) and the opportunity cost of the capital employed, calculated as WACC × Capitale Investito, where WACC is the *Weighted Average Cost of Capital*.

Taken together, these metrics allow the regulatory and managerial perspectives to be integrated, steering pricing, capital allocation, and performance assessment decisions toward criteria of efficiency and risk sustainability.

Below-cost transactions must be identified, justified, and approved in accordance with **dedicated internal policies**, with the recording of exceptions and traceability of the decision. Such cases are permitted only if justified by strategic, relationship, or portfolio considerations, and are subject to periodic monitoring and reporting. Finally, the **pricing** system must ensure consistency with **fairness** rules (procedural fairness, distributive fairness, transparency and explainability, accountability and traceability, continuous monitoring).

12.5 Collateral valuation and management of *haircuts*

12.5.1 Eligibility and principles

The eligibility of guarantees requires compliance with legal, economic, and operational requirements that ensure their full prudential and managerial effectiveness. Pursuant to Articles 194–214 of the **Regulation (EU) No. 575/2013 (CRR)** and the **EBA Guidelines on Credit Risk Mitigation for IRB Approaches (EBA/GL/2020/05)**, a guarantee is considered eligible only if:

- it is **legally valid and enforceable** in all relevant jurisdictions, with formally correct documentation, absence of adverse encumbrances, and the possibility of timely enforcement;
- it can be **valued independently and verifiably**, based on objective data and transparent methods, suitable to estimate the current value and its volatility;
- it is **liquidable within an appropriate timeframe**, i.e. realizable on the market or through enforcement procedures within a prudential horizon compatible with recovery processes;
- it exhibits **transparency and operational segregation**, with recording, retention, and custody that ensure title and independence from other exposures;
- it complies with **internal limits on concentration and correlation**, avoiding excessive exposure to specific guarantors, assets, or underlying markets.

These criteria apply to all recognized forms of guarantee, both financial (collateral, deposits, securities) and personal (sureties, bank guarantees). The risk function is responsible for defining the eligibility criteria, verifying compliance with quality requirements, and periodically updating the lists of eligible guarantees and counterparties.

12.5.2 Valuation and *haircut*

The **valuation** of collateral is independent of origination processes and must ensure neutrality, accuracy, and traceability. Consistent with Articles 208 and 229 of the CRR, valuations must be based on prudent criteria, reflect current market value, and take into account any economic, legal, or operational deterioration factors. The valuation process includes:

- **documentary verification** of ownership of the asset, the absence of encumbrances or liens, enforcement conditions, and insurance coverage;
- **independent appraisal** for real estate, entrusted to qualified appraisers, independent from the line of business and operating according to recognized standards, with mandatory traceability and retention of evidence;
- **valuation at observable prices** for market collateral (securities, financial instruments), using official sources, validated valuation models, and consistency checks;
- **prudent determination of haircuts**, reflecting risks of volatility, liquidity, maturity, correlation, and time to realization, as well as legal and operational factors.

Haircuts represent a prudent reduction of the gross value of the asset to cover market and realization risks. The determination of haircuts follows the criteria of the *Comprehensive Approach*[2], with levels calibrated based on the nature of the asset, the frequency of valuation updates, historical volatility, and the degree of correlation with the obligor. An operational formulation of the *loan-to-value* (LTV) ratio is as follows:

$$\mathrm{LTV}_t = \frac{E_t}{V_t^{\mathrm{adj}}}, \qquad V_t^{\mathrm{adj}} = V_t^{\mathrm{lordo}} \cdot (1 - h_t),$$

where:

[2]The Comprehensive Approach is a regulatory approach to credit risk mitigation techniques that reduces the exposure through the value of the collateral, applying haircuts for volatility, currency, and maturity, in order to obtain an adjusted exposure to be risk-weighted for capital purposes.

- E_t is the relevant credit exposure at date t;
- V_t^{lordo} is the market value of the collateral gross of prudential adjustments;
- h_t is the overall haircut, which incorporates the components for volatility (H_{vol}), liquidity (H_{liq}), foreign-exchange risk (H_{fx}) and maturity (H_{mat}).

The relationship ensures that the prudential benefit does not exceed the value of the asset that can actually be realized under stress conditions. The bank must maintain documentary evidence on the assumptions, models, and data used for the estimate, as well as periodically review haircut levels based on market dynamics, stress tests, and the results of independent valuations.

12.5.3 Valuation standards, models and frequencies

Valuation and revaluation activities of collateral must be carried out according to **internationally recognized professional standards**, in compliance with the provisions of Articles 208 and 229 of **Regulation (EU) No 575/2013 (CRR)** and the **EBA Guidelines on loan origination and monitoring (EBA/GL/2020/06)**.

The **appraisers** must be *independent* of the origination and decision-making units, operate on a rotational basis, and belong to *accredited panels*, subject to periodic monitoring of skills, integrity, and appraisal quality. The bank must ensure that the appraisers have an adequate level of specialization with respect to the asset type (real estate, movable, financial) and that no conflicts of interest exist. Valuations must be supported by **detailed and traceable reports**, containing: applied methodology, information sources, market assumptions, comparison parameters, and analysis of impairment risks.

The **backtesting** process is an integral part of the internal control system and serves to verify, at least annually, the consistency between estimated values and actual realization prices, identifying any systematic deviations or valuation bias. In the event of significant anomalies, model review actions and, if necessary, the replacement of appraisers must be implemented.

The use of **statistical or automated models** for the estimation or revaluation of values is permitted only where stringent conditions are met concerning the quality, size, and timeliness of input data, the robustness of the methodologies, and the availability of *uncertainty measures* associated with the estimates. Such models must be subject to methodological validation and continuous performance monitoring.

The **valuation frequencies** are defined according to a principle of proportionality with respect to the nature and volatility of the asset:

- **Continuous updates** for assets with market-observable prices (e.g., listed securities, liquid financial instruments), through automated *mark-to-market* procedures;
- **Periodic updates** for non-traded assets, such as real estate, at least every three years for residential assets and annually for commercial or special-purpose ones, accompanied by intermediate *valuation review* based on market indicators or analysis of comparables;
- **Event-driven revaluations** in the presence of *triggers* such as material market changes, legal or zoning changes, deterioration in the debtor's conditions, breaches of contractual covenants, or force majeure events.

As part of the valuations, **ESG factors** must also be taken into account where they have a material impact on the value of the asset or on its marketability. By way of example, the energy efficiency of a property or the risk of environmental obsolescence may affect the estimate of residual value, the time to realization and, consequently, the level of prudential *haircut* applied.

12.6 Monitoring, KPI, early warning and reviews

12.6.1 Framework and indicators

The **monitoring framework** must be proportionate to the risk, the size, and the complexity of the portfolio. It integrates managerial and regulatory sources, enables aggregations by lines of business, products, sectors, and geographies, and supports the identification of concentrations and interdependencies. The covered areas include: payment behavior, borrower and transaction risks, ultimate exposures and line utilization, *impairment/write-off* dynamics, and valuation decisions.

KPIs enable integrated monitoring of portfolio performance, counterparties' credit quality, and the consistency of the risk profile with managerial and regulatory objectives. They include:

- the evolution of risk parameters — probability of default (PD), rating and score — with analysis of changes, migrations, and upgrade/downgrade rates;
- the trend in arrears, limit breaches, and the utilization rate of credit lines;
- compliance with contractual *covenants* and financial sustainability indicators;
- leverage ratios and debt service coverage ratios;
- collateral indicators, such as *Loan-to-Value* - LTV and *Loan-to-Cost* - LTC;
- concentration measures and large exposures;
- impairment and reversal flows (*impairment/reversal*);
- risk-adjusted profitability metrics.

Alert thresholds, materiality levels, and **data quality** rules are formalized in corporate *policies* and supported by *ex ante* and *ex post* controls, with *data lineage* and *audit trail* systems that ensure verifiability and consistency of the information.

12.6.2 Collateral: monitoring, triggers and escalation

Collateral monitoring is continuous and integrates performance and market indicators in order to promptly assess the effective protection capacity of the collateral and the possible emergence of residual risks. The bank oversees at least: (i) *loan-to-value* ratio (LTV), (ii) *coverage ratio* of non-performing exposures[3], (iii) volatility and liquidity of the underlying asset, (iv) legal and enforceability risks, (v) correlation with the borrower and concentration by guarantor/asset. Prudential *haircut*s are dynamically reviewed in light of market conditions, the evolution of the exposure, and the results of *stress test*s consistent with the *RAF*.

Stress testing and haircut review. The adjusted value of the collateral is subjected to *stress tests* for market, liquidity, and foreign exchange; adverse scenarios inform the periodic review of haircuts and the possible application of *add-ons* for concentration or adverse correlation. Intervention thresholds are calibrated by segment/asset class and are consistent with the limits of the *RAF*.

[3]The Coverage Ratio is the ratio between provisions and NPLs.

Triggers and mitigating measures. The **early warning** system provides for *Early Warning Indicators* (EWI) aligned with the RAF and calibrated by segment and portfolio. The **EWI** cover **performance signals** (recurring arrears, increased utilization, deterioration in DSCR/ICR), **creditworthiness signals** (rating/score downgrade, increase in expected PD), **collateral signals** (deteriorating LTV/LTC, loss of enforceability), and **qualitative signals** (legal event, *covenant breach*, material ESG risks). When thresholds are breached, **escalation workflows** are triggered with action plans: collateral enhancement, *margin call*[4], facility re-sizing, renegotiation of covenants/pricing, or initiation of the recovery process.

Periodic reviews are proportionate to the counterparty's risk, size, and complexity, with increased frequency in the event of deterioration or material changes in the information profile. The review includes documentation updates, reassessment of solvency, verification of compliance with *covenants*, reconsideration of risk parameters and *risk-adjusted* metrics.

Escalation and reporting. **Escalation** to the risk function is timely, structured, and accompanied by: case classification, analysis of the *cause*, remediation plan, and implementation timelines. **Reporting** is aligned with **RDARR** principles and ensures end-to-end **data lineage** and **audit trail**, with aggregations by portfolio and position and reconciliation with regulatory flows. Monitoring outcomes are communicated to the competent bodies and used to update limits and collateral policies.

12.7 Outsourcing, operational continuity and group governance

The **outsourcing policy** governs the entire life cycle of third-party arrangements, in line with the governance framework set out in Articles 74 and 88 of the **CRD** and with the prudential safeguards provided for by the **CRR**. It sets criteria for the classification of *critical or important functions*, *due diligence* and risk assessment processes, minimum contractual requirements, methods for ongoing monitoring, and rules on *exit* and *sub-outsourcing.* The aim is to preserve **business continuity** and the **adequacy of controls** even when processes or services are outsourced, ensuring that the institution retains at all times *accountability*, audit rights and governance capability.

Lifecycle and risk controls. Before outsourcing, the bank carries out a systematic analysis of:

- **operational, legal and compliance risk**, including liability profiles, data ownership, personal data protection and cross-border transfers;
- **ICT/IT risk** (information security, availability, integrity and confidentiality), with resilience requirements, *incident management* and notification obligations;
- **reputational and concentration risk**, considering exposures to individual providers, *single points of failure* and chains of *sub-outsourcing*;
- **contractual compatibility** with the rights of the control function, the governing bodies and the competent authority (access, inspection, audit, *on-site/off-site*).

The contract defines scope and *service level agreement*, security standards, *key performance indicators* (KPI), *key risk indicators* (KRI), *reporting* obligations, restrictions on *sub-outsourcing*, audit rights and exit conditions. The institution keeps a **centralized outsourcing register** with criticality classification, data location, chain of sub-providers and results of controls.

[4] *Margin calls* are requests between counterparties to top up collateral in order to maintain collateral balance in the face of market movements.

Business continuity and emergency plans. The **Business Continuity Management** (BCM) covers internal and outsourced processes and provides for *Business Impact Analysis* (BIA), continuity and disaster recovery plans with measurable objectives.

The plans define *runbooks*, *fallbacks*, and escalation points; they are tested with a frequency proportionate to the criticality of the service and to the outcomes of the *stress scenarios*. For services in *outsourcing* or *cloud*, ***switch-over* capabilities**[5], **data portability**[6] and **exit plan** are required, all executable within timeframes compatible with the limits of the *RAF* and with liquidity/collateral constraints.

Monitoring, SLA and *incident management*. Continuous monitoring verifies adherence to *SLA, KPI*, residual risks, and compliance with security and *data governance* requirements. **Incidents** are classified by severity, with *root cause analysis*, *remediation* plans, and defined timelines; relevant events are notified to the competent bodies and, where applicable, to the supervisory authority. Information flows adhere to **RDARR** principles and ensure end-to-end **data lineage** and **audit trail**.

Group governance and consolidated perimeter. At consolidated level, the **parent entity** ensures consistency of policies, methodologies, and controls across all group entities, including **third countries**. Subsidiaries adopt minimum risk standards, with uniform metrics. **Intra-group outsourcing** is subject to the same assessment and control criteria as outsourcing to third parties, so as to avoid *blind spots* and uncontrolled critical dependencies. The parent entity maintains **visibility over the sub-supplier chain** and an effective audit capability, including through group-level agreements and standardized contractual clauses.

Accountability, roles and reporting. The bodies with a **strategic oversight** function approve the *outsourcing* policy, the criticality classification criteria, and the concentration limits. The *first line of defense* is responsible for the operational management of outsourcing and for compliance with the *Service Level Agreement* (SLA); the *second line* oversees the methodologies adopted, residual risks, and regulatory compliance; the *third line* assesses the overall effectiveness of the control framework and the implementation of *remediation* plans. The **reporting** to the corporate bodies and the competent authority must comprehensively document the decisions taken, performance and risk indicators, the outcomes of continuity tests, and the status of *exit* plans, ensuring **traceability**, **completeness**, and **proportionality** in relation to the criticality of the outsourced services.

[5]Switch-over is the capability of a system (technical, organizational, or operational) to transfer its functions from a primary asset/environment to an alternative one in a timely and controlled manner, ensuring continuity and stability of the service.

[6]Data portability is the ability to transfer information securely, completely, and in a format readable by one information system to another.

Chapter 13

ESG and climate risks

13.1 Purpose and scope

The objectives of integrating **ESG** factors, and in particular **climate risks**, into the credit process are multiple. First, it is necessary to proceed with the systematic **identification** of **sensitive exposures**, taking into account size, tenor, economic sector, geographical area, and characteristics of the **collateral**. Second, it is necessary to **measure** physical and transition impacts through **forward-looking** approaches, based on **scenarios** and **sensitivity analyses**, consistent with the time horizons of the exposures and with their uses in internal models. Third, the assessments produced must translate into **risk adjustments** in the **credit analysis** and **pricing** phases, so as to correctly reflect the risk profile of the borrower and the collateral.

A further objective concerns the **documentation** of **ESG** considerations and decisions taken: assessments, assumptions, parameters, and outcomes must be **traceable**, **verifiable**, and **reproducible**, so as to support compliance, control, and validation functions. Finally, it is necessary to define **monitoring metrics** and **reporting** processes (internal and external) that make it possible to consistently represent exposure to **ESG** and climate risks, the impacts on credit portfolios, and the evolution of risk management policies.

In this context, two main categories of climate risk are distinguished:

- **Physical risks**: effects arising from acute events (e.g., extreme weather events) or from chronic trends (sea-level rise, desertification, structural changes in precipitation) with impacts on counterparties and *collateral*;
- **Transition risks**: potential losses linked to **regulatory**, **technological**, and **market** changes associated with decarbonization processes and the transition to a low-carbon economy.

13.2 Identification of sensitive exposures

Identifying exposures most affected by **climate risks** and, more broadly, by **ESG** factors, relies on a structured process that integrates **sectoral**, **geographical** and **counterparty-specific** analyses. The first phase consists of a **sectoral and geographical mapping** aimed at identifying areas characterized by high vulnerability to **physical risks** (extreme weather events, adverse climate trends) or to **transition risks** (**regulatory**, **technological** or **market** changes).

Alongside the mapping is the definition of **materiality thresholds**, calibrated as a function of size, exposure duration, level of **concentration** and the economic relevance of the **collateral**. These thresholds make it possible to proportionately select the positions to be further analysed, ensuring a targeted use of analytical resources. Exposures are organized using **internal taxonomies**, consistent with those adopted at the group level.

Finally, the process requires a **clear assignment of roles and responsibilities** for the construction, updating and periodic maintenance of the lists of sensitive exposures, ensuring continuity of oversight and a clear organizational framework for managing **climate risk**.

13.3 Integration of ESG and climate factors into credit assessment

The integration of **ESG factors** and **climate risks** into credit assessment spans all the main stages of the process.

13.3.1 Due diligence

Due diligence includes the structured collection of relevant **ESG** information (i.e. governance, environmental policies, transition strategies, exposure to physical risks, and commitments undertaken by the entity). The information sources must be **verified**, **documented** and assessed using **ESG scorecards** that measure the **materiality** of the factors observed and their consistency with the characteristics of the transaction and the counterparty.

13.3.2 Risk modelling

In **risk modelling**, ESG and climate factors are integrated through the inclusion of **variables** and **segmentations** that reflect **physical risks** and **transition risks**. Where appropriate, the following can be introduced:

- **adjustments** to regulatory and managerial risk parameters (PD, LGD, EAD), supported by quantitative evidence, forward-looking analyses, and climate scenarios;
- **overrides** of standard assessments, based on formalised rationales and accompanied by **governance** safeguards, consistency checks, and adequate documentation of decisions.

All modelling choices must be **consistent** with the regulatory framework, based on **explicit assumptions**, supported by **traceable data sources**, and subject to **independent controls** by the relevant functions. This approach ensures methodological robustness and informational continuity in the treatment of climate and ESG risks throughout the credit process.

13.3.3 Credit decision-making

In the **credit decision-making** phase, *ESG* profiles are integrated through the inclusion of dedicated **contractual conditions** and **covenants**, intended to oversee the aspects relevant to the sustainability of the transaction. The most significant elements include:

- the company's **environmental commitments**, including emissions-reduction policies and climate-neutrality targets;
- the **transition milestones**, defined on a periodic basis and monitored against the decarbonization or adaptation plans declared by the counterparty;
- the **disclosure obligations** and **ESG reporting** requirements, necessary to ensure transparency, verifiability, and the continuous updating of relevant information.

Failure to comply with the commitments undertaken may trigger **contractual adjustment** mechanisms, such as: revision of the economic terms, updating of collateral levels, suspension of further disbursements, or activation of **escalation** procedures. These measures must be consistent with the **risk governance framework**.

13.4 Integration into pricing

The integration of *ESG* factors and **climate risks** into *pricing* follows a fully **risk–based** approach, aimed at consistently reflecting the overall risk profile of the transaction. In particular, the price incorporates:

- **additional components** to cover physical and transition risks, determined on the basis of potential losses, expected costs, and the risk tolerance set out in the *Risk Appetite Framework*;
- **haircut adjustments** on collateral exposed to physical risks, so as to reflect the volatility, liquidity, and vulnerability of the underlying asset;

13.5 Risk adjustments and operational limits

At the **portfolio** level, **climate risks** are incorporated through the introduction of **operational limits** and **risk appetite metrics** (*risk appetite*) referring to **sectors** and **geographical areas** characterized by high sensitivity to *physical risks* and *transition risks.* In this framework, institutions:

- define **thresholds** and **containment objectives** for exposures to segments with high climate risk, systematically monitoring their evolution over time;
- assess the presence of **material risk concentrations** (by sector, geographical area, counterparty, or type of collateral) and plan **roadmaps for reduction or realignment** consistent with the institution's **sustainable transition** strategy;
- use **climate scenarios** and **stress tests** to verify the **portfolio's resilience** and the **robustness of the limits set**, ensuring consistency between forward-looking analysis, portfolio policies, and internal capital and risk requirements.

13.6 Modelling, validation and governance

13.6.1 Modelling

Risk modelling integrates *ESG* factors and **climate risks** through the use of **adjusted values** (e.g., parameters and valuations that take into account exposure to physical and transition risks) and **haircuts** in risk metrics and loss estimates. **Assumptions**, **parameters**, and **results** must be mutually consistent, documented, and verifiable, so as to ensure a **solid methodological basis**.

13.6.2 Validation and backtesting

The **independent validation function** verifies the **methodologies** adopted for the integration of *ESG* and climate factors; the **databases** and related quality controls; the **assumptions** and **parameters** used in the models; the **performance** of the models in capturing the effects of *physical risks* and *transition risks.* **Backtesting** activities ensure **comparability** between **model estimates** and **observed results**, allowing the assessment of the adequacy of the methodological choices.

13.6.3 Governance

The **governance framework** provides for a clear **separation of roles** between:

- functions of **origination** and **credit assessment**, responsible for the analysis of individual transactions;
- functions of **risk management**, tasked with defining methodologies, limits, and monitoring metrics;
- functions of **internal control**, which verify the overall consistency of the system.

The **policies** are updated periodically to incorporate **regulatory, technical, and methodological developments**.

13.7 Monitoring and reporting

The process entails the definition of **structured monitoring metrics** for **climate risks**, consistent with the institution's **strategy** and **risk limits**.

Internal and external **reporting** streams must support the **compliance**, **risk management**, and **validation** functions, ensuring **completeness** of informational content; **timeliness** in the production and dissemination of information; and **traceability** of sources, processing, and results.

The **documentation** of evidence, analyses, and determinations made must be suitable to support decisions, as well as kept up to date and **auditable**, so as to ensure **full transparency** and **verifiability** throughout the entire **climate and ESG risk management** cycle.

Chapter 14

IFRS 9 – Financial Instruments

14.1 Introduction and regulatory context

The *International Financial Reporting Standard 9* (**IFRS 9**) constitutes one of the pillars of the reform of international accounting standards following the 2007–2009 global financial crisis. It replaces *International Accounting Standard 39* (**IAS 39**), with the exception of certain hedge accounting cases (in particular, portfolio interest rate risk hedges), which continue to be governed by IAS 39 within the European Union.

IFRS 9 applies to a wide range of financial assets and liabilities held both by supervised intermediaries (banks, financial companies, insurance companies) and by industrial and commercial firms, significantly affecting the structure of the balance sheet, the dynamics of the income statement, and, indirectly, risk management choices.

14.1.1 Limitations of the IAS 39 model

IAS 39 has long been criticized for its conceptual and operational complexity. First, the **classification** of financial assets was divided into numerous categories, often characterized by stringent formal rules and extensive use of *fair value*[1], with sometimes opaque effects on volatility in profit or loss and equity. Second, the **impairment**[2] model based on *incurred loss* (*incurred loss model*) allowed the recognition of impairment losses only upon the occurrence of an objective *loss event*, generating a systematic delay in capturing the evolution of credit risk and making the financial statements poorly responsive to early warning signals of deterioration.

Added to these issues was the difficulty of interpreting the **hedge accounting**[3] rules, the fragmentation between accounting approaches and managerial risk measurement models, and the heterogeneity of practices across jurisdictions and among intermediaries. The financial crisis made these limitations evident, prompting the development of a framework more consistent with the underlying economics and more sensitive to forward-looking risk profiles.

14.1.2 Objectives of IFRS 9

IFRS 9 was developed with the primary objective of improving the **informational quality** of the financial statements in relation to financial instruments. The new standard pursues three core aims:

[1] Fair value is the price that would be received to sell an asset or paid to transfer a liability in an orderly transaction between market participants at the measurement date.

[2] Impairment is the value adjustment recognized when an asset (mostly receivables, but also financial instruments, property, goodwill, etc.) is overstated relative to its recoverable amount.

[3] Hedge accounting is an accounting approach (IFRS 9) that allows the accounting treatment of the hedging instrument (derivative, etc.) and the hedged item to be aligned, so that the gains/losses of the hedge offset those of the hedged item.

- to simplify and streamline the **classification and measurement of financial assets and liabilities**, anchoring it to the *business model* and the characteristics of contractual cash flows;
- to introduce an **impairment model based on expected losses** (*Expected Credit Loss model*) capable of bringing forward the recognition of credit losses and of systematically incorporating forward-looking information.

  The shift from the **incurred loss** approach to the **ECL model** introduces a **proactive** assessment of credit risk, based on 12-month horizons or on the entire remaining lifetime depending on the deterioration in credit quality. Estimates must systematically integrate **forward-looking information** and probability-weighted macroeconomic scenarios, increasing the cyclical sensitivity of the financial statements;
- to realign **hedge accounting** with the actual management of financial risks, reducing asymmetries between accounting representation and the hedging strategies adopted by the entity, and introducing greater flexibility in the designation of hedging instruments and hedged items.

14.1.3 Scope of application and structure of the standard

IFRS 9 applies, in general, to all **financial assets** and **financial liabilities** as defined by IAS 32, including loans and advances to customers and banks, debt securities, equity instruments, derivatives, and certain forms of hybrid contracts.

Excluded from the standard are investments in subsidiaries, associates, and joint ventures accounted for in accordance with IFRS 10, IFRS 11, and IAS 28, as well as certain non-financial liabilities governed by specific standards.

14.2 Classification and measurement of financial instruments

14.2.1 Guiding principles

IFRS 9 bases the **classification of financial instruments** on two conceptual pillars: the **business model** and the **SPPI test**.

The **business model** describes how the entity manages, at the portfolio level, financial assets to generate cash flows (hold to collect, hold with possible sales, active management for trading). It is inferred from *objective evidence* such as management policies, risk management systems, and sales history, and not from an *ad hoc* choice for the individual instrument.

In parallel, the **SPPI test** (*Solely Payments of Principal and Interest*) requires that the contractual cash flows are *solely payments of principal and interest*, reflecting an investment in a simple debt instrument. The presence of components linked to equity indices, non-linear derivatives, or complex forms of leverage may invalidate the SPPI requirement, requiring classification at **fair value through profit or loss (FVTPL)**.

Measurement categories for financial assets

From the combination of the **business model** and passing the **SPPI test**, three measurement categories for financial assets arise:

- **Amortised cost**: applicable when the management objective is to *hold the instruments to collect the contractual cash flows* and such flows are solely payments of principal and interest (SPPI).

Financial assets at **amortised cost** mainly include **amounts due from customers**, **loans** and **debt securities** held for investment to maturity. They are initially recognised at **fair value** including transaction costs, premiums and discounts, and subsequently measured at amortised cost using the **effective interest method**, net of **allowances for expected credit losses** under the ECL model.

- **Fair value through Other Comprehensive Income (FVOCI)**: used when the business model is *hold to collect and sell*, that is, it combines the collection of contractual cash flows with the possibility of sale, provided the flows meet the SPPI test.

  Assets classified as **FVOCI** are typically **debt securities** managed in a mixed *hold to collect and sell* model. **Interest** calculated using the effective interest rate and **expected credit losses** are recognised in profit or loss, while **fair value changes** (net of allowances) are recognised in *Other Comprehensive Income* and recycled to profit or loss upon sale or derecognition.

- **Fair value through profit or loss (FVTPL)**: residual category, which includes assets held for trading.

  Financial assets classified at **FVTPL** include instruments held primarily for **trading** purposes, namely **complex instruments** that do not pass the SPPI test and assets for which the *fair value option* has been exercised. **Fair value changes** and the related income and expenses are recognised directly in profit or loss, consistent with a management approach oriented to **market value**.

14.2.2 Amortised cost, effective interest rate and derecognition

Effective interest rate and amortisation. The **effective interest rate** is the rate that exactly discounts the estimated future cash flows over the expected life of the instrument to its **initial carrying amount**. It determines the **amortised cost**, allowing the allocation over time, in a systematic and consistent manner, of interest, premiums, discounts, transaction costs, and fees. This mechanism ensures an economically meaningful representation of the instrument's **effective return**.

Contract modifications and derecognition. In the presence of changes to the **contractual terms** of a financial asset or liability, the entity must assess whether the modification is **substantial**. Substantial modifications result in the derecognition (*derecognition*[4]) of the original instrument and the recognition of the new contract as a separate instrument, with any difference recognized in profit or loss. If the modification is not substantial, there is no derecognition: the carrying amount is adjusted and the new effective interest rate incorporates the updated terms.

14.3 The expected credit loss (ECL) impairment model

IFRS 9 replaces the impairment model based on the *incurred loss* with a model based on **expected credit losses** (*Expected Credit Losses*, ECL), in order to accelerate the recognition of impairments and make the financial statements more responsive to the forward-looking evolution of credit risk.

[4] **Derecognition** of a financial asset occurs when the **contractual rights** are extinguished or transferred to third parties. Similarly, a **financial liability** is derecognised when the contractual obligation is extinguished, cancelled, or replaced with another substantially different obligation.

14.3.1 From the incurred loss model to the ECL model

IFRS 9 introduces an **expected loss** model that requires estimating, from the initial recognition of the instrument, the expected credit losses over a horizon consistent with the degree of credit deterioration. Two main measures are distinguished:

- **12-month ECL**: expected losses arising from default events that are possible within the next 12 months;
- **Lifetime ECL**: expected losses over the instrument's entire remaining life.

The move from 12-month to Lifetime ECL is driven by evidence of a *significant increase in credit risk* (SICR). The approach is explicitly **forward-looking**, requiring the use of probability-weighted macroeconomic scenarios.

14.3.2 The three-stage model

Stages. The impairment model is structured into three **stages** that reflect the degree of deterioration in credit risk:

- **Stage 1**: performing exposures for which no significant increase in credit risk has been observed compared with initial recognition; 12-month ECLs are calculated.
- **Stage 2**: exposures that have exhibited a **SICR**, while not being in default; the calculation of ECLs over the entire remaining lifetime is required.
- **Stage 3**: **credit-impaired** exposures, for which a default event or objective evidence of deterioration has occurred; Lifetime ECLs are estimated and interest is, as a general rule, calculated on the net carrying amount, to avoid recognizing income on amounts that are expected not to be collected.

Rules for transitions between stages and cures. The transition between stages is driven by observing changes in credit risk relative to *origination*. The transfer from Stage 1 to Stage 2 occurs when the increase in risk is deemed significant; the move to Stage 3 is tied to classification as *credit-impaired*. IFRS 9 allows the **return to less deteriorated stages** (cures) when the conditions that had led to the deterioration no longer hold, subject to an adequate observation period and a structural improvement in the risk profile.

14.3.3 Calculation of expected credit losses

Fundamental components. ECLs are calculated as the *present value* of expected credit losses, typically based on three components:

- **Probability of Default (PD)**: probability that a default occurs over a given horizon;
- **Loss Given Default (LGD)**: portion of the exposure expected to be lost in the event of default, net of expected recoveries;
- **Exposure at Default (EAD)**: amount of the exposure expected at the time of default.

The combination PD $\times$ LGD $\times$ EAD, discounted at the effective interest rate, provides the measure of ECLs over one or more time horizons.

Time horizon and simplified approach. For Stage 1 exposures, 12-month ECLs are used; for Stage 2 and Stage 3, lifetime ECLs are used. IFRS 9 provides a **simplified approach** for trade receivables, receivables arising from contracts with customers (IFRS 15), and lease receivables (IFRS 16), for which Lifetime ECLs are applied directly without staging, often using impairment matrices for past-due *buckets*.

Macroeconomic and forward-looking scenarios. ECLs must reflect **reasonable and supportable macroeconomic scenarios**, probability-weighted (e.g., baseline, adverse, and favourable scenarios). Estimates of PD, LGD, and EAD are therefore conditioned on these scenarios, in order to capture the effect of expected economic conditions on credit risk. The use of forward-looking information is one of the most significant departures from the IAS 39 model.

14.3.4 Identification of SICR and default

Quantitative and qualitative indicators. The identification of **SICR** is based on a set of indicators:

- **quantitative**: increase in PD relative to origination, deterioration of internal rating, widening of spreads;
- **qualitative**: granting of *forbearance* measures, signs of financial difficulty, inclusion in *watchlist* or *early warning* lists.

As a *backstop*, the principle provides that exceeding 30 days past due is at least a strong indicator of SICR, unless there is evidence to the contrary.

Accounting default and discretionary judgments. The definition of **default** for the purposes of IFRS 9 is generally consistent with that used for regulatory purposes (e.g., 90 days past due or *unlikely to pay* indicators). The entity must document its definition of default and ensure consistency in its application. The model allows the use of management **judgement**[5] and **overlays**[6] to supplement quantitative results (e.g., in the presence of shocks not fully captured by historical data).

14.3.5 Write-offs and recoveries

The **write-off** represents the accounting derecognition of the exposure (or of a portion thereof) when no further contractual recoveries are expected, even if legal rights persist. The amount written off is recognised in profit or loss and reduces the gross carrying amount. Any **subsequent recoveries** are recognised as reversals of impairment in profit or loss in the period in which they occur, without reinstating the instrument's original value.

14.3.6 Links with prudential regulation

The parameters used for IFRS 9 are generally of a **point-in-time** nature and incorporate macroeconomic scenarios, whereas regulatory parameters (e.g., in IRB models) are often *through-the-cycle* and include different margins of conservatism. This can generate divergences between accounting ECL and capital requirements for credit risk. IFRS 9 adjustments affect **equity** and, indirectly, capital adequacy ratios. For banks, this requires consistency with risk management processes and with the **ICAAP**, **RAF** and **stress test** frameworks, so that the expected

[5] Judgement is a reasoned professional assessment that interprets context, emerging risks, and scenario consistency.

[6] Overlays are quantitative adjustments to model outputs when the latter do not adequately capture risks, shocks, or forward-looking information.

loss estimates used for accounting purposes are integrated into decision-making mechanisms, while respecting the different informational objectives between the accounting and prudential frameworks.

14.4 Hedge accounting under IFRS 9

IFRS 9 introduces a **hedge accounting** model aimed at aligning the accounting representation of hedges with the actual management of financial risks. Although it introduces significant changes compared with IAS 39, the standard allows, on an optional basis, the continued application of the previous hedge accounting rules in certain contexts, particularly for the hedging of interest rate portfolios (macro-hedging), pending completion of the specific project on *dynamic risk management*.

14.4.1 Objectives of the new hedge accounting

Alignment with economic risk management. IFRS 9's main objective is to **bring hedge accounting closer to the actual management of risks** undertaken by the intermediary. The standard aims to ensure that the financial statements more faithfully reflect the **hedging strategies** adopted (on interest rate risk, foreign exchange risk, commodity price risk, etc.), reducing the gap between accounting representation and the underlying economic logic. The designation of hedging relationships is therefore more tightly anchored to the instruments and positions actually used in internal *risk management*.

The standard also allows the designation of a wide range of **risk components** (e.g., benchmark components of interest rates or foreign exchange risk) and the hedging of **portions** or *layers* of positions, making it possible to represent in the financial statements hedging strategies that are more sophisticated and widespread in operational practice.

Option to continue applying IAS 39. IFRS 9 allows entities that deem it appropriate to **continue to apply the IAS 39 hedge accounting rules** for specific cases, in particular for *fair value macro hedges* of interest rate risk on the banking book. This option is transitional in nature and is linked to the still-evolving project for a comprehensive portfolio-level *macro-hedging* framework.

14.4.2 Types of hedging and hedging instruments

Types of hedging relationships. IFRS 9 confirms three fundamental types of **hedging relationships**:

- **Fair value hedge**: hedging of exposure to changes in the fair value of an asset, liability or firm commitment attributable to a specific risk;
- **Cash flow hedge**: hedging of exposure to variability in highly probable future cash flows, attributable to a particular risk associated with a recognized asset or liability or a forecast transaction;
- **Hedge of a net investment** in a foreign operation, aimed at mitigating foreign exchange risk on investments in different currencies.

Hedging instruments. The **hedging instruments** are primarily **derivatives** (swaps, forwards, options, futures), which allow targeted management of market risks. IFRS 9 also permits, in certain cases, the use of **non-derivative instruments** as hedging instruments, e.g., for hedging foreign exchange risk, provided they meet the designation and measurement requirements prescribed by the standard.

14.4.3 Hedged items and hedge effectiveness

Hedged items. The **hedged items** may include assets or liabilities recognized in the statement of financial position; highly probable forecast transactions (such as future purchases or sales in a foreign currency); firm commitments. It is also possible to designate a specific **risk component** as the hedged item, e.g., the benchmark interest-rate risk or the foreign exchange risk relating to a particular currency.

Designation and documentation requirements. To apply hedge accounting, the entity must prepare **formal documentation** at the designation date, including the identification of the hedging instrument and the hedged item, the nature of the risk being hedged, the risk management strategy, and the methods for assessing hedge effectiveness. Such documentation provides the basis for demonstrating that the hedging relationship is consistent with the *risk management* policies and meets the criteria of the standard.

Assessment of effectiveness. IFRS 9 requires that the hedging relationship be **economically coherent** and that the value of the hedging instrument and the hedged item move in opposite directions in response to the hedged risk. The numerical constraint is no longer required, but the entity must establish quantitative and qualitative criteria to demonstrate that:

- there is an **economic relationship** between the **hedged item** and the **hedging instrument**;
- the effect of **credit risk does not dominate the changes in value** attributable to the hedged risk;
- the **hedge ratio** is **consistent** with the **risk management strategy**.

14.5 Operational, implementation, and governance aspects

The adoption of IFRS 9 does not end with understanding the accounting principles, but requires a structured **implementation project** that involves processes, information systems, risk models, and governance frameworks. In particular, for supervised intermediaries, the introduction of the expected credit loss (ECL) model entails an integrated rethinking of data flows, measurement methodologies, and organizational responsibilities throughout the entire credit life cycle.

14.5.1 IFRS 9 implementation project

Project phases: analysis, design, development, go-live. An IFRS 9 implementation project typically unfolds in four main phases. In the **analysis** phase, the entity performs a *gap assessment* between the existing framework (IAS 39, credit processes, systems) and the requirements of the new standard, identifying the areas of impact (classification, impairment, hedge accounting). The **design** phase defines the target architecture: business model, SPPI criteria, SICR metrics, data flows, ECL models, operational responsibilities. In the **development** phase, models are implemented, accounting and risk applications are adapted, and data marts and interfaces are configured. Finally, the **go-live** is preceded by integrated testing (user acceptance test, parallel run) and by a period of joint operation under the old/new regime to verify the consistency of results and calibrate the impacts on the financial statements.

Definition of business model, SPPI criteria, SICR and default. A central element of the design is the formalization of the **business model** for portfolio management (hold to collect, hold to collect and sell, trading), from which the measurement categories derive. At the same time, it is necessary to define operational criteria for the application of the **SPPI test**, including guidelines for the assessment of optional clauses, indexation features, and structured components. For impairment, the entity must establish an **SICR policy** (significant increase in credit risk) that combines quantitative and qualitative indicators, as well as align and document the **definition of default** for accounting purposes, as a rule consistent with the regulatory definition, specifying any differences.

Policies on write-off, forbearance and non-performing exposures. Implementation also requires clear policies regarding **write-off**, **forbearance**, and **management of non-performing exposures**. The criteria for writing off the exposure from the financial statements (absence of reasonable expectations of recovery, even where residual legal rights remain), the types of *concessions* that qualify as forbearance for IFRS 9 purposes, and the exit criteria from non-performing status (cures) must be defined. These policies must be consistent with supervisory regulation and credit risk management practices, in order to avoid arbitrage between the accounting and prudential frameworks.

14.5.2 Data, systems and models

Information requirements. The ECL model is highly **data-driven**. Historical data on borrower behavior (collection patterns, rating migrations, defaults), information on collateral and recoveries, together with **macroeconomic data** and scenario data (GDP, interest rates, unemployment, real estate prices, etc.), are required. The quality, historical depth, and granularity of such data directly affect the reliability of PD, LGD, and EAD estimates, as well as the robustness of forward-looking analyses.

Internal models for accounting purposes. Many entities, particularly banks, rely on **internal models** to estimate risk parameters for IFRS 9 purposes. While they may draw on the methodological foundation of regulatory models, the accounting parameters must reflect a *point-in-time* approach, incorporate macroeconomic scenarios, and be consistent with the ECL time horizon (12 months or lifetime).

Integration with management, regulatory, and risk systems. At the application level, IFRS 9 solutions must be integrated with **management systems** (general ledger, master data, front-office systems), with **regulatory systems** (RWA calculation, supervisory reporting) and with **risk management systems** (credit risk, ALM, stress testing). The goal is to avoid redundancies and inconsistencies, fostering a synergistic use of information and, where possible, a convergence of databases and calculation processes.

14.5.3 Governance, controls, and the role of corporate functions

Role of management and operational functions. The proper application of IFRS 9 requires a clear **allocation of responsibilities**. The **management** defines accounting and risk policies, approves ECL models, and oversees P&L and balance sheet impacts. The **risk function** contributes to methodological design, the calculation of estimates, and the monitoring of parameters; the **accounting function** ensures correct financial statement presentation, reconciliation with the accounting ledgers, and consistency with other IFRS standards. Business operating units provide qualitative inputs and manage frontline decisions on forbearance and classifications.

Model validation, backtesting, and monitoring. IFRS 9 models must be subject to periodic **independent validation** and **backtesting**, in order to verify their predictive capability and adequacy over time. This includes analyses of deviations between expected and realized losses, sensitivity testing to macroeconomic scenarios, and assessment of the stability of parameters and SICR choices. Monitoring results feed any methodological revisions and the calibration of management *overlays*, to be documented and approved according to structured governance processes. Finally, the **internal audit** function assesses the adequacy of the IFRS 9 framework as a whole (processes, controls, systems, models), verifying compliance with internal policies and regulatory requirements.

14.5.4 Financial communication and impacts on the financial statements

IFRS 7 requires **detailed disclosure** on the ECL methodologies adopted, on segmentation by stages, on movements in impairment allowances (opening/closing reconciliations), and on the main financial risks (credit, liquidity, market), in order to make the entity's risk profile transparent to users of the financial statements. IFRS 9 impairment allowances have a direct impact on **equity** and can increase **earnings volatility**, depending on the dynamics of risk parameters and macroeconomic scenarios. This makes clear financial communication on the effects of the ECL model and its sensitivities crucial.

Chapter 15

Supervisory Review and Evaluation Process SREP

15.1 Purpose and scope

The **Supervisory Review and Evaluation Process (SREP)** is the cornerstone process of **European banking supervision**, conducted within the framework of the **Single Supervisory Mechanism (SSM)**. It adopts a **holistic** approach to assessing an institution's risks, integrating **analysis**, **assessment** and **intervention** into a single framework. The main objective is to assess the **soundness** and **sustainability** of the **risk profile**, the **consistency of the business model** and the **adequacy** of **governance**, **management** and **control** arrangements.

During the SREP, the supervisory authority examines a set of **risks** (*credit*, *operational*, *liquidity*, *market*, *counterparty*, etc.) and, at the end of the process, issues a **formal decision** containing **remedial measures**, **objectives** and **timelines** to address the identified deficiencies, including through additional capital and qualitative requirements.

15.1.1 Assessment areas

The **SREP** assesses banks along four **main perspectives** that are mutually consistent and interdependent:

- **Business model analysis (Business Model Analysis, BMA)**: examines the model's **viability**[1] over a horizon of at least **12 months**, with particular focus on the **ability to generate profits** and to maintain an **adequate level of capital**. At the same time, it assesses the strategy's **sustainability** over a forward-looking horizon of at least **three years**, taking into account factors such as competitive positioning, revenue structure and resilience to adverse scenarios.
- **Risks to capital and internal parameters**: consists of the **quantitative** assessment of **material risks** for capital purposes, with reference to the measurement of exposures via **risk-weighted assets** (*Risk-Weighted Assets*, **RWA**). This area also includes the assessment of **internal models** and consistency with the **ICAAP** and **ILAAP** processes.
- **Governance and risk management**: assesses the **adequacy of the corporate governance system** and **internal controls** relative to the institution's **nature**, **size** and **complexity**. This area covers the organization and effectiveness of **committees** and **management bodies**, the quality of **risk management processes**, the **traceability of decisions** and the existence of structured **monitoring** processes (of models and profiles).

[1] The viability of a model is the model's ability to remain usable and fit for purpose over time.

- **Data and IT infrastructure**: focuses on the **robustness** of **risk data aggregation** capabilities and **reporting** practices, both under **normal** conditions and in **stress** situations. The analysis includes checking **data quality**, as well as identifying any **organizational** or **cross-border** obstacles that could compromise the **distribution**, **consistency** and **usability** of information.

15.1.2 Indicators and assessment methods

The **indicators** and **assessment techniques** used by the supervisory authorities are structured around the four main areas of the **SREP**, in line with the **risk-based** and **proportionate** approach.

- **Business model and sustainability**: **profitability indicators** (net interest margin, operating margin, ROE, RORAC) and the **ability to maintain an adequate level of capital** in the short term (**viability** at 12 months) are analyzed. In parallel, the **sustainability** of the strategy is assessed over a horizon of at least **three years**, from a **forward-looking** perspective that includes **scenario analysis** and **stress tests** on the evolution of the business model, the revenue base and the risk profiles.

- **Risk quantification and RWA**: the supervisory authority uses measures based on **Risk-Weighted Assets (RWA)** for **credit risk**, **counterparty risk** and **market risk**, analyzing the level and dynamics of capital consumption. Checks are carried out on the **consistency** and **stability** of **internal parameters** and on the behavior of **exposures** across the different portfolio segments, also through comparisons over time and against peers.

- **Performance of internal models**: for banks that use **internal models**, the authority analyzes the **distribution of defaults** versus **non-defaults** by **grade** or **segment**, making use of **graphical analyses** (such as **ROC curves**, **empirical cumulative distribution functions** and other discrimination metrics) and **statistical tests** based on the data requested during the inspection. At the same time, a **review of the documentation** (completeness of modeling choices, **change log**, methodological justifications) and a verification of the **consistency of definitions** (default, forbearance, NPE, etc.) are carried out.

- **Governance, data and IT**: the **processes** established for **monitoring**, **validation** and **controls** over models and risk data, the **quality** and **timeliness** of **reporting**, as well as the capacity of the **Management Information System (MIS)** to **aggregate** and **distribute** risk data within the group, are evaluated. Particular attention is paid to identifying any **constraints** to **aggregation** and **usability** of data in **cross-border** contexts, which may hinder an integrated view of group risks.

15.1.3 SREP outcomes and prudential impacts

The **SREP** concludes with a **formal decision** notified to the institution, in which the supervisory authority defines **priority objectives**, **measures** and **deadlines** for the **remediation** of the shortcomings identified. The outcomes of the process affect, in particular:

- **own funds requirements**, through the setting of **Pillar 2** components (e.g. **P2R** and, as a complement, **P2G**);

- the **action plans** required of the bank, which include **corrective measures** proportionate to the **vulnerabilities** identified (strengthening of controls, review of internal models, adjustment of processes);

- **model governance**, with possible **limitations on the use** of internal models, requests for **methodological revision** and **strengthening** of the **validation** and **audit** functions.

In parallel, the **Targeted Review of Internal Models (TRIM)** (2016–2021) constituted a **targeted review** of the **internal models** used by significant banks in the euro area. Over the course of the TRIM project, the supervisor conducted **in-depth inspections** on the models, producing a structured list of **findings** and **corrective actions** to be incorporated into methodologies and control processes. The results of TRIM integrate with the outcomes of the SREP, contributing to the definition of **prudential requirements** and to the overall strengthening of the **quality of internal models**.

15.1.4 Requested information, reporting and internal responsibilities

Supervisory authorities may request, including in an **inspection** context, a structured set of **information** and **evidence** to support the assessment of models and internal processes. By way of example, the most frequent requests include:

- **complete model documentation**, including an explanation of the **methodological choices**, a description of the underlying **assumptions** and the **version history** (*change log*); an **overview** of the set of models used, with a **mapping of weaknesses** and of the main areas of methodological risk;
- **input and output datasets**, accompanied by analyses of the **distribution of defaults** by **grade** or **segment** and of the **realized rates**, including the provision of **separate files** to enable **independent recalculations** on samples or cases selected by the inspection team;
- **walkthroughs of internal calculation processes**, with an end-to-end illustration of the phases of **data acquisition**, **transformation**, **parameter calculation** and **aggregation of results**; where necessary, the ability to perform **independent replications** of the results starting from institutional databases is required;
- **analyses of special cases** for **LGD** estimates (e.g., **cured** exposures, values close to zero, negative or very high, **collateralized**, **restructured**, **sold** transactions, with possible **currency mismatch**), in order to verify the consistency of modeling rules with recovery and credit management practices.

The **reporting timelines** and implementation of **remedial actions** are defined in the **supervisory decision** and, for aspects related to **data** and **IT**, are often agreed with **management** in order to realistically plan the **improvements** to be implemented.

15.1.5 Operational management of inspections and remediation plans

Effective management of supervisory **inspections** and subsequent **remediation plans** requires a sequence of codified operational steps and a clear assignment of responsibilities.

- **Preparation**: phase of **documentation consolidation** (policies, methodological manuals, reports), definition of **scopes** and **operational definitions**, preparation of an **inventory of models** and relevant **datasets**. In this phase it is good practice to perform an **internal review** of **key metrics** (e.g., grade distributions, **ROC curves**, **empirical CDFs**) and collect evidence on existing **monitoring processes** and **controls**.
- **Conduct of the inspection**: the institution must be able to present **end-to-end walkthroughs** of the **calculations** and ensure **traceability of data transformations**

along the entire chain (from the source to reporting systems). It is also essential to enable **support for independent recalculations** by the supervisory authority and provide **timely responses** on implementation, methodological and organizational aspects.

- **Closure and remediation**: following the inspection findings, an **action plan** is drawn up that specifies **actions**, **responsibilities**, **milestones**, **dependencies** and **effectiveness metrics**. The findings are **prioritized** based on their materiality and the underlying risk, and the actions are integrated into the **model governance processes** and **internal reporting**. **Ongoing monitoring** continues until the **formal closure** of the findings by the supervisory authority, in compliance with the **set deadlines**.

For aspects related to **data** and **IT**, supervisors assess the **robustness** of **aggregation** and **reporting** capabilities both under **business-as-usual** conditions and in **stress** situations. In coordination with the **institution's management**, **strategies** and **timelines** are agreed for **infrastructure enhancements**.

15.2 Change policy, significance metrics and modification process

15.2.1 Change policy and classification

Institutions adopting the *Internal Ratings Based* (**IRB**) approach must have a formally approved **change management policy** (*change policy*) consistent with **Commission Delegated Regulation (EU) No. 529/2014** and **Commission Delegated Regulation (EU) 2022/439**, which define **modalities** and **uniform criteria** for the **classification of changes** to internal models. This policy forms an integral part of the **model governance** framework, as it sets out **responsibilities**, **decision-making flows** and **control safeguards** aimed at ensuring that each change is adequately **tracked**, **assessed** and **approved** according to its **prudential materiality**.

In accordance with Article 84 of the **RTS 2022/439**, the *change policy* must include:

- **Roles and responsibilities:** precise identification of the **parties involved** in the process (**development** function, *model validation*, **risk management**, *internal audit*, **decision-making bodies**) and clear definition of the related **reporting lines** up to the **management body**;

- **Metrics and materiality levels:** definition of **quantitative** and **qualitative** criteria for the **classification of changes**, consistent with the **relevance** of their effects on the **regulatory parameters** (**PD**, **LGD**, **CCF**, **EAD**) and on overall **capital requirements**;

- **Approval and documentation process:** description of the **operational phases** (proposal, **assessment**, **validation**, **approval**, **communication** to the competent authorities), including the rules for **prior** or **subsequent notification** depending on the **materiality category**;

- **Classification criteria:** definition of the different categories of changes — **material** and **non-material** — according to the standardized methodology set out by the **RTS** and consistent with Article **143** of the **CRR**.

Classification principles and prevention of arbitrage. The general principle laid down by **RTS 2022/439** provides that the **classification** of a change must be carried out on the basis of the **highest potential level of materiality**, even when the quantitative impact is not yet fully observable. It is expressly prohibited to **split** a significant intervention into several smaller changes solely to **circumvent regulatory thresholds** or to **simplify** the authorisation process (**prohibition on "splitting" material changes**). Where there is **interpretative doubt** or **uncertainty** about the overall impact, the change must be classified in the **highest materiality category**, according to a principle of **prudence** and **regulatory conservatism**.

Four-eyes principle. The **European Central Bank (ECB)**, in its *Guide to Internal Models*, requires the adoption of a principle of **independent dual control**, which entails:

- the **assessment of the change** by the **model development or maintenance function**, responsible for proposing a **preliminary classification** of the change in terms of materiality;
- the **independent confirmation** by a **separate control function** (validation or compliance), tasked with verifying the **correctness of the classification**, the **methodological consistency** and the **traceability** of the criteria applied.

This **cross-checking** mechanism is designed to prevent **regulatory arbitrage**, strengthen the **operational accountability** of the development functions and ensure the **transparency of the decision-making process**. The outcome of the **dual control**, including the **rationale**, **regulatory references** and **supporting evidence**, must be **formally documented** and retained for the purposes of **internal audit** and **supervisory review**.

Chapter 16

Disclosure & Reporting (Pillar 3, COREP and FINREP)

16.1 Purpose and scope

The **Third Pillar disclosure** (*Pillar 3*) governs **prudential transparency** as **market discipline**, in a **complementary** manner to the **minimum capital requirements** (*Pillar 1*) and the **risk management** and **internal control safeguards** (*Pillar 2*). Within the **European Union** framework, the Third Pillar disclosure is integrated with **COREP** and **FINREP reporting**, as defined by the **Implementing Technical Standards** (**ITS**) issued by the **EBA**. The main **purposes** include:

- the **comparability** of information across institutions, facilitating cross-analysis by investors, counterparties, and other interested parties;
- the **reduction of information asymmetries** between supervised intermediaries and the market, through a minimum set of public data on **risks**, **capital**, and **liquidity**;
- the **strengthening of confidence** of **stakeholders** in institutions' soundness and prudent risk management.

The **scope** of application of the Third Pillar disclosure is defined by **Part Eight of the CRR** and refers to the **prudential consolidation perimeter**, which may differ from the **accounting consolidation** under **IFRS**. In this context, institutions must clarify the consolidation scope, the main entities **included** and **excluded**, and the underlying **regulatory reasons**. The **implementation principles** of the Third Pillar disclosure include, among others:

- **materiality**, understood as the publication of information relevant to the assessment of the institution's risk and capital profile;
- **proportionality**, with a level of detail calibrated according to **size**, **nature**, and **complexity** of the institution;
- **completeness** and **clarity**, so that the information provided is sufficiently **comprehensive** and readily **understandable**;
- **timeliness**, through **timely** publication and, where applicable, **more than once a year** of key prudential metrics;
- **comparability**, through the use of **tables** and **standardized templates** prepared by the EBA, which facilitate comparisons across institutions and over time.

The disclosures cover both **qualitative content** and **quantitative content**, organized according to **common formats** and **regulatory templates** where applicable, with the aim of providing a consistent representation of the institution's **risk profile**, **capital**, and **liquidity position**.

16.2 Structure of the Pillar 3 disclosure

The **Pillar 3** disclosure, required by **Part Eight of the CRR** and the related **Implementing Technical Standards** (ITS EBA), is organized so as to combine **descriptive elements** and **quantitative information**. The overall structure must enable **investors**, **counterparties** and other **stakeholders** to understand:

- the **objectives** and **risk management policies**;
- the **capital profile** (levels of CET1, Tier 1 and Total) and the **liquidity position**;
- the **linkage** between the published data, the **supervisory reports** and the key financial statement figures.

16.2.1 Qualitative content

The **qualitative sections** describe the **risk governance and management** framework, with particular attention to **risk objectives**, **risk-taking and mitigation** policies (limits, thresholds, diversification strategies), the functioning of the **internal control system**, and the allocation of **responsibilities** between corporate bodies and control functions.

They set out the principles of the **risk appetite framework (RAF)**, the responsibilities of the **management bodies** and **control functions**, as well as the **organizational structure** by risk type: **credit risk**, **market risk**, **counterparty risk**, **interest rate risk in the banking book (IRRBB)**, **liquidity risk**, **operational risk**, **ESG risks** and **reputational risk**.

The qualitative disclosure also outlines the **prudential scope of consolidation** and any **differences with respect to the accounting consolidation**, highlighting the main **entities included and excluded** and the **regulatory reasons**. The following are described:

- the use of **internal models** (e.g. IRB, IMM, IMA) and **standardized approaches** for the various risk types;
- the **validation** and **approval** criteria by the **competent authority**;
- the **integration** between **internal measurements**, **regulatory requirements** and **business processes**.

Also part of the qualitative content are **capital management policies** (including **capital buffers**), **funding strategies** and **liquidity management**, the main **risk mitigation techniques** (**guarantees**, **collateral**, **hedging derivatives**, **securitizations**) and how these instruments are **governed**, **measured** and **included** in the **overall risk profile**. The **documentation** must ensure **consistency** with the information disclosed in other areas of **external reporting**.

16.2.2 Quantitative content

The **quantitative sections** present, predominantly in **tabular** form and in accordance with the *templates* set out by the **technical standards**, the main **capital metrics** and **risk metrics**, with an appropriate level of **detail** by **portfolio**, **exposure class**, **calculation methodology** and **jurisdiction**. By way of example, this includes information on:

- **own funds** (Comprehensive Own Funds), with a distinction among **CET1**, **AT1** and **Tier 2**;
- **capital requirements** for the different types of risk;
- **risk-weighted assets (RWA)**;
- **capital buffers** (conservation buffer, countercyclical buffer, buffer for O-SII/G-SII and systemic risk buffer);
- **key indicators** such as the ***Total Capital Ratio***, the ***CET1 Ratio*** and the ***Leverage Ratio***.

The main **quantitative tables** cover:

- the **breakdown of portfolios** by **counterparty type** (e.g. retail customers, corporates, financial institutions, public administrations);
- the **breakdown of portfolios** by **instrument type** (mortgages, credit lines, debt securities, derivatives);
- the **distinction by measurement approach** (**standardised**, internal models);
- the **presentation of exposures** subject to **risk mitigation techniques** (guarantees, **collateral**, **hedging derivatives**), to **securitisations** or to other forms of **risk transfer**;
- the **concentration** by **economic sector** and **geographic area**;
- **credit quality** (e.g. distribution by **rating classes** or **risk classes**);
- the levels of **non-performing exposure** (NPE) and **performing exposure**, with any information on **forbearance** and **coverage** (*coverage ratio*).

A central element of the **quantitative content** is the **numerical reconciliation** between the **public disclosure** and the **supervisory reports**, carried out through specific **"bridge" tables** that link the accounting **balance sheet** and **income statement** to **own funds** and **regulatory RWA**. These reconciliations must highlight in a **transparent** manner:

- **classification adjustments** (differences between the accounting and prudential perimeters);
- **valuation adjustments** (prudential filters, deductions, other regulatory adjustments).

In this way, **consistency** is ensured between **Pillar 3** information, **COREP** and **FINREP** data and other **regulatory disclosures** made to the **competent authority**, allowing users to understand how **accounting** figures are transformed into **prudential metrics**.

16.3 Harmonized reporting

16.3.1 Structural differences between COREP and FINREP

Although both fall within the framework of **EBA harmonised reporting** and share a set of **common definitions** through the **Single Rulebook**, **COREP** and **FINREP** address different **purposes**, **information bases**, **scopes** and **reporting logics**.

Scope	**COREP**	**FINREP**
Purpose	Predominantly **prudential** in nature; provides the supervisory authority with information on **capital strength**, **own funds**, **capital requirements** and **RWA**, **credit risk** (**SA/IRB**), **market**, **operational**, **counterparty** risk, **leverage ratio**, **LCR/NSFR**, **large exposures**.	Predominantly **accounting-based** in nature; transmits **harmonised IAS/IFRS accounting data** (**balance sheet**, **income statement**), **credit quality** indicators, **NPEs**, **movements in impairment allowances**, information on **derivatives**, **equity instruments**, **cash flows**.
Information basis	Data built on **prudential** logics: **risk-weighted exposures**, **CCF**, **prudential filters**, **deductions from CET1**, **supervisory haircuts** and other regulatory adjustments.	**IFRS consolidated accounting values**, incorporating **fair value**, **IFRS 9 impairment**, **accounting classifications** and **recognition and measurement** criteria under **IAS/IFRS**.
Scope of consolidation	**Prudential** scope under **CRR**: **exclusion** of certain **non-financial entities**, specific treatment of **insurance undertakings**, use of **proportional consolidation** or **deduction**.	**Accounting** scope under **IFRS 10/11/12**; may **include** entities **not relevant for prudential purposes** or **exclude** entities included in the regulatory consolidation.
Units of measure and granularity	Segmentation by **regulatory exposure classes** (*corporate*, *retail*, *institutions*, *sovereigns*), **IRB** segmentations, **risk parameters** (**PD**, **LGD**, **EAD**), **capital requirements** and regulatory **buffers**.	**Accounting** classification (*loans and advances*, *debt securities*, *derivatives*, etc.), **breakdown of NPEs** and detail of **movements in allowances**, distinction between **performing/non-performing** exposures according to the **EBA NPE** definition.
Reported values	**Regulatory figures**: **CET1**, **AT1**, **Tier 2**, **RWA**, **minimum requirements**, **prudential ratios**, **IRB parameters**.	**Accounting figures**: **financial statement values**, **fair value**, **IFRS 9 impairment**, **profit/loss** for the period and related components; basis for **reconciliation** to regulatory values.
Periodicity and frequency	Reports typically **monthly** and/or **quarterly**, with sometimes **higher** frequencies for **liquidity** indicators (e.g. **LCR**) and the **leverage ratio**, depending on the requests of the **competent authority** and the entity's **size**.	Reports generally **quarterly**, aligned with **IAS/IFRS** reporting cycles, with potentially higher frequencies for intermediaries of greater systemic relevance.
Impact on capital and supervision	Directly feeds **SREP**, **monitoring** of **minimum requirements** and **supervisory thresholds** (Pillar 1, **buffers**, additional **Pillar 2** requirements), providing an up-to-date view of **RWA**, **own funds**, **prudential ratios**.	Supports **structural analyses** of **asset quality**, **NPEs**, the **impairment process** (IFRS 9) and the **correlations** between **accounting figures** and **prudential profiles**, integrating the information framework for **micro-** and **macroprudential** supervision.

16.3.2 Definitions, databases and interoperability

Although they follow **different logic**, **COREP** and **FINREP** share a **harmonized** set of **definitions**, **information hierarchies** and **reporting instructions** thanks to the **Data Point Model (DPM)** and the **validation rules** of the **EBA**. These tools ensure **semantic** and **numerical** consistency across **accounting data**, **managerial data** and **prudential data**, enabling an **integrated** and **traceable** flow from the **internal source** to the *supervisory templates*.

16.3.3 Data Point Model (DPM)

The **Data Point Model (DPM)** represents the **logical–semantic structure of the data**: it defines **domains**, **attributes**, **constraints**, **hierarchical relationships**, **consistency formulas** and **aggregation rules**. It provides the basis for the **transformation** from **internal data** (accounting and risk) to the individual **cells** of the reporting *templates*. The **DPM** ensures that **COREP** and **FINREP** share **common concepts** (**exposure**, **collateral**, **counterparty**, **sector**, **country**, **instrument type**), while tailoring them according to **specific logics** (**prudential** vs **accounting**).

16.3.4 XBRL: mapping and validations

Reports are generated in **XBRL** format, in which each **cell** of the *template* corresponds to a **data point** defined in the **DPM**. **XBRL mapping** requires:

- the **reconciliation** with the applicable **accounting** or **prudential** scope;
- the **tracking of transformations** (*data lineage*) from **source systems** to the **reported data**;
- the **application** of **formal validation rules** (data type, domain) and of **consistency checks** (**cross-template** between **COREP** and **FINREP**);
- the **management** of **error**, **warning** and **rejection** outcomes returned by the **competent authority**.

16.3.5 Expected end-to-end flow

The **end-to-end process** for producing regulatory reports can be outlined as follows:

1. **extraction** of data from **accounting**, **management** and **risk** systems;
2. **preliminary checks** on **quality**, **consistency** and **completeness**;
3. **enrichments**, **aggregations** and **transformations** according to internal **metadata** and **DPM**;
4. **generation** of **XBRL** files via updated **EBA taxonomies**;
5. **submission** to the supervisory authority, management of **receipts** and **analysis of validation outcomes**;
6. **final archiving** with a complete **audit trail**, **versioning** and **reproducibility** checks on the reported data.

This framework ensures **transparency**, **traceability** and **reliability** across the entire lifecycle of regulatory information, from the **source data** to the **final report** to the supervisory authority.

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